

The terms proportional to $\log(\sqrt{s}/k^+)$ and the BFKL evolution of the leading-order cross section

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1 Question and answer

Question. Do for the terms proportional to $\log(\sqrt{s}/k^+)$ what was done for the terms proportional to $\log(\sqrt{s}/\Lambda)$:

1. collect, group by group and row by row, all four- ρ , three- ρ and two- ρ terms proportional to $\log(\sqrt{s}/k^+)$, using your rules $\log(\Lambda/k^+) = -\log(\sqrt{s}/\Lambda) + \log(\sqrt{s}/k^+)$ and $\log((\sqrt{s} - k^+)/\Lambda) = \log(\sqrt{s}/\Lambda) + \log(1 - k^+/\sqrt{s})$;
2. reorder the three- ρ terms into their symmetrized part plus two- ρ terms, and check that the symmetrized three- ρ (and four- ρ) terms cancel;
3. combine all two- ρ terms and compare them with the BFKL evolution of the leading-order cross section (Appendix E, Eq. (LOBFKL));
4. find and fix the reasons for any mismatch, and write the BFKL terms out explicitly.

Answer.

1. **What multiplies $\log(\sqrt{s}/k^+)$.** With your rules, the coefficient of $\log(\sqrt{s}/k^+)$ is

$$\boxed{[\text{coefficient of } \log(\sqrt{s}/k^+)] = \text{BFKL} \otimes \text{LO} - \text{JIMWLK} \otimes \text{LO}} \quad (1.1)$$

not $\text{BFKL} \otimes \text{LO}$ alone (Sec. 3). The reason is that $\sqrt{s}$ appears only as the upper limit of the p^+ integrals. So the coefficient of $\log \sqrt{s}$, which is the sum of the coefficients of $\log(\sqrt{s}/\Lambda)$ and $\log(\sqrt{s}/k^+)$, comes from $p^+ \gg k^+$ alone. That region is the evolution of the projectile, i.e. BFKL. The $\log(\sqrt{s}/\Lambda)$ terms were shown to give $\text{JIMWLK} \otimes \text{LO}$ in the previous note (`TwoRho_evolution`). Equivalently,

$$\begin{aligned} \frac{d^3 N}{d^3 k} \Big|_{\text{LL}} &= \log \frac{\sqrt{s}}{\Lambda} [\text{JIMWLK} \otimes \text{LO}] + \log \frac{\sqrt{s}}{k^+} [\text{BFKL} \otimes \text{LO} - \text{JIMWLK} \otimes \text{LO}] \\ &= \log \frac{k^+}{\Lambda} [\text{JIMWLK} \otimes \text{LO}] + \log \frac{\sqrt{s}}{k^+} [\text{BFKL} \otimes \text{LO}]. \end{aligned} \quad (1.2)$$

2. **Where the $\log(\sqrt{s}/k^+)$ terms are.** For every row, the coefficient of $\log(\sqrt{s}/k^+)$ is $B = P - A$. Here A is the row's coefficient of $\log(\sqrt{s}/\Lambda)$ (as in `Evolution_three_rho.tex` and `Evolution_two_rho.tex`). P is the coefficient of $\log \sqrt{s}$, i.e. the large- p^+ limit of the row's integrand. Many of these terms are explicit in the notes, through $\log(\Lambda/k^+)$, $\log(\sqrt{s}/(\Lambda + k^+))$ and $\log(\sqrt{s}/k^+ - 1)$. Others are hidden inside logarithms of transverse distances, e.g. $\log \frac{(\sqrt{s} - k^+)(x-z)^2 + k^+(x-w)^2}{\Lambda(x-z)^2 + k^+(x-w)^2}$, or inside the plus-prescription integrals. All of them are listed row by row in Sec. 4.
3. **Four- ρ .** There are no four- ρ terms proportional to $\log(\sqrt{s}/k^+)$. All four- ρ terms of the notes multiply $\log(\sqrt{s}/\Lambda)$ only (Sec. 5). The four- ρ terms are the same in the two regions $p^+ \ll k^+$ and $p^+ \gg k^+$ (checked).
4. **As written, the $\log(\sqrt{s}/k^+)$ terms do not work.** Counted pointwise at random points as in Table 4, 30 of the 42 symmetrized three- ρ structures are left over, and in the two- ρ terms 35 of 49 structures differ from $\text{BFKL} \otimes \text{LO} - \text{JIMWLK} \otimes \text{LO}$.
5. **Groups II and III are correct.** Their region- P parts agree row by row with an independent leading-log computation (Table 3). The overall sign of Group III, Rows III, V and VI is the one of `Evolution_three_rho.tex` (your correction of `Group_III.tex`). It applies to the $\log(\sqrt{s}/k^+)$ terms as well, and to Row III part 2 (fix F5 of the previous note).
6. **Fixes.** Group I needs one new fix, and three fixes of the previous note carry over:
 - P1 (**new**) Group I, Row III. The large- p^+ limit of the term with $\bar{\mathbb{B}}_3^\dagger$ integrated over $k^+ + \Lambda < p^+ < \sqrt{s}$ must be $+2K_h$. The notes give $-2K_h$, where $K_h = \mathcal{K}(x' - w') \cdot \mathcal{K}(z - w) K(x, y, z)$ (in units of $\frac{ig^4 f^{abd}}{16\pi^5 k^+}$). Your p^+ integration is correct (checked numerically). The sign error is in the integrand for $p^+ > k^+$. As a result, the $\log(\sqrt{s}/k^+)$ terms of Row III change by $+4i K_h$.
 - F3 The $\log(\sqrt{s}/k^+)$ terms of a row contain minus its $\log(\sqrt{s}/\Lambda)$ terms. The replacement of Row III part 2 (the Φ terms) by $\mathcal{D}$ therefore carries over with the opposite sign.
 - F5 Group III, Row III, part 2: the overall sign is reversed, for $\log(\sqrt{s}/k^+)$ as for $\log(\sqrt{s}/\Lambda)$.
 - F6 Group I, Row II, part 1: the bracket is $\frac{1}{\pi} \int_z [K(x, w, z) - K(w, w, z)]$, as before.

The fix F4 of the previous note (the α part of Row II) does *not* change the $\log(\vee/k^+)$ terms. That part multiplies $\log(\vee/\Lambda)$ only, so it is removed from A and from P together. The region- P comparison confirms F4 independently. F1 and F2 (`4rho.tex`) do not enter, because four- ρ terms have no $\log(\vee/k^+)$.

7. After the fixes everything cancels or matches exactly.

- The symmetrized three- ρ terms proportional to $\log(\vee/k^+)$ cancel exactly, in all 6 kernel structures and all 68 colour networks (Sec. 7). BFKL $\otimes$ LO has no three- ρ part.
- The two- ρ terms equal BFKL $\otimes$ LO–JIMWLK $\otimes$ LO exactly: pointwise in z , in all 18 kernel structures, with exact coefficients (Sec. 8). No two- ρ term is left over.

8. **The BFKL terms.** Sec. 9 gives BFKL $\otimes$ LO in operator form and fully written out. In the colour-singlet projection it equals your Eq. (LOBFKL) exactly (Table 2), so Appendix E is correct as written.

Everything uses your convention $U^{bc} = U^{cb}$.

2 Conventions

$$\mathcal{N} \equiv \frac{1}{(2\pi)^3} \frac{g^4}{16\pi^5} \frac{1}{k^+} \log \frac{\vee}{\Lambda}, \quad \mathcal{N}_P \equiv \frac{1}{(2\pi)^3} \frac{g^4}{16\pi^5} \frac{1}{k^+} \log \frac{\vee}{k^+}, \quad \mathcal{K}(X) \equiv \frac{X}{X^2}, \quad K(a, b, z) \equiv \mathcal{K}(a-z) \cdot \mathcal{K}(b-z). \quad (2.1)$$

All transverse positions except w, w' are integrated, and the phase $e^{-ik \cdot (w' - w)}$ is always present. The measured gluon is at w (amplitude) and w' (conjugate). The unobserved gluon is at z . The labels of the charges are those of your notes. A coefficient written as a number, e.g. $2i$, is in units of $\mathcal{N}$ (for $\log(\vee/\Lambda)$ terms) or $\mathcal{N}_P$ (for $\log(\vee/k^+)$ terms). Thus $\frac{ig^4}{8\pi^5 k^+} \rightarrow 2i$, $\frac{ig^4}{16\pi^5 k^+} \rightarrow i$, $\frac{g^4 N_c}{32\pi^4 k^+} \rightarrow \frac{\pi N_c}{2}$, and $\frac{g^4}{8\pi^4 k^+} \rightarrow 2\pi$.

Charges.

$$[\rho^a(x), \rho^b(y)] = if^{abc} \rho^c(x) \delta^{(2)}(x - y). \quad (2.2)$$

The reordering identities of the three- ρ note, with S the fully symmetrized product and one- ρ terms dropped, are

$$ABC = S(ABC) + \frac{1}{4} \left(\{[A, B], C\} + \{[A, C], B\} + \{[B, C], A\} \right), \quad (2.3)$$

$$A\{B, C\} = 2S(ABC) + \frac{1}{2} \{[A, B], C\} + \frac{1}{2} \{[A, C], B\}, \quad (2.4)$$

$$\{B, C\}A = 2S(ABC) - \frac{1}{2} \{[A, B], C\} - \frac{1}{2} \{[A, C], B\}. \quad (2.5)$$

After reordering, the two charges of a two- ρ term form a symmetric product and behave as commuting variables. A two- ρ term has a real coefficient, so its complex conjugate is the same term with $w \leftrightarrow w'$.

3 Which terms multiply $\log(\vee/k^+)$

The rule. Every longitudinal function L in the rows is a function of Λ, k^+ and $\vee$ (and of transverse distances). With your rules it is written uniquely as

$$L = A_L \log \frac{\vee}{\Lambda} + B_L \log \frac{\vee}{k^+} + (\text{no large logarithm}), \quad (3.1)$$

where “no large logarithm” means a function that stays finite for $\Lambda \rightarrow 0, \vee \rightarrow \infty$ (e.g. $\log(1 - k^+/\vee)$, $\log(1 - \Lambda/k^+)$, $\log \frac{(x-z)^2}{(x-w)^2}$). Table 1 lists every function that appears in the rows.

Table 1: The longitudinal functions of the rows. $a = (x - w)^2$, $b = (x - z)^2$ (primed for x', w'). A : coefficient of $\log(\vee/\Lambda)$. B : coefficient of $\log(\vee/k^+)$. $P = A + B$: coefficient of $\log \vee$.

function L	A	B	remark
$\log \frac{\vee}{\Lambda}$	1	0	
$\log \frac{\vee - k^+}{\Lambda} = \log \frac{\vee}{\Lambda} + \log(1 - \frac{k^+}{\vee})$	1	0	your second rule
$\log \frac{\Lambda}{k^+}$	-1	1	your first rule
$\log \frac{\vee}{k^+}, \log(\frac{\vee}{k^+} - 1)$	0	1	
$\log \frac{\vee}{\Lambda + k^+}, \log \frac{\vee}{k^+ - \Lambda}$	0	1	
$\log \frac{k^+ - \Lambda}{\vee}, \log \frac{k^+}{\vee}$	0	-1	
$\log \frac{\vee}{\Lambda} + \log \frac{k^+ - \Lambda}{\vee} = \log \frac{k^+ - \Lambda}{\Lambda}$	1	-1	Group I Rows III (3a), V; Group II Row IV
$L_P(a, b) \equiv \log \frac{(\vee - k^+)b + k^+a}{\Lambda b + k^+a}$	0	1	$\rightarrow \log \frac{\vee}{k^+} + \log \frac{b}{a}$

function L	A	B	remark
$L_o(a, b) \equiv \log \left \frac{\sqrt{b-k^+}a}{(k^++\Lambda)b-k^+a} \right $	0	1	$\rightarrow \log \frac{\sqrt{b}}{k^+} + \log \left \frac{b}{b-a} \right $
$L_{o'}(a, b) \equiv \log \left \frac{k^+a-\sqrt{b}}{k^+a-\Lambda b} \right $	0	1	$\rightarrow \log \frac{\sqrt{b}}{k^+} + \log \frac{b}{a}$
$\log \left \frac{(k^+-\Lambda)b-k^+a}{\Lambda b-k^+a} \right , \mathcal{A} \text{ (arctan)},$	0	0	finite
$\log \frac{[k^+(x-w)-(k^+-\Lambda)(x-z)]^2}{[k^+(x-w)-\Lambda(x-z)]^2}$			
$\log \frac{k^+p^2}{\sqrt{k^2+k^+p^2}}$ (momentum space)	0	-1	Group I Row II
$\log \left[1 + \left(\frac{\sqrt{b}}{k^+} - 1 \right) \frac{k^2}{(k-p)^2} \right]$ (momentum space)	0	1	Group I Row II
$\int_0^{\sqrt{b}-k^+} dp^+ \left[\frac{1}{p^+} \right]_+ e^{-i\beta p^+} G(p^+)$	0	$-G(0)$	hidden; Eq. (3.5)
$\int^{\sqrt{b}-k^+} dp^+ \frac{1}{p^+} G$ in the variable w'	0	$G(\infty)$	hidden; Sec. 4.3.2

$B = P - A$. In every row, $\sqrt{b}$ appears only as the upper limit of the p^+ integral, $\int^{\sqrt{b}-k^+} dp^+$ or $\int^{\sqrt{b}} dp^+$. Hence the coefficient of $\log \sqrt{b}$ is the large- p^+ limit of the integrand:

$$P_{\text{row}} \equiv A_{\text{row}} + B_{\text{row}} = \frac{\partial}{\partial \log \sqrt{b}} \frac{d^3 N_{\text{row}}}{d^3 k} = \lim_{p^+ \rightarrow \infty} p^+ \times (\text{integrand}). \quad (3.2)$$

So the $\log(\sqrt{b}/k^+)$ terms of a row are

$$B_{\text{row}} = P_{\text{row}} - A_{\text{row}} \quad (3.3)$$

For most rows B is read off directly from the notes with Table 1. For the rows left as plus-prescription integrals (Group II Row IV, Group III Rows III, V, VI) P is obtained from Eq. (3.2) before the p^+ integration.

Which evolution is $\sum P$. For $p^+ \gg k^+$ the unobserved gluon is harder than the measured one. It is part of the projectile wave function: the measured gluon is emitted by the valence charges or by the gluon at z , and both scatter on the target. Integrating it out is one step of the evolution of the projectile, which is BFKL:

$$\sum_{\text{rows}} P_{\text{row}} = \text{BFKL} \otimes \text{LO}, \quad \sum_{\text{rows}} A_{\text{row}} = \text{JIMWLK} \otimes \text{LO}, \quad \sum_{\text{rows}} B_{\text{row}} = \text{BFKL} \otimes \text{LO} - \text{JIMWLK} \otimes \text{LO}. \quad (3.4)$$

The second equation is the result of **TwoRho evolution**. The first is checked here row by row against an independent leading-log computation of the region $p^+ \gg k^+$ (Sec. 10). Both hold after the fixes.

Remark on A . A_{row} is not always the $p^+ \rightarrow 0$ limit of the integrand. A pole at $p^+ = k^+$, cut off at $k^+ \pm \Lambda$, also gives $\log(1/\Lambda)$. These “edge” logarithms appear in Group I Row III (the Φ terms). Eq. (3.3) holds regardless, because P only involves the upper limit.

The plus prescription hides a $\log(\sqrt{b}/k^+)$. For the rows written with $[1/p^+]_+$ and a phase,

$$\int_0^P dp^+ \frac{e^{-i\beta p^+} - 1}{p^+} = -\log(|\beta|P) - \gamma_E - \frac{i\pi}{2} \text{sgn } \beta + \mathcal{O}\left(\frac{1}{\beta P}\right), \quad (3.5)$$

with $P = \sqrt{b} - k^+$ and $\beta \propto 1/k^+$. This contains $-\log(\sqrt{b}/k^+)$, so $\int_0^{\sqrt{b}-k^+} dp^+ \left[\frac{1}{p^+} \right]_+ (\dots)$ contains $-\log(\sqrt{b}/k^+) \times (\text{the residue at } p^+ = 0)$, i.e. $-A$.

4 The $\log(\sqrt{b}/k^+)$ terms, row by row

For each row: the $\log(\sqrt{b}/\Lambda)$ terms A (as in your **Evolution** files), the coefficient of $\log \sqrt{b}$, P , and the $\log(\sqrt{b}/k^+)$ terms $B = P - A$. Notation used in several rows (Group I Row IV, Group II Row I, Group III Rows I, II), with $X = x' - w'$ and $Y = y - z$:

$$K_{\mathfrak{B}} = \mathcal{K}(x' - w') \cdot \mathcal{K}(x - w) \left[\mathcal{K}(z - w) \cdot \mathcal{K}(y - z) + \frac{1}{2} K(x, y, z) \right] \quad (\text{your } \log(\sqrt{b}/\Lambda) \text{ bracket}), \quad (4.1)$$

$$K_{\mathfrak{B}}^P = \mathcal{K}(x' - w') \cdot \mathcal{K}(z - w) K(x, y, z) - \frac{1}{2} \mathcal{K}(x' - w') \cdot \mathcal{K}(x - w) K(x, y, z) \quad (\text{coefficient of } \log \sqrt{b}), \quad (4.2)$$

$$\begin{aligned} K_{\mathfrak{B}}^L &= K_{\mathfrak{B}}^P - K_{\mathfrak{B}} \\ &= \mathcal{K}(x' - w') \cdot \mathcal{K}(z - w) K(x, y, z) - \mathcal{K}(x' - w') \cdot \mathcal{K}(x - w) \mathcal{K}(y - z) \cdot \mathcal{K}(z - w) \\ &\quad - \mathcal{K}(x' - w') \cdot \mathcal{K}(x - w) K(x, y, z). \end{aligned} \quad (\text{coefficient of } \log(\sqrt{b}/k^+)) \quad (4.3)$$

$K_{\mathfrak{B}}^L = K^i(x' - w') \mathcal{K}^j(y - z) \left[J_1^{ij} - \frac{\delta^{ij}}{2(z-w)^2} \right]$, where J_1^{ij} is the bracket multiplying $\log \frac{(\sqrt{b}-k^+)(x-z)^2+k^+(x-w)^2}{\Lambda(x-z)^2+k^+(x-w)^2}$ in your notes. In $K_{\mathfrak{B}}^P$ the measured gluon is emitted by the harder gluon at z , which is emitted by x (first term). In $K_{\mathfrak{B}}$ the softer gluon at z is emitted by the measured gluon (first term). $\bar{K}_{\mathfrak{B}}, \bar{K}_{\mathfrak{B}}^P, \bar{K}_{\mathfrak{B}}^L$ are the same with $w \leftrightarrow w'$ in the second factor and $(x' - w') \rightarrow (x' - w)$ (Group II Row I, second term).

4.1 Group I (with c.c.)

4.1.1 Row I

Only four- ρ terms, all $\propto \log(\vee/\Lambda)$ with $A = P$: $B = 0$.

4.1.2 Row II

Part 1 (two- ρ). Your bracket is $\left[\frac{11}{3} + \log \frac{\vee}{k^+} - 2 \log \frac{\vee}{\Lambda} + \log\left(\frac{\vee}{k^+} - 1\right)\right]$, and $\log \frac{\vee}{k^+} + \log\left(\frac{\vee}{k^+} - 1\right) = 2 \log \frac{\vee}{k^+} + \log\left(1 - \frac{k^+}{\vee}\right)$. The prefactor $\frac{g^4 N_c}{32\pi^4 k^+}$ is $\frac{\pi N_c}{2}$, so

$$\begin{aligned} A_{\text{I,II},1} &= -\pi N_c \mathcal{N} \int \mathcal{K}(x' - w') \cdot \mathcal{K}(x - w) B(x, w) [U^{ab'}(x') - U^{ab'}(w')] \rho^{b'}(x') [U^{ab}(w) - U^{ab}(x)] \rho^b(x) + \text{c.c.}, \\ B_{\text{I,II},1} &= +\pi N_c \mathcal{N}_P \int \mathcal{K}(x' - w') \cdot \mathcal{K}(x - w) B(x, w) [U^{ab'}(x') - U^{ab'}(w')] \rho^{b'}(x') [U^{ab}(w) - U^{ab}(x)] \rho^b(x) + \text{c.c.}, \end{aligned} \quad (4.4)$$

and $P_{\text{I,II},1} = 0$. As in the previous note (fix F6), the bracket must be $B(x, w) = \frac{1}{\pi} \int_z [K(x, w, z) - K(w, w, z)] = \frac{1}{\epsilon} - \log(4\pi^2 \mu^2 (x - w)^2)$ instead of $-\frac{2}{\epsilon} - 2\gamma - \log \frac{(x-w)^2 \mu^2}{4}$.

Part 2 (three- ρ , momentum space). In your bracket (with $\mathbf{k}$ the Fourier variable inside the bracket), Table 1 gives, term by term:

$$\begin{aligned} \log \frac{\Lambda}{k^+} \frac{k^2 p^i - k \cdot p k^i}{(k-p)^2} &\rightarrow A : -\frac{k^2 p^i - k \cdot p k^i}{(k-p)^2}, \quad B : +\frac{k^2 p^i - k \cdot p k^i}{(k-p)^2}, \\ -2k^i \log \frac{\vee}{\Lambda} &\rightarrow A : -2k^i, \\ -2k^i \log \frac{k^+ p^2}{\vee k^2 + k^+ p^2} &\rightarrow B : +2k^i, \\ \frac{(2p^2 - 2k \cdot p + k^2) k^i + k^2 p^i}{(k-p)^2} \log \left[1 + \left(\frac{\vee}{k^+} - 1\right) \frac{k^2}{(k-p)^2}\right] &\rightarrow B : +\frac{(2p^2 - 2k \cdot p + k^2) k^i + k^2 p^i}{(k-p)^2}, \\ -\frac{p^2 k^i + k^2 p^i}{(k-p)^2} \log \frac{\vee - k^+}{\Lambda} &\rightarrow A : -\frac{p^2 k^i + k^2 p^i}{(k-p)^2}, \\ -k^i \log \frac{\vee}{k^+} &\rightarrow B : -k^i. \end{aligned}$$

So the $\log(\vee/k^+)$ bracket is

$$B\text{-bracket} = \frac{2k^2 p^i + (3p^2 - 5k \cdot p + 2k^2) k^i}{(k-p)^2}, \quad A\text{-bracket} = -\frac{2k^2 p^i + (p^2 - k \cdot p) k^i}{(k-p)^2} - 2k^i, \quad (4.5)$$

and their sum (the coefficient of $\log \vee$) is $\frac{2(k-p)^2 k^i}{(k-p)^2} - 2k^i = 0$. Hence $P_{\text{I,II},2} = 0$ and

$$\begin{aligned} B_{\text{I,II},2} &= -A_{\text{I,II},2} = +\frac{i}{2} \mathcal{N}_P \int \mathcal{K}^i(x' - w') \mathfrak{F}^i(w; x, y) [U^{ab'}(x') - U^{ab'}(w')] \rho^{b'}(x') \\ &\quad \times \left[U^{ah}(w) f^{hbc} \{ \rho^b(x), \rho^c(y) \} - f^{abc} U^{bd}(x) U^{ce}(y) \{ \rho^d(x), \rho^e(y) \} \right] + \text{c.c.}, \end{aligned} \quad (4.6)$$

with $\mathfrak{F}^i = \mathfrak{F}_\alpha^i + \mathfrak{F}_\beta^i$ the position-space kernel of the three- ρ note (App. A there).

Fix F4 does not change B . The previous note removed the α part $\mathfrak{F}_\alpha$ from the $\log(\vee/\Lambda)$ terms of the $U(w)\mathbb{A}^{(3)}$ sub-terms ($A \rightarrow A - \alpha$). The region- P comparison (Sec. 10) shows that the same α part must also be removed from P ($P = 0 \rightarrow -\alpha$). So the α part multiplies $\log(\vee/\Lambda)$ only, and $B = P - A$ is unchanged: the $\log(\vee/k^+)$ terms of Row II are correct as written.

4.1.3 Row III

Prefactor $\frac{ig^4 f^{abd}}{16\pi^5 k^+} \rightarrow i f^{abd}$ and colour operator

$$\mathcal{O}_{\text{III}} = [U^{ab'}(x') \rho^{b'}(x') - U^{ab'}(w') \rho^{b'}(x')] [U^{de}(x) U^{bc}(z) \rho^e(x) \rho^c(y) - U^{de}(x) U^{bc}(y) \rho^e(x) \rho^c(y)].$$

With $a = (x - w)^2$, $b = (x - z)^2$, $X = x' - w'$, $Y = y - z$, $Z_w = z - w$, the kernels of your parts are

$$\begin{aligned} K_a &= \frac{X \cdot Y (a + b)}{X^2 Y^2 (b - a)^2} = -\mathcal{K}(x' - w') \cdot \mathcal{K}(y - z) \Phi(x), \quad K_b = \mathcal{K}(x' - w') \cdot \mathcal{K}(x - w) K(x, y, z), \\ K_c &= \mathcal{K}(x' - w') \cdot \mathcal{K}(x - w) \mathcal{K}(y - z) \cdot \mathcal{K}(z - w), \quad K_h = \mathcal{K}(x' - w') \cdot \mathcal{K}(z - w) K(x, y, z), \\ K_d &= \frac{X \cdot Y Z_w \cdot (x - z) a}{X^2 Y^2 Z_w^2 (a - b)^2}, \quad K_e = \frac{X \cdot Y Z_w \cdot (x - w) b}{X^2 Y^2 Z_w^2 (a - b)^2}, \quad K_f = \frac{X \cdot Y}{X^2 Y^2 Z_w^2}, \end{aligned}$$

and $g_1, \dots, g_4$ the four terms with $\frac{1}{a-b}$ in your part 2 (multiplying $L_o(a, b)$, in order: $+\frac{X \cdot Z_w Y \cdot (x - w)}{(a - b)}$, $-\frac{X \cdot (x - w) Y \cdot Z_w b}{(a - b)a}$, $-\frac{X \cdot Z_w Y \cdot (x - z) a}{(a - b)b}$, $+\frac{X \cdot (x - z) Y \cdot Z_w}{(a - b)}$, all over $X^2 Y^2 Z_w^2$). Your longitudinal functions give (units $i f^{abd}$, Table 1):

part	your term	A	B
1	$-\left[\log \frac{\vee}{\Lambda} + \log\left(1 - \frac{k^+}{\vee}\right)\right] K_a$	$-K_a$	0
1	$+L_o(a, b) K_a$	0	$+K_a$
1	$-\left[\log \frac{\vee}{k^+ - \Lambda} - \log \frac{\vee}{\Lambda}\right] K_a$	$+K_a$	$-K_a$
1	$+\log \frac{\vee}{\Lambda} K_b$	$+K_b$	0
1	$-2 L_o'(a, b) K_b$	0	$-2K_b$
2	$-2 \log \frac{\vee}{\Lambda + k^+} K_c$	0	$-2K_c$
2	$+2\left[\log \frac{\vee}{\Lambda} + \log\left(1 - \frac{k^+}{\vee}\right)\right] (K_d - K_e)$	$2(K_d - K_e)$	0
2	$+2L_o(a, b) [g_1 + g_2 + g_3 + g_4 + K_e - K_d]$	0	$2(g_1 + g_2 + g_3 + g_4) + 2(K_e - K_d)$
3a	$\left[\log \frac{\vee}{\Lambda} + \log \frac{k^+ - \Lambda}{\vee}\right] (2K_d - 2K_e - 2K_f)$	$2(K_d - K_e - K_f)$	$-2(K_d - K_e - K_f)$
3b	$\mathcal{A}, \log \frac{[k^+(x-w) - (k^+ - \Lambda)(x-z)]^2}{[k^+(x-w) - \Lambda(x-z)]^2}$ terms	0	0

Two identities simplify the sums: $4(K_d - K_e) - 2K_f = -2K_a$ (use $Z_w = (x-w) - (x-z)$ and $Z_w^2 = a+b-2(x-w) \cdot (x-z)$), and $-K_c + g_1 + g_2 + g_3 + g_4 = -K_h$. Then

$$\begin{aligned}
A_{\text{I,III}} &= i(K_b - 2K_a) f^{abd} \mathcal{O}_{\text{III}} + \text{c.c.} \quad (\text{your Evolution_three_rho, with } -2K_a = 2\mathcal{K}(x' - w') \cdot \mathcal{K}(y - z)\Phi(x)), \\
B_{\text{I,III}} &= i(-2K_b - 2K_h + 2K_a) f^{abd} \mathcal{O}_{\text{III}} + \text{c.c.}, \\
P_{\text{I,III}} &= A + B = i(-K_b - 2K_h) f^{abd} \mathcal{O}_{\text{III}} + \text{c.c.}
\end{aligned} \tag{4.7}$$

Direct check of P . Only two of your p^+ integrals reach $\vee$. The one with $\frac{[k^+(x-w)^2 + p^+(x-z)^2]}{[p^+(x-z)^2 - k^+(x-w)^2]} \frac{1}{p^+ k^+}$ tends to $\frac{1}{p^+ k^+}$ and gives $-K_b$. The one with $\frac{[k^+(x-w)^m - p^+(x-z)^m]}{[k^+(x-w)^2 - p^+(x-z)^2]} \frac{1}{k^+ - p^+} [\delta_{ik} \delta_{jm} - \dots]$ tends to $-\frac{1}{p^+} \frac{(x-z)^m}{(x-z)^2} \delta_{ik} \delta_{jm}$ and gives $-2K_h$. This is the same P , so your p^+ integration is correct (also checked by numerical integration).

Fix P1. The region- P comparison (Sec. 10) requires $P_{\text{I,III}} = i(-K_b + 2K_h)$. The term $-K_b$ is right, but the $\mathbb{B}_3^+$ term over $k^+ + \Lambda < p^+ < \vee$ has the wrong sign at large p^+ . In the comparison, the structures with K_h match the independent computation exactly once the sign is reversed, and they are off by exactly a factor -1 otherwise.

Fix F3 carried over. In the previous note the Φ terms of A ($-2iK_a$) were replaced by $\mathcal{D}$,

$$\mathcal{D} = -2i \int \mathcal{K}(x' - w') \cdot \mathcal{K}(y - w) K(x, w, z) f^{abd} [U^{ab'}(x') - U^{ab'}(w')] \rho^{b'}(x') [U^{de}(x) U^{bc}(w) - U^{de}(x) U^{bc}(y)] \rho^e(x) \rho^c(y) + \text{c.c.}$$

(the soft gluon exchanged between the measured gluon and the rotated charge x). Hence, corrected,

$$\begin{aligned}
A_{\text{I,III}} &= iK_b f^{abd} \mathcal{O}_{\text{III}} + \mathcal{D}, \quad P_{\text{I,III}} = i(-K_b + 2K_h) f^{abd} \mathcal{O}_{\text{III}} \quad (+\text{c.c.}), \\
\boxed{B_{\text{I,III}} &= i(-2K_b + 2K_h) f^{abd} \mathcal{O}_{\text{III}} - \mathcal{D} \quad (+\text{c.c.})}
\end{aligned} \tag{4.8}$$

The terms $\pm K_a$ (the edge kernel Φ) are absent. In your notes they come from the poles at $p^+ = k^+$ and multiply $\log \frac{k^+}{\Lambda} = \log \frac{\vee}{\Lambda} - \log \frac{\vee}{k^+}$.

4.1.4 Row IV

Prefactor $\frac{ig^4}{8\pi^2 k^+} \rightarrow 2i$. Colour operator

$$\mathcal{O}_{\text{IV}} = [U^{ab'}(x') \rho^{b'}(x') - U^{ab'}(w') \rho^{b'}(x')] [f^{cde'} U^{ac}(w) U^{bd}(z) U^{be}(y) \rho^e(y) \rho^{c'}(x) - f^{abc} U^{be}(y) U^{cd}(x) \rho^e(y) \rho^d(x)].$$

Your result has $L_P(a, b) \frac{(x' - w')^i (y - z)^j}{(x' - w')^2 (y - z)^2} J_1^{ij}$ ($B = 1$), $-\frac{1}{2} \log \frac{\vee}{\Lambda + k^+} \frac{(x' - w') \cdot (y - z)}{(x' - w')^2 (y - z)^2 (z - w)^2}$ ($B = 1$) and the blue $[\log \frac{\vee}{\Lambda} + \log(1 - \frac{k^+}{\vee})] K_{\mathfrak{B}}$ ($A = 1$). Therefore

$$\begin{aligned}
A_{\text{I,IV}} &= 2i K_{\mathfrak{B}} \mathcal{O}_{\text{IV}} + \text{c.c.}, \quad P_{\text{I,IV}} = 2i K_{\mathfrak{B}}^P \mathcal{O}_{\text{IV}} + \text{c.c.}, \\
\boxed{B_{\text{I,IV}} &= 2i K_{\mathfrak{B}}^L \mathcal{O}_{\text{IV}} + \text{c.c.}}
\end{aligned} \tag{4.9}$$

P agrees with the large- p^+ limit of your integrand, $\frac{1}{k^+} \left[\frac{(x-z)^j (z-w)^i}{(z-w)^2} - \frac{(x-z)^j (x-w)^i}{2(x-w)^2} \right] \frac{1}{p^+ (x-z)^2}$. Row IV is correct.

4.1.5 Row V

Here $0 < p^+ < k^+$, so $P_{\text{I,V}} = 0$. All your logarithms are $\log \frac{\vee}{\Lambda} + \log \frac{k^+ - \Lambda}{\vee} = \log \frac{k^+ - \Lambda}{\Lambda}$ ($A = 1$, $B = -1$). Hence, for parts 1, 2a and 2b,

$$\boxed{B_{\text{I,V}} = -A_{\text{I,V}}} \tag{4.10}$$

with $A_{\text{I,V}}$ the $\log(\vee/\Lambda)$ terms of your Evolution files:

$$B_{\text{I,V},1} = +\frac{i}{2} \mathcal{N}_P \int \left[e^{-ik \cdot (w' - z)} \mathcal{K}(x' - w') \cdot \mathcal{K}(y - z) \mathcal{K}(z - x) \cdot \mathcal{K}(y' - x) \right]$$

$$\begin{aligned}
& + e^{-ik \cdot (w' - x)} \mathcal{K}(x' - w') \cdot \mathcal{K}(y' - x) \mathcal{K}(z - x) \cdot \mathcal{K}(y - z) \Big] \\
& \times [U^{ab'}(x') - U^{ab'}(w')] \rho^{b'}(x') \Big[f^{adc} U^{db}(x) U^{ce}(z) \{ \rho^e(y), \rho^b(y') \} - f^{abc} U^{cd}(y) U^{be}(y') \{ \rho^d(y), \rho^e(y') \} \Big] + \text{c.c.},
\end{aligned} \tag{4.11}$$

and $B_{\text{I.V},2}$ is minus Eq. (4.6) of the previous note (parts 2a and 2b together, with $\mathcal{N} \rightarrow \mathcal{N}_P$).

4.2 Group II (no c.c.)

4.2.1 Row I

Part 1 (three- ρ). The two terms have the structure of Group I Row IV, with prefactors $-\frac{ig^4}{16\pi^5 k^+} \rightarrow -i$ and $+i$:

$$\begin{aligned}
B_{\text{II.I},1} &= -i \mathcal{N}_P \int K_{\mathfrak{B}}^L \mathcal{O}^{(1)} + i \mathcal{N}_P \int \bar{K}_{\mathfrak{B}}^L \mathcal{O}^{(2)}, \\
\mathcal{O}^{(1)} &= \left[U^{ac'}(w') \{ \rho^{c'}(x'), \rho^b(y) \} - U^{bc'}(z) U^{ad'}(x') U^{c'e'}(y) \{ \rho^{d'}(x'), \rho^{e'}(y) \} \right] \\
&\quad \times \left[f^{cbd} U^{ac}(w) \rho^d(x) - f^{acd} U^{bc}(z) U^{de}(x) \rho^e(x) \right], \\
\mathcal{O}^{(2)} &= \left[f^{c'bd'} U^{ac'}(w') \rho^{d'}(x) - f^{ac'd'} U^{bc'}(z) U^{d'e'}(x) \rho^{e'}(x) \right] \\
&\quad \times \left[U^{ac}(w) \{ \rho^c(x'), \rho^b(y) \} - U^{bc}(z) U^{ad}(x') U^{ce}(y) \{ \rho^d(x'), \rho^e(y) \} \right].
\end{aligned} \tag{4.12}$$

A and P are the same with $K^L \rightarrow K$ and $K^L \rightarrow K^P$.

Part 2 (two- ρ). Colour operator (your reduced form)

$$\begin{aligned}
\mathcal{O}_{\text{II.I}} &= f^{c'bd'} f^{cbd} U^{ac'}(w') U^{ac}(w) \rho^{d'}(x') \rho^d(x) - f^{c'bd'} f^{acd} U^{ac'}(w') U^{bc}(z) U^{de}(x) \rho^{d'}(x') \rho^e(x) \\
&\quad - f^{ac'd'} f^{cbd} U^{ac}(w) U^{bc'}(z) U^{e'd'}(x') \rho^{e'}(x') \rho^d(x) + N_c U^{e'd}(x') U^{de}(x) \rho^{e'}(x') \rho^e(x).
\end{aligned}$$

With a, b, a', b' as in Table 1 and $D = a'b - ab'$, the $\log(\vee/k^+)$ content of your six parts 2a–2f is:

- 2a: $L_P(a, b) - L_P(a', b')$, so $B = 1 - 1 = 0$;
- 2b: blue $\log \frac{\vee - k^+}{\Lambda}$ ($B = 0$; it gives A), and $-\frac{b}{aD} L_P(a, b) + \frac{b'}{a'D} L_P(a', b')$, so $B = \frac{-a'b + ab'}{aa'D} = -\frac{1}{aa'}$ times K_B ;
- 2c: $\frac{a'}{b'D} L_P(a', b') - \frac{a}{bD} L_P(a, b)$, so $B = \frac{a'b - ab'}{bb'D} = \frac{1}{bb'}$ times K_C ;
- 2d, 2e, 2f: the coefficients of $\log \frac{\vee}{\Lambda + k^+}$ and of the two L_P cancel, $B = 0$.

Here K_B and K_C are your brackets multiplying $\frac{(p^+ + k^+)^2}{(p^+)^2}$ and $\frac{(p^+ + k^+)^2}{(k^+)^2}$. They factorize:

$$\begin{aligned}
\frac{K_B}{aa'} &= \mathcal{K}(x - w) \cdot \mathcal{K}(x' - w') \left[\mathcal{K}(w - z) - \frac{1}{2} \mathcal{K}(x - z) \right] \cdot \left[\mathcal{K}(w' - z) - \frac{1}{2} \mathcal{K}(x' - z) \right], \\
\frac{K_C}{bb'} &= \mathcal{K}(x, x', z) \left[\mathcal{K}(z - w) - \frac{1}{2} \mathcal{K}(x - w) \right] \cdot \left[\mathcal{K}(z - w') - \frac{1}{2} \mathcal{K}(x' - w') \right].
\end{aligned} \tag{4.13}$$

With the prefactor $\frac{g^4}{8\pi^5 k^+} \rightarrow 2$:

$$A_{\text{II.I},2} = 2 \int \frac{K_B}{aa'} \mathcal{O}_{\text{II.I}}, \quad P_{\text{II.I},2} = 2 \int \frac{K_C}{bb'} \mathcal{O}_{\text{II.I}}, \quad \boxed{B_{\text{II.I},2} = 2 \int \left[\frac{K_C}{bb'} - \frac{K_B}{aa'} \right] \mathcal{O}_{\text{II.I}}} \tag{4.14}$$

In A the softer gluon z is emitted by the measured gluon (or by x , with weight $-\frac{1}{2}$). In P the measured gluon is emitted by the harder gluon z (or by x , with weight $-\frac{1}{2}$). The two are related by exchanging the roles of z and w, w' .

4.2.2 Rows II and III

Only four- ρ terms $\propto \log(\vee/\Lambda)$: $B = 0$.

4.2.3 Row IV

Colour operator $\mathcal{O}_{\text{II.IV}}(p', p) = [U^{e'c}(p') - U^{e'c}(x')] [U^{ec}(p) - U^{ce}(x)] \rho^{e'}(x') \rho^e(x)$. Your blue factor is $\log \frac{\vee}{\Lambda} + \log \frac{k^+}{\vee}$ ($A = 1, B = -1$). The ξ integral $\int_1^{k^+/\vee} \frac{d\xi}{\xi^2}$ hides a $+\log(\vee/k^+)$ at its lower end $\xi \rightarrow k^+/\vee$, i.e. $p^+ \rightarrow \vee$. It is easiest

to read off before the w , w' and z integrations. For $p^+ \gg k^+$ the product of your two brackets is $(\frac{p^++k^+}{k^+})^2 \delta_{mm'} \delta_{kk'}$, times $\frac{p^+k^+}{(p^++k^+)^4}$. This gives $\frac{1}{k^+p^+}$ with $y, y' \rightarrow z$. Hence $(\frac{g^4 N_c}{8\pi^4 k^+} \rightarrow 2\pi N_c$ and $\frac{g^4 N_c}{8\pi^5 k^+} \rightarrow 2N_c$)

$$\begin{aligned} A_{\text{II.IV}} &= 2N_c \int_{x,x'} \left[\int_z K(w, w', z) \right] \mathcal{K}(x-w) \cdot \mathcal{K}(x'-w') \mathcal{O}_{\text{II.IV}}(w', w), \\ P_{\text{II.IV}} &= 2N_c \int_{x,x',z} K(w, w', z) K(x, x', z) \mathcal{O}_{\text{II.IV}}(z, z), \\ B_{\text{II.IV}} &= P_{\text{II.IV}} - A_{\text{II.IV}}. \end{aligned} \quad (4.15)$$

Here $\int_z K(w, w', z) = \pi[\frac{1}{\epsilon} - \log(4\pi^2 \mu^2 (w-w')^2)]$ is your bracket. In P both measured gluons are emitted by the harder gluon at z , which comes from x and x' .

4.3 Group III (with c.c.)

The overall sign of Rows III, V and VI is that of `Evolution_three_rho.tex` $(-\frac{4}{(2\pi)^3}, -\frac{2}{(2\pi)^3})$, which is opposite to `Group_III.tex`. This applies to the $\log(\sqrt{v}/k^+)$ terms as well, and to part 2 of Row III (fix F5).

4.3.1 Rows I and II

Same structure as Group I Row IV, with prefactor $-\frac{ig^4}{8\pi^5 k^+} \rightarrow -2i$:

$$B_{\text{III.I(II)}} = -2i \mathcal{N}_P \int K_{\text{B}}^L \mathcal{O}_{\text{III.I(II)}} + \text{c.c.}, \quad (4.16)$$

with

$$\begin{aligned} \mathcal{O}_{\text{III.I}} &= [U^{ab'}(x') U^{c'd'}(y) U^{c'b}(z) \rho^{b'}(x') \rho^{d'}(y) - U^{bd'}(z) U^{d'e'}(y) U^{ac'}(w') \rho^{c'}(x') \rho^{e'}(y)] \\ &\quad \times [f^{cbd} U^{ac}(w) \rho^d(x) - f^{acd} U^{bc}(z) U^{de}(x) \rho^e(x)], \\ \mathcal{O}_{\text{III.II}} &= [U^{c'd'}(y) U^{ab'}(x') U^{bc'}(z) \rho^{d'}(y) \rho^{b'}(x') - U^{ab'}(x') \rho^b(y) \rho^{b'}(x')] \\ &\quad \times [f^{cbd} U^{ac}(w) \rho^d(x) - f^{acd} U^{bc}(z) U^{de}(x) \rho^e(x)]. \end{aligned}$$

4.3.2 Row III

Part 1 (three- ρ). Your result is the plus-prescription integral $\mathcal{I}_1$ plus the blue term $[\log \frac{\sqrt{v}}{\Lambda} + \log(1 - \frac{k^+}{\sqrt{v}})]$. The blue term has $A = 1$, $B = 0$, and gives

$$A_{\text{III.III,1}} = -i \mathcal{N} \int e^{-ik \cdot (y'-w)} K(y', x, z) \mathcal{K}(x' - y') \cdot \mathcal{K}(y - w) f^{c'd'a} \mathcal{O}_4 \mathcal{O}_5 + \text{c.c.}$$

with

$$\begin{aligned} \mathcal{O}_4 &= U^{e'c'}(y') U^{d'b}(z) \rho^{e'}(x') - U^{bd'}(z) U^{c'e'}(x') \rho^{e'}(x'), \\ \mathcal{O}_5 &= U^{ac}(w) \{\rho^c(y), \rho^b(x)\} - U^{bc}(z) U^{ad}(y) U^{ce}(x) \{\rho^d(y), \rho^e(x)\}. \end{aligned}$$

The integral $\mathcal{I}_1 = \int_0^{\sqrt{v}-k^+} \frac{dp^+}{2\pi} (\dots)$ contains two hidden $\log(\sqrt{v}/k^+)$:

1. the $[\frac{1}{p^+}]_+$ term with its phase gives $-\log(\sqrt{v}/k^+) \times A_{\text{III.III,1}}$ (Eq. (3.5));
2. the term $-\frac{1}{k^+} \delta_{im} \delta_{jk'}$ behaves as $\frac{1}{k^+p^+}$ in the variable w' , where $y' = z + \xi(w' - z)$, $\xi = \frac{k^+}{k^++p^+}$, $d^2 y' = \xi^2 d^2 w'$, and the phase becomes $e^{-ik \cdot (w'-w)}$. This gives $+\log(\sqrt{v}/k^+) \times P_{\text{III.III,1}}$ with

$$P_{\text{III.III,1}} = -i \mathcal{N}_P \int e^{-ik \cdot (w'-w)} K(x', x, z) \mathcal{K}(w' - z) \cdot \mathcal{K}(y - w) f^{c'd'a} \mathcal{O}_4|_{y' \rightarrow z} \mathcal{O}_5 + \text{c.c.}$$

So $B_{\text{III.III,1}} = P_{\text{III.III,1}} - A_{\text{III.III,1}}$. In P the measured gluon w' is emitted by the harder gluon z , which is exchanged between x' and x .

Part 2 (two- ρ). In the same way, from the blue term and the plus-prescription integral (with the sign of fix F5; prefactor $\frac{g^4}{8\pi^5 k^+} \rightarrow 2$):

$$\begin{aligned} A_{\text{III.III,2}} &= -2\mathcal{N} \int e^{-ik \cdot (y'-w)} \mathcal{K}(x-w) \cdot \mathcal{K}(x'-y') \left[\mathcal{K}(z-w) \cdot \mathcal{K}(y'-z) + \frac{1}{2} K(x, y', z) \right] \mathcal{O}_6 \mathcal{O}_7 + \text{c.c.}, \\ P_{\text{III.III,2}} &= -2\mathcal{N}_P \int e^{-ik \cdot (w'-w)} K(x', x, z) \mathcal{K}(w' - z) \cdot \left[\mathcal{K}(z-w) - \frac{1}{2} K(x-w) \right] \mathcal{O}_6|_{y' \rightarrow z} \mathcal{O}_7 + \text{c.c.}, \end{aligned} \quad (4.17)$$

with $\mathcal{O}_6 = f^{c'd'a} [U^{e'c'}(y') U^{d'b}(z) - U^{bd'}(z) U^{c'e'}(x')] \rho^{e'}(x')$ and $\mathcal{O}_7 = f^{cbd} U^{ac}(w) \rho^d(x) - f^{acd} U^{bc}(z) U^{de}(x) \rho^e(x)$. P comes from the term $-\frac{p^+}{k^++2} \left[\frac{(x-z)^{k'} (z-w)^m}{(z-w)^2} - \frac{(x-z)^{k'} (x-w)^m}{2(x-w)^2} \right]$ of your bracket, which in the variable w' behaves as $\frac{1}{k^++p^+(x-z)^2} [\dots]$. $B_{\text{III.III,2}} = P - A$.

4.3.3 Row IV

Only four- ρ terms $\propto \log(\vee/\Lambda)$: $B = 0$.

4.3.4 Rows V and VI

Both have the longitudinal structure of Row III part 1, with twice its prefactor:

$$\begin{aligned} A_{\text{III.V(VI)}} &= -2i\mathcal{N} \int e^{-ik \cdot (y' - w)} K(y', x, z) \mathcal{K}(x' - y') \cdot \mathcal{K}(y - w) \mathcal{C}_{\text{V(VI)}} + \text{c.c.}, \\ P_{\text{III.V(VI)}} &= -2i\mathcal{N}_P \int e^{-ik \cdot (w' - w)} K(x', x, z) \mathcal{K}(w' - z) \cdot \mathcal{K}(y - w) \mathcal{C}_{\text{V(VI)}}|_{y' \rightarrow z} + \text{c.c.}, \end{aligned} \quad (4.18)$$

with $\mathcal{C}_V = f^{c'ca}[U^{e'e'}(y')\rho^{e'}(x') - U^{c'e'}(x')\rho^{e'}(x')][U^{ce}(x)U^{ad}(y)\rho^e(x)\rho^d(y) - U^{ad}(w)U^{ce}(x)\rho^e(x)\rho^d(y)]$ and $\mathcal{C}_{VI} = f^{c'd'a}[U^{e'e'}(y')U^{d'b}(z)\rho^{e'}(x') - U^{bd'}(z)U^{c'e'}(x')\rho^{e'}(x')][U^{bc}(z)U^{ad}(y)U^{ce}(x)\rho^d(y)\rho^e(x) - U^{ad}(y)\rho^d(y)\rho^b(x)]$. $B = P - A$.

4.4 Summary of the rows

row	$\log(\vee/k^+)$ terms B	as written	status
I.I, II.II, II.III, III.IV, Eq. (4rho)	0	–	correct
I.II part 1	$-A$	explicit	F6 (bracket)
I.II part 2	$-A$	explicit	correct (F4 affects A only)
I.III	$P - A$	explicit	P1 ($+2K_h$), F3 ($\mathcal{D}$)
I.IV	$2iK_{\frac{L}{2}}$	explicit	correct
I.V	$-A$	explicit	correct
II.I parts 1, 2	$P - A$	explicit	correct
II.IV	$P - A$	$-A$ explicit, P hidden	correct
III.I, III.II	$-2iK_{\frac{L}{2}}$	explicit	correct
III.III part 1, III.V, III.VI	$P - A$	hidden	correct (sign of Evolution_three_rho)
III.III part 2	$P - A$	hidden	F5 (sign)

5 Four- ρ terms

All four- ρ terms of the notes (Eq. (4rho): Group I Row I, Group II Rows II and III, Group III Row IV and the four- ρ parts of the other rows) multiply $\log(\vee/\Lambda)$, with $A = P$. Their B is zero. The same holds for their three- ρ and two- ρ remainders (A1–B2 and the lists of `4rho.tex`). Physically, in these terms both gluons are emitted independently by the valence charges, and the p^+ integrand is $\propto 1/p^+$ over the whole range $\Lambda < p^+ < \vee$. As a check, the symmetrized four- ρ part of Eq. (4rho) equals the four- ρ part of the independent region- P computation, and both vanish in total (12 kernel structures).

6 The three- ρ terms: reordering

The three- ρ terms proportional to $\log(\vee/k^+)$ are $B = P - A$ for the rows of Sec. 4 with three charges. The reordering is linear, so the two- ρ terms of B are those of P minus those of A :

- The two- ρ terms of A (the $\log(\vee/\Lambda)$ three- ρ terms) are worked out step by step in `ThreeRho_to_TwoRho`, Secs. 4–7 (with the corrections of `ThreeRho_cancellation`: $\mathcal{D}$, and the α part of Row II). They enter here with a minus sign.
- The two- ρ terms of P are worked out below in the same way, for every P source. There the coefficient of $\log \vee$ is written as $P|_{\text{source}}$, in units of $\mathcal{N}_P$, like B . The colour structures are those of the corresponding A source, only the kernel changes (for Group III Rows III, V, VI also $U(y') \rightarrow U(z)$).

Group I Rows II and V have $P = 0$, and their $B = -A$. The fix P1 is included in the source “Group I, Row III (region P)” below: its kernel is $K_{\text{I.III}}^P = -K_b + 2K_h$, i.e. the corrected one. The kernels at the coincident points are listed after each source.

6.1 Group I, Row III ($\bar{\mathbb{B}}_3^{\dagger}\mathbb{A}$) (region P)

The three- ρ term.

$$\begin{aligned} P \Big|_{\text{GI.III.P}} &= i\mathcal{N}_P \int e^{-ik \cdot (w' - w)} K_{\text{I.III}}^P(x', x, y) \Big\{ \\ &\quad + f^{abd} U^{ab'}(x') U^{de}(x) U^{bc}(z) \rho^{b'}(x') \rho^e(x) \rho^c(y) \\ &\quad - f^{abd} U^{ab'}(x') U^{de}(x) U^{bc}(y) \rho^{b'}(x') \rho^e(x) \rho^c(y) \end{aligned}$$

$$\begin{aligned}
& - f^{abd} U^{ab'}(w') U^{de}(x) U^{bc}(z) \rho^{b'}(x') \rho^e(x) \rho^c(y) \\
& + f^{abd} U^{ab'}(w') U^{de}(x) U^{bc}(y) \rho^{b'}(x') \rho^e(x) \rho^c(y) \} + \text{c.c.}
\end{aligned} \tag{6.1}$$

with $K_{\text{I.III}}^P(x', x, y) = \left[-\mathcal{K}(x' - w') \cdot \mathcal{K}(x - w) \mathcal{K}(x - z) \cdot \mathcal{K}(y - z) + 2\mathcal{K}(x' - w') \cdot \mathcal{K}(z - w) \mathcal{K}(y - z) \cdot \mathcal{K}(x - z) \right]$.

Reordering. In every sub-term (one product of the two brackets) the charges appear in the order shown, so we apply Eq. (2.3) with A, B, C the charges at x', x, y . The sub-terms are numbered 1–4 in the order above. For each sub-term we list the commutators, and each resulting two- ρ piece before and after the colour algebra.

Sub-term 1: $+ f^{abd} U^{ab'}(x') U^{de}(x) U^{bc}(z) \rho^{b'}(x') \rho^e(x) \rho^c(y)$.

$$\begin{aligned}
[A, B] : [\rho^{b'}(x'), \rho^e(x)] &= i f^{b'eh} \rho^h(x') \delta^{(2)}(x' - x), \quad x \rightarrow x' \\
\frac{1}{4} \{ [A, B], C \} &= \frac{1}{4} i K_{\text{I.III}}^P(x', x', y) f^{abd} U^{ab'}(x') U^{de}(x') U^{bc}(z) f^{b'eh} \{ \rho^h(x'), \rho^e(y) \} \\
&= -\frac{1}{4} i N_c K_{\text{I.III}}^P(x', x', y) U^{ab}(z) U^{ac}(x') \{ \rho^c(x'), \rho^b(y) \} \quad (\text{U2}), \quad (\text{F})
\end{aligned} \tag{6.2}$$

$$\begin{aligned}
[A, C] : [\rho^{b'}(x'), \rho^c(y)] &= i f^{b'ch} \rho^h(x') \delta^{(2)}(x' - y), \quad y \rightarrow x' \\
\frac{1}{4} \{ [A, C], B \} &= \frac{1}{4} i K_{\text{I.III}}^P(x', x, x') f^{abd} U^{ab'}(x') U^{de}(x) U^{bc}(z) f^{b'ch} \{ \rho^h(x'), \rho^e(x) \} \\
&= \frac{1}{4} i K_{\text{I.III}}^P(x', x, x') f^{abc} f^{deh} U^{ad}(x') U^{cl}(x) U^{be}(z) \{ \rho^h(x'), \rho^l(x) \}
\end{aligned} \tag{6.3}$$

$$\begin{aligned}
[B, C] : [\rho^e(x), \rho^c(y)] &= i f^{ech} \rho^h(x) \delta^{(2)}(x - y), \quad y \rightarrow x \\
\frac{1}{4} \{ [B, C], A \} &= \frac{1}{4} i K_{\text{I.III}}^P(x', x, x) f^{abd} U^{ab'}(x') U^{de}(x) U^{bc}(z) f^{ech} \{ \rho^h(x), \rho^{b'}(x') \} \\
&= \frac{1}{4} i K_{\text{I.III}}^P(x', x, x) f^{abc} f^{deh} U^{al}(x') U^{cd}(x) U^{be}(z) \{ \rho^h(x), \rho^l(x') \}
\end{aligned} \tag{6.4}$$

Sub-term 2: $- f^{abd} U^{ab'}(x') U^{de}(x) U^{bc}(y) \rho^{b'}(x') \rho^e(x) \rho^c(y)$.

$$\begin{aligned}
[A, B] : [\rho^{b'}(x'), \rho^e(x)] &= i f^{b'eh} \rho^h(x') \delta^{(2)}(x' - x), \quad x \rightarrow x' \\
\frac{1}{4} \{ [A, B], C \} &= -\frac{1}{4} i K_{\text{I.III}}^P(x', x', y) f^{abd} U^{ab'}(x') U^{de}(x') U^{bc}(y) f^{b'eh} \{ \rho^h(x'), \rho^e(y) \} \\
&= \frac{1}{4} i N_c K_{\text{I.III}}^P(x', x', y) U^{ab}(y) U^{ac}(x') \{ \rho^c(x'), \rho^b(y) \} \quad (\text{U2}), \quad (\text{F})
\end{aligned} \tag{6.5}$$

$$\begin{aligned}
[A, C] : [\rho^{b'}(x'), \rho^c(y)] &= i f^{b'ch} \rho^h(x') \delta^{(2)}(x' - y), \quad y \rightarrow x' \\
\frac{1}{4} \{ [A, C], B \} &= -\frac{1}{4} i K_{\text{I.III}}^P(x', x, x') f^{abd} U^{ab'}(x') U^{de}(x) U^{bc}(x') f^{b'ch} \{ \rho^h(x'), \rho^e(x) \} \\
&= -\frac{1}{4} i N_c K_{\text{I.III}}^P(x', x, x') U^{ab}(x) U^{ac}(x') \{ \rho^c(x'), \rho^b(x) \} \quad (\text{U2}), \quad (\text{F})
\end{aligned} \tag{6.6}$$

$$\begin{aligned}
[B, C] : [\rho^e(x), \rho^c(y)] &= i f^{ech} \rho^h(x) \delta^{(2)}(x - y), \quad y \rightarrow x \\
\frac{1}{4} \{ [B, C], A \} &= -\frac{1}{4} i K_{\text{I.III}}^P(x', x, x) f^{abd} U^{ab'}(x') U^{de}(x) U^{bc}(x) f^{ech} \{ \rho^h(x), \rho^{b'}(x') \} \\
&= \frac{1}{4} i N_c K_{\text{I.III}}^P(x', x, x) U^{ab}(x') U^{ac}(x) \{ \rho^c(x), \rho^b(x') \} \quad (\text{U2}), \quad (\text{F})
\end{aligned} \tag{6.7}$$

Sub-term 3: $- f^{abd} U^{ab'}(w') U^{de}(x) U^{bc}(z) \rho^{b'}(x') \rho^e(x) \rho^c(y)$.

$$\begin{aligned}
[A, B] : [\rho^{b'}(x'), \rho^e(x)] &= i f^{b'eh} \rho^h(x') \delta^{(2)}(x' - x), \quad x \rightarrow x' \\
\frac{1}{4} \{ [A, B], C \} &= -\frac{1}{4} i K_{\text{I.III}}^P(x', x', y) f^{abd} U^{ab'}(w') U^{de}(x') U^{bc}(z) f^{b'eh} \{ \rho^h(x'), \rho^e(y) \} \\
&= -\frac{1}{4} i K_{\text{I.III}}^P(x', x', y) f^{abc} f^{deh} U^{ad}(w') U^{ce}(x') U^{bl}(z) \{ \rho^h(x'), \rho^l(y) \}
\end{aligned} \tag{6.8}$$

$$\begin{aligned}
[A, C] : [\rho^{b'}(x'), \rho^c(y)] &= i f^{b'ch} \rho^h(x') \delta^{(2)}(x' - y), \quad y \rightarrow x' \\
\frac{1}{4} \{ [A, C], B \} &= -\frac{1}{4} i K_{\text{I.III}}^P(x', x, x') f^{abd} U^{ab'}(w') U^{de}(x) U^{bc}(z) f^{b'ch} \{ \rho^h(x'), \rho^e(x) \} \\
&= -\frac{1}{4} i K_{\text{I.III}}^P(x', x, x') f^{abc} f^{deh} U^{ad}(w') U^{cl}(x) U^{be}(z) \{ \rho^h(x'), \rho^l(x) \}
\end{aligned} \tag{6.9}$$

$$\begin{aligned}
[B, C] : [\rho^e(x), \rho^c(y)] &= i f^{ech} \rho^h(x) \delta^{(2)}(x - y), \quad y \rightarrow x \\
\frac{1}{4} \{ [B, C], A \} &= -\frac{1}{4} i K_{\text{I.III}}^P(x', x, x) f^{abd} U^{ab'}(w') U^{de}(x) U^{bc}(z) f^{ech} \{ \rho^h(x), \rho^{b'}(x') \} \\
&= -\frac{1}{4} i K_{\text{I.III}}^P(x', x, x) f^{abc} f^{deh} U^{al}(w') U^{cd}(x) U^{be}(z) \{ \rho^h(x), \rho^l(x') \}
\end{aligned} \tag{6.10}$$

Sub-term 4: $+ f^{abd} U^{ab'}(w') U^{de}(x) U^{bc}(y) \rho^{b'}(x') \rho^e(x) \rho^c(y)$.

$$\begin{aligned}
[A, B] : [\rho^{b'}(x'), \rho^e(x)] &= i f^{b'eh} \rho^h(x') \delta^{(2)}(x' - x), \quad x \rightarrow x' \\
\frac{1}{4} \{ [A, B], C \} &= \frac{1}{4} i K_{\text{I.III}}^P(x', x', y) f^{abd} U^{ab'}(w') U^{de}(x') U^{bc}(y) f^{b'eh} \{ \rho^h(x'), \rho^e(y) \} \\
&= \frac{1}{4} i K_{\text{I.III}}^P(x', x', y) f^{abc} f^{deh} U^{ad}(w') U^{ce}(x') U^{bl}(y) \{ \rho^h(x'), \rho^l(y) \}
\end{aligned} \tag{6.11}$$

$$\begin{aligned}
[A, C] : [\rho^{b'}(x'), \rho^c(y)] &= i f^{b'ch} \rho^h(x') \delta^{(2)}(x' - y), \quad y \rightarrow x' \\
\frac{1}{4} \{[A, C], B\} &= \frac{1}{4} i K_{\text{I.III}}^P(x', x, x') f^{abd} U^{ab'}(w') U^{de}(x) U^{bc}(x') f^{b'ch} \{\rho^h(x'), \rho^e(x)\} \\
&= \frac{1}{4} i K_{\text{I.III}}^P(x', x, x') f^{abc} f^{deh} U^{ad}(w') U^{cl}(x) U^{be}(x') \{\rho^h(x'), \rho^l(x)\}
\end{aligned} \tag{6.12}$$

$$\begin{aligned}
[B, C] : [\rho^e(x), \rho^c(y)] &= i f^{ech} \rho^h(x) \delta^{(2)}(x - y), \quad y \rightarrow x \\
\frac{1}{4} \{[B, C], A\} &= \frac{1}{4} i K_{\text{I.III}}^P(x', x, x) f^{abd} U^{ab'}(w') U^{de}(x) U^{bc}(x) f^{ech} \{\rho^h(x), \rho^{b'}(x')\} \\
&= -\frac{1}{4} i N_c K_{\text{I.III}}^P(x', x, x) U^{ab}(w') U^{ac}(x) \{\rho^c(x), \rho^b(x')\} \quad (\text{U2}), \quad (\text{F})
\end{aligned} \tag{6.13}$$

Two- ρ terms of this source. Multiplying by the prefactor $i\mathcal{N}_P$ and collecting equal terms (dummy colour indices renamed),

$$\begin{aligned}
P^{(2\rho)} \Big|_{\text{GI.III.P}} &= \mathcal{N}_P \int e^{-ik \cdot (w' - w)} \left\{ \right. \\
&\quad + \frac{1}{4} N_c K_{\text{I.III}}^P(x', x', y) U^{ab}(z) U^{ac}(x') \{\rho^c(x'), \rho^b(y)\} \\
&\quad - \frac{1}{4} K_{\text{I.III}}^P(x', x, x') f^{abc} f^{deh} U^{ad}(x') U^{cl}(x) U^{be}(z) \{\rho^h(x'), \rho^l(x)\} \\
&\quad - \frac{1}{4} K_{\text{I.III}}^P(x', x, x) f^{abc} f^{deh} U^{al}(x') U^{cd}(x) U^{be}(z) \{\rho^h(x), \rho^l(x')\} \\
&\quad - \frac{1}{4} N_c K_{\text{I.III}}^P(x', x', y) U^{ab}(y) U^{ac}(x') \{\rho^c(x'), \rho^b(y)\} \\
&\quad + \frac{1}{4} N_c K_{\text{I.III}}^P(x', x, x') U^{ab}(x) U^{ac}(x') \{\rho^c(x'), \rho^b(x)\} \\
&\quad - \frac{1}{4} N_c K_{\text{I.III}}^P(x', x, x) U^{ab}(x') U^{ac}(x) \{\rho^c(x), \rho^b(x')\} \\
&\quad + \frac{1}{4} K_{\text{I.III}}^P(x', x', y) f^{abc} f^{deh} U^{ad}(w') U^{ce}(x') U^{bl}(z) \{\rho^h(x'), \rho^l(y)\} \\
&\quad + \frac{1}{4} K_{\text{I.III}}^P(x', x, x') f^{abc} f^{deh} U^{ad}(w') U^{cl}(x) U^{be}(z) \{\rho^h(x'), \rho^l(x)\} \\
&\quad + \frac{1}{4} K_{\text{I.III}}^P(x', x, x) f^{abc} f^{deh} U^{al}(w') U^{cd}(x) U^{be}(z) \{\rho^h(x), \rho^l(x')\} \\
&\quad - \frac{1}{4} K_{\text{I.III}}^P(x', x', y) f^{abc} f^{deh} U^{ad}(w') U^{ce}(x') U^{bl}(y) \{\rho^h(x'), \rho^l(y)\} \\
&\quad - \frac{1}{4} K_{\text{I.III}}^P(x', x, x') f^{abc} f^{deh} U^{ad}(w') U^{cl}(x) U^{be}(x') \{\rho^h(x'), \rho^l(x)\} \\
&\quad \left. + \frac{1}{4} N_c K_{\text{I.III}}^P(x', x, x) U^{ab}(w') U^{ac}(x) \{\rho^c(x), \rho^b(x')\} \right\} + (w' \leftrightarrow w)
\end{aligned} \tag{6.14}$$

The kernel at the coincident points:

$$\begin{aligned}
K_{\text{I.III}}^P(x', x', y) &= \left[-\mathcal{K}(x' - w') \cdot \mathcal{K}(x' - w) \mathcal{K}(x' - z) \cdot \mathcal{K}(y - z) + 2\mathcal{K}(x' - w') \cdot \mathcal{K}(z - w) \mathcal{K}(y - z) \cdot \mathcal{K}(x' - z) \right] \\
K_{\text{I.III}}^P(x', x, x') &= \left[-\mathcal{K}(x' - w') \cdot \mathcal{K}(x - w) \mathcal{K}(x - z) \cdot \mathcal{K}(x' - z) + 2\mathcal{K}(x' - w') \cdot \mathcal{K}(z - w) \mathcal{K}(x' - z) \cdot \mathcal{K}(x - z) \right] \\
K_{\text{I.III}}^P(x', x, x) &= \frac{1}{(x - z)^2} \left[-\mathcal{K}(x' - w') \cdot \mathcal{K}(x - w) + 2\mathcal{K}(x' - w') \cdot \mathcal{K}(z - w) \right]
\end{aligned}$$

6.2 Group I, Row IV ($\bar{\mathbb{A}}^\dagger \mathbb{B}_2$) (region P)

The three- ρ term.

$$\begin{aligned}
P \Big|_{\text{GI.IV.P}} &= 2i\mathcal{N}_P \int e^{-ik \cdot (w' - w)} K_{\mathfrak{B}}^P(x', x, y) \left\{ \right. \\
&\quad + U^{ab'}(x') f^{cdc'} U^{ac}(w) U^{bd}(z) U^{be}(y) \rho^{b'}(x') \rho^e(y) \rho^{c'}(x) \\
&\quad - U^{ab'}(x') f^{abc} U^{be}(y) U^{cd}(x) \rho^{b'}(x') \rho^e(y) \rho^d(x) \\
&\quad - U^{ab'}(w') f^{cdc'} U^{ac}(w) U^{bd}(z) U^{be}(y) \rho^{b'}(x') \rho^e(y) \rho^{c'}(x) \\
&\quad \left. + U^{ab'}(w') f^{abc} U^{be}(y) U^{cd}(x) \rho^{b'}(x') \rho^e(y) \rho^d(x) \right\} + \text{c.c.}
\end{aligned} \tag{6.15}$$

$$\text{with } K_{\mathfrak{B}}^P(x', x, y) = \left[\mathcal{K}(x' - w') \cdot \mathcal{K}(z - w) \mathcal{K}(y - z) \cdot \mathcal{K}(x - z) - \frac{1}{2} \mathcal{K}(x' - w') \cdot \mathcal{K}(x - w) \mathcal{K}(x - z) \cdot \mathcal{K}(y - z) \right].$$

Reordering. In every sub-term (one product of the two brackets) the charges appear in the order shown, so we apply Eq. (2.3) with A, B, C the charges at x', y, x . The sub-terms are numbered 1–4 in the order above. For each sub-term we list the commutators, and each resulting two- ρ piece before and after the colour algebra.

Sub-term 1: $+ U^{ab'}(x') f^{cdc'} U^{ac}(w) U^{bd}(z) U^{be}(y) \rho^{b'}(x') \rho^e(y) \rho^{c'}(x)$.

$$\begin{aligned}
[A, B] : [\rho^{b'}(x'), \rho^e(y)] &= i f^{b'eh} \rho^h(x') \delta^{(2)}(x' - y), \quad y \rightarrow x' \\
\frac{1}{4} \{[A, B], C\} &= \frac{1}{4} i K_{\mathfrak{B}}^P(x', x, x') U^{ab'}(x') f^{cdc'} U^{ac}(w) U^{bd}(z) U^{be}(x') f^{b'eh} \{\rho^h(x'), \rho^{c'}(x)\}
\end{aligned}$$

$$= \frac{1}{4} i K_{\mathfrak{B}}^P(x', x, x') f^{abc} f^{deh} U^{da}(w) U^{eb}(z) U^{hl}(x') \{\rho^l(x'), \rho^c(x)\} \quad (\text{U2}) \quad (6.16)$$

$$\begin{aligned} [A, C] : [\rho^{b'}(x'), \rho^c(x)] &= i f^{b'c'h} \rho^h(x') \delta^{(2)}(x' - x), \quad x \rightarrow x' \\ \frac{1}{4} \{[A, C], B\} &= \frac{1}{4} i K_{\mathfrak{B}}^P(x', x', y) U^{ab'}(x') f^{cdc'} U^{ac}(w) U^{bd}(z) U^{be}(y) f^{b'c'h} \{\rho^h(x'), \rho^e(y)\} \\ &= \frac{1}{4} i K_{\mathfrak{B}}^P(x', x', y) f^{abc} f^{dce} U^{hd}(x') U^{ha}(w) U^{lb}(z) U^{lm}(y) \{\rho^e(x'), \rho^m(y)\} \end{aligned} \quad (6.17)$$

$$\begin{aligned} [B, C] : [\rho^e(y), \rho^c(x)] &= i f^{ec'h} \rho^h(y) \delta^{(2)}(y - x), \quad x \rightarrow y \\ \frac{1}{4} \{[B, C], A\} &= \frac{1}{4} i K_{\mathfrak{B}}^P(x', y, y) U^{ab'}(x') f^{cdc'} U^{ac}(w) U^{bd}(z) U^{be}(y) f^{ec'h} \{\rho^h(y), \rho^{b'}(x')\} \\ &= \frac{1}{4} i K_{\mathfrak{B}}^P(x', y, y) f^{abc} f^{dce} U^{hl}(x') U^{ha}(w) U^{mb}(z) U^{md}(y) \{\rho^e(y), \rho^l(x')\} \end{aligned} \quad (6.18)$$

Sub-term 2: $- U^{ab'}(x') f^{abc} U^{be}(y) U^{cd}(x) \rho^{b'}(x') \rho^e(y) \rho^d(x)$.

$$\begin{aligned} [A, B] : [\rho^{b'}(x'), \rho^e(y)] &= i f^{b'eh} \rho^h(x') \delta^{(2)}(x' - y), \quad y \rightarrow x' \\ \frac{1}{4} \{[A, B], C\} &= -\frac{1}{4} i K_{\mathfrak{B}}^P(x', x, x') U^{ab'}(x') f^{abc} U^{be}(x') U^{cd}(x) f^{b'eh} \{\rho^h(x'), \rho^d(x)\} \\ &= -\frac{1}{4} i N_c K_{\mathfrak{B}}^P(x', x, x') U^{ab}(x) U^{ac}(x') \{\rho^c(x'), \rho^b(x)\} \quad (\text{U2}), \quad (\text{F}) \end{aligned} \quad (6.19)$$

$$\begin{aligned} [A, C] : [\rho^{b'}(x'), \rho^d(x)] &= i f^{b'dh} \rho^h(x') \delta^{(2)}(x' - x), \quad x \rightarrow x' \\ \frac{1}{4} \{[A, C], B\} &= -\frac{1}{4} i K_{\mathfrak{B}}^P(x', x', y) U^{ab'}(x') f^{abc} U^{be}(y) U^{cd}(x') f^{b'dh} \{\rho^h(x'), \rho^e(y)\} \\ &= \frac{1}{4} i N_c K_{\mathfrak{B}}^P(x', x', y) U^{ab}(y) U^{ac}(x') \{\rho^c(x'), \rho^b(y)\} \quad (\text{U2}), \quad (\text{F}) \end{aligned} \quad (6.20)$$

$$\begin{aligned} [B, C] : [\rho^e(y), \rho^d(x)] &= i f^{edh} \rho^h(y) \delta^{(2)}(y - x), \quad x \rightarrow y \\ \frac{1}{4} \{[B, C], A\} &= -\frac{1}{4} i K_{\mathfrak{B}}^P(x', y, y) U^{ab'}(x') f^{abc} U^{be}(y) U^{cd}(y) f^{edh} \{\rho^h(y), \rho^{b'}(x')\} \\ &= -\frac{1}{4} i N_c K_{\mathfrak{B}}^P(x', y, y) U^{ab}(x') U^{ac}(y) \{\rho^c(y), \rho^b(x')\} \quad (\text{U2}), \quad (\text{F}) \end{aligned} \quad (6.21)$$

Sub-term 3: $- U^{ab'}(w') f^{cdc'} U^{ac}(w) U^{bd}(z) U^{be}(y) \rho^{b'}(x') \rho^e(y) \rho^c(x)$.

$$\begin{aligned} [A, B] : [\rho^{b'}(x'), \rho^e(y)] &= i f^{b'eh} \rho^h(x') \delta^{(2)}(x' - y), \quad y \rightarrow x' \\ \frac{1}{4} \{[A, B], C\} &= -\frac{1}{4} i K_{\mathfrak{B}}^P(x', x, x') U^{ab'}(w') f^{cdc'} U^{ac}(w) U^{bd}(z) U^{be}(x') f^{b'eh} \{\rho^h(x'), \rho^c(x)\} \\ &= -\frac{1}{4} i K_{\mathfrak{B}}^P(x', x, x') f^{abc} f^{deh} U^{ld}(w') U^{la}(w) U^{mb}(z) U^{me}(x') \{\rho^h(x'), \rho^c(x)\} \end{aligned} \quad (6.22)$$

$$\begin{aligned} [A, C] : [\rho^{b'}(x'), \rho^c(x)] &= i f^{b'c'h} \rho^h(x') \delta^{(2)}(x' - x), \quad x \rightarrow x' \\ \frac{1}{4} \{[A, C], B\} &= -\frac{1}{4} i K_{\mathfrak{B}}^P(x', x', y) U^{ab'}(w') f^{cdc'} U^{ac}(w) U^{bd}(z) U^{be}(y) f^{b'c'h} \{\rho^h(x'), \rho^e(y)\} \\ &= -\frac{1}{4} i K_{\mathfrak{B}}^P(x', x', y) f^{abc} f^{dce} U^{hd}(w') U^{ha}(w) U^{lb}(z) U^{lm}(y) \{\rho^e(x'), \rho^m(y)\} \end{aligned} \quad (6.23)$$

$$\begin{aligned} [B, C] : [\rho^e(y), \rho^c(x)] &= i f^{ec'h} \rho^h(y) \delta^{(2)}(y - x), \quad x \rightarrow y \\ \frac{1}{4} \{[B, C], A\} &= -\frac{1}{4} i K_{\mathfrak{B}}^P(x', y, y) U^{ab'}(w') f^{cdc'} U^{ac}(w) U^{bd}(z) U^{be}(y) f^{ec'h} \{\rho^h(y), \rho^{b'}(x')\} \\ &= -\frac{1}{4} i K_{\mathfrak{B}}^P(x', y, y) f^{abc} f^{dce} U^{hl}(w') U^{ha}(w) U^{mb}(z) U^{md}(y) \{\rho^e(y), \rho^l(x')\} \end{aligned} \quad (6.24)$$

Sub-term 4: $+ U^{ab'}(w') f^{abc} U^{be}(y) U^{cd}(x) \rho^{b'}(x') \rho^e(y) \rho^d(x)$.

$$\begin{aligned} [A, B] : [\rho^{b'}(x'), \rho^e(y)] &= i f^{b'eh} \rho^h(x') \delta^{(2)}(x' - y), \quad y \rightarrow x' \\ \frac{1}{4} \{[A, B], C\} &= \frac{1}{4} i K_{\mathfrak{B}}^P(x', x, x') U^{ab'}(w') f^{abc} U^{be}(x') U^{cd}(x) f^{b'eh} \{\rho^h(x'), \rho^d(x)\} \\ &= \frac{1}{4} i K_{\mathfrak{B}}^P(x', x, x') f^{abc} f^{deh} U^{ad}(w') U^{be}(x') U^{cl}(x) \{\rho^h(x'), \rho^l(x)\} \end{aligned} \quad (6.25)$$

$$\begin{aligned} [A, C] : [\rho^{b'}(x'), \rho^d(x)] &= i f^{b'dh} \rho^h(x') \delta^{(2)}(x' - x), \quad x \rightarrow x' \\ \frac{1}{4} \{[A, C], B\} &= \frac{1}{4} i K_{\mathfrak{B}}^P(x', x', y) U^{ab'}(w') f^{abc} U^{be}(y) U^{cd}(x') f^{b'dh} \{\rho^h(x'), \rho^e(y)\} \\ &= \frac{1}{4} i K_{\mathfrak{B}}^P(x', x', y) f^{abc} f^{deh} U^{ad}(w') U^{bl}(y) U^{ce}(x') \{\rho^h(x'), \rho^l(y)\} \end{aligned} \quad (6.26)$$

$$\begin{aligned} [B, C] : [\rho^e(y), \rho^d(x)] &= i f^{edh} \rho^h(y) \delta^{(2)}(y - x), \quad x \rightarrow y \\ \frac{1}{4} \{[B, C], A\} &= \frac{1}{4} i K_{\mathfrak{B}}^P(x', y, y) U^{ab'}(w') f^{abc} U^{be}(y) U^{cd}(y) f^{edh} \{\rho^h(y), \rho^{b'}(x')\} \\ &= \frac{1}{4} i N_c K_{\mathfrak{B}}^P(x', y, y) U^{ab}(w') U^{ac}(y) \{\rho^c(y), \rho^b(x')\} \quad (\text{U2}), \quad (\text{F}) \end{aligned} \quad (6.27)$$

Two- ρ terms of this source. Multiplying by the prefactor $2i\mathcal{N}_P$ and collecting equal terms (dummy colour indices renamed),

$$\begin{aligned} P^{(2\rho)} \Big|_{\text{GI.IV.P}} &= \mathcal{N}_P \int e^{-ik \cdot (w' - w)} \left\{ \right. \\ &\quad \left. - \frac{1}{2} K_{\mathfrak{B}}^P(x', x, x') f^{abc} f^{deh} U^{da}(w) U^{eb}(z) U^{hl}(x') \{\rho^l(x'), \rho^c(x)\} \right\} \end{aligned}$$

$$\begin{aligned}
& -\frac{1}{2} K_{\mathfrak{B}}^P(x', x', y) f^{abc} f^{dce} U^{hd}(x') U^{ha}(w) U^{lb}(z) U^{lm}(y) \{\rho^e(x'), \rho^m(y)\} \\
& -\frac{1}{2} K_{\mathfrak{B}}^P(x', y, y) f^{abc} f^{dce} U^{hl}(x') U^{ha}(w) U^{mb}(z) U^{md}(y) \{\rho^e(y), \rho^l(x')\} \\
& +\frac{1}{2} N_c K_{\mathfrak{B}}^P(x', x, x') U^{ab}(x) U^{ac}(x') \{\rho^c(x'), \rho^b(x)\} \\
& -\frac{1}{2} N_c K_{\mathfrak{B}}^P(x', x', y) U^{ab}(y) U^{ac}(x') \{\rho^c(x'), \rho^b(y)\} \\
& +\frac{1}{2} N_c K_{\mathfrak{B}}^P(x', y, y) U^{ab}(x') U^{ac}(y) \{\rho^c(y), \rho^b(x')\} \\
& +\frac{1}{2} K_{\mathfrak{B}}^P(x', x, x') f^{abc} f^{deh} U^{ld}(w') U^{la}(w) U^{mb}(z) U^{me}(x') \{\rho^h(x'), \rho^e(x)\} \\
& +\frac{1}{2} K_{\mathfrak{B}}^P(x', x', y) f^{abc} f^{dce} U^{hd}(w') U^{ha}(w) U^{lb}(z) U^{lm}(y) \{\rho^e(x'), \rho^m(y)\} \\
& +\frac{1}{2} K_{\mathfrak{B}}^P(x', y, y) f^{abc} f^{dce} U^{hl}(w') U^{ha}(w) U^{mb}(z) U^{md}(y) \{\rho^e(y), \rho^l(x')\} \\
& -\frac{1}{2} K_{\mathfrak{B}}^P(x', x, x') f^{abc} f^{deh} U^{ad}(w') U^{be}(x') U^{cl}(x) \{\rho^h(x'), \rho^l(x)\} \\
& -\frac{1}{2} K_{\mathfrak{B}}^P(x', x', y) f^{abc} f^{deh} U^{ad}(w') U^{bl}(y) U^{ce}(x') \{\rho^h(x'), \rho^l(y)\} \\
& -\frac{1}{2} N_c K_{\mathfrak{B}}^P(x', y, y) U^{ab}(w') U^{ac}(y) \{\rho^c(y), \rho^b(x')\} \} + (w' \leftrightarrow w)
\end{aligned} \tag{6.28}$$

The kernel at the coincident points:

$$\begin{aligned}
K_{\mathfrak{B}}^P(x', x, x') &= \left[\mathcal{K}(x' - w') \cdot \mathcal{K}(z - w) \mathcal{K}(x' - z) \cdot \mathcal{K}(x - z) - \frac{1}{2} \mathcal{K}(x' - w') \cdot \mathcal{K}(x - w) \mathcal{K}(x - z) \cdot \mathcal{K}(x' - z) \right] \\
K_{\mathfrak{B}}^P(x', x', y) &= \left[\mathcal{K}(x' - w') \cdot \mathcal{K}(z - w) \mathcal{K}(y - z) \cdot \mathcal{K}(x' - z) - \frac{1}{2} \mathcal{K}(x' - w') \cdot \mathcal{K}(x' - w) \mathcal{K}(x' - z) \cdot \mathcal{K}(y - z) \right] \\
K_{\mathfrak{B}}^P(x', y, y) &= \frac{1}{(y - z)^2} \left[\mathcal{K}(x' - w') \cdot \mathcal{K}(z - w) - \frac{1}{2} \mathcal{K}(x' - w') \cdot \mathcal{K}(y - w) \right]
\end{aligned}$$

6.3 Group II, Row I, first term (region P)

The three- ρ term.

$$\begin{aligned}
P \Big|_{\text{GII.II.P}} &= -i N_P \int e^{-ik \cdot (w' - w)} K_{\mathfrak{B}}^P(x', x, y) \left\{ \right. \\
& + U^{ac'}(w') f^{cbd} U^{ac}(w) \{\rho^{c'}(x'), \rho^b(y)\} \rho^d(x) \\
& - U^{ac'}(w') f^{acd} U^{bc}(z) U^{de}(x) \{\rho^{c'}(x'), \rho^b(y)\} \rho^e(x) \\
& - U^{bc'}(z) U^{ad'}(x') U^{c'e'}(y) f^{cbd} U^{ac}(w) \{\rho^{d'}(x'), \rho^{e'}(y)\} \rho^d(x) \\
& \left. + U^{bc'}(z) U^{ad'}(x') U^{c'e'}(y) f^{acd} U^{bc}(z) U^{de}(x) \{\rho^{d'}(x'), \rho^{e'}(y)\} \rho^e(x) \right\}
\end{aligned} \tag{6.29}$$

$$\text{with } K_{\mathfrak{B}}^P(x', x, y) = \left[\mathcal{K}(x' - w') \cdot \mathcal{K}(z - w) \mathcal{K}(y - z) \cdot \mathcal{K}(x - z) - \frac{1}{2} \mathcal{K}(x' - w') \cdot \mathcal{K}(x - w) \mathcal{K}(x - z) \cdot \mathcal{K}(y - z) \right].$$

Reordering. In every sub-term (one product of the two brackets) the charges appear in the order shown, so we apply Eq. (2.5) with A, B, C the charges at x, x', y . The sub-terms are numbered 1–4 in the order above. For each sub-term we list the commutators, and each resulting two- ρ piece before and after the colour algebra.

Sub-term 1: $+ U^{ac'}(w') f^{cbd} U^{ac}(w) \{\rho^{c'}(x'), \rho^b(y)\} \rho^d(x)$.

$$\begin{aligned}
[A, B] : [\rho^d(x), \rho^{c'}(x')] &= i f^{dc'h} \rho^h(x) \delta^{(2)}(x - x'), \quad x' \rightarrow x \\
-\frac{1}{2} \{[A, B], C\} &= -\frac{1}{2} i K_{\mathfrak{B}}^P(x, x, y) U^{ac'}(w') f^{cbd} U^{ac}(w) f^{dc'h} \{\rho^h(x), \rho^b(y)\} \\
&= -\frac{1}{2} i K_{\mathfrak{B}}^P(x, x, y) f^{abc} f^{cde} U^{hd}(w') U^{ha}(w) \{\rho^e(x), \rho^b(y)\}
\end{aligned} \tag{6.30}$$

$$\begin{aligned}
[A, C] : [\rho^d(x), \rho^b(y)] &= i f^{dbh} \rho^h(x) \delta^{(2)}(x - y), \quad y \rightarrow x \\
-\frac{1}{2} \{[A, C], B\} &= -\frac{1}{2} i K_{\mathfrak{B}}^P(x', x, x) U^{ac'}(w') f^{cbd} U^{ac}(w) f^{dbh} \{\rho^h(x), \rho^{c'}(x')\} \\
&= \frac{1}{2} i N_c K_{\mathfrak{B}}^P(x', x, x) U^{ab}(w') U^{ac}(w) \{\rho^c(x), \rho^b(x')\} \quad (\text{F})
\end{aligned} \tag{6.31}$$

Sub-term 2: $- U^{ac'}(w') f^{acd} U^{bc}(z) U^{de}(x) \{\rho^{c'}(x'), \rho^b(y)\} \rho^e(x)$.

$$\begin{aligned}
[A, B] : [\rho^e(x), \rho^{c'}(x')] &= i f^{ec'h} \rho^h(x) \delta^{(2)}(x - x'), \quad x' \rightarrow x \\
-\frac{1}{2} \{[A, B], C\} &= \frac{1}{2} i K_{\mathfrak{B}}^P(x, x, y) U^{ac'}(w') f^{acd} U^{bc}(z) U^{de}(x) f^{ec'h} \{\rho^h(x), \rho^b(y)\} \\
&= \frac{1}{2} i K_{\mathfrak{B}}^P(x, x, y) f^{abc} f^{deh} U^{ae}(w') U^{lb}(z) U^{cd}(x) \{\rho^h(x), \rho^l(y)\}
\end{aligned} \tag{6.32}$$

$$\begin{aligned}
[A, C] : [\rho^e(x), \rho^b(y)] &= i f^{ebh} \rho^h(x) \delta^{(2)}(x - y), \quad y \rightarrow x \\
-\frac{1}{2} \{[A, C], B\} &= \frac{1}{2} i K_{\mathfrak{B}}^P(x', x, x) U^{ac'}(w') f^{acd} U^{bc}(z) U^{de}(x) f^{ebh} \{\rho^h(x), \rho^{c'}(x')\} \\
&= \frac{1}{2} i K_{\mathfrak{B}}^P(x', x, x) f^{abc} f^{deh} U^{al}(w') U^{eb}(z) U^{cd}(x) \{\rho^h(x), \rho^l(x')\}
\end{aligned} \tag{6.33}$$

Sub-term 3: $-U^{bc'}(z)U^{ad'}(x')U^{c'e'}(y)f^{cbd}U^{ac}(w)\{\rho^{d'}(x'),\rho^{e'}(y)\}\rho^d(x).$

$$\begin{aligned} [A, B] : [\rho^d(x), \rho^{d'}(x')] &= if^{dd'h}\rho^h(x)\delta^{(2)}(x-x'), \quad x' \rightarrow x \\ -\frac{1}{2}\{[A, B], C\} &= \frac{1}{2}i K_{\mathfrak{B}}^P(x, x, y)U^{bc'}(z)U^{ad'}(x)U^{c'e'}(y)f^{cbd}U^{ac}(w)f^{dd'h}\{\rho^h(x), \rho^{e'}(y)\} \\ &= \frac{1}{2}i K_{\mathfrak{B}}^P(x, x, y)f^{abc}f^{cde}U^{bh}(z)U^{ld}(x)U^{hm}(y)U^{la}(w)\{\rho^e(x), \rho^m(y)\} \end{aligned} \quad (6.34)$$

$$\begin{aligned} [A, C] : [\rho^d(x), \rho^{e'}(y)] &= if^{de'h}\rho^h(x)\delta^{(2)}(x-y), \quad y \rightarrow x \\ -\frac{1}{2}\{[A, C], B\} &= \frac{1}{2}i K_{\mathfrak{B}}^P(x', x, x)U^{bc'}(z)U^{ad'}(x')U^{c'e'}(x)f^{cbd}U^{ac}(w)f^{de'h}\{\rho^h(x), \rho^{d'}(x')\} \\ &= \frac{1}{2}i K_{\mathfrak{B}}^P(x', x, x)f^{abc}f^{cde}U^{bh}(z)U^{lm}(x')U^{hd}(x)U^{la}(w)\{\rho^e(x), \rho^m(x')\} \end{aligned} \quad (6.35)$$

Sub-term 4: $+U^{bc'}(z)U^{ad'}(x')U^{c'e'}(y)f^{acd}U^{bc}(z)U^{de}(x)\{\rho^{d'}(x'),\rho^{e'}(y)\}\rho^e(x).$

$$\begin{aligned} [A, B] : [\rho^e(x), \rho^{d'}(x')] &= if^{ed'h}\rho^h(x)\delta^{(2)}(x-x'), \quad x' \rightarrow x \\ -\frac{1}{2}\{[A, B], C\} &= -\frac{1}{2}i K_{\mathfrak{B}}^P(x, x, y)U^{bc'}(z)U^{ad'}(x)U^{c'e'}(y)f^{acd}U^{bc}(z)U^{de}(x)f^{ed'h}\{\rho^h(x), \rho^{e'}(y)\} \\ &= -\frac{1}{2}i N_c K_{\mathfrak{B}}^P(x, x, y)U^{ab}(y)U^{ac}(x)\{\rho^c(x), \rho^b(y)\} \quad (\text{U1}), (\text{U2}), (\text{F}) \end{aligned} \quad (6.36)$$

$$\begin{aligned} [A, C] : [\rho^e(x), \rho^{e'}(y)] &= if^{ee'h}\rho^h(x)\delta^{(2)}(x-y), \quad y \rightarrow x \\ -\frac{1}{2}\{[A, C], B\} &= -\frac{1}{2}i K_{\mathfrak{B}}^P(x', x, x)U^{bc'}(z)U^{ad'}(x')U^{c'e'}(x)f^{acd}U^{bc}(z)U^{de}(x)f^{ee'h}\{\rho^h(x), \rho^{d'}(x')\} \\ &= \frac{1}{2}i N_c K_{\mathfrak{B}}^P(x', x, x)U^{ab}(x')U^{ac}(x)\{\rho^c(x), \rho^b(x')\} \quad (\text{U1}), (\text{U2}), (\text{F}) \end{aligned} \quad (6.37)$$

Two- ρ terms of this source. Multiplying by the prefactor $-i\mathcal{N}_P$ and collecting equal terms (dummy colour indices renamed),

$$\begin{aligned} P^{(2\rho)} \Big|_{\text{GH.II.P}} &= \mathcal{N}_P \int e^{-ik \cdot (w'-w)} \left\{ \right. \\ &\quad -\frac{1}{2} K_{\mathfrak{B}}^P(x, x, y) f^{abc} f^{cde} U^{hd}(w') U^{ha}(w) \{\rho^e(x), \rho^b(y)\} \\ &\quad +\frac{1}{2} N_c K_{\mathfrak{B}}^P(x', x, x) U^{ab}(w') U^{ac}(w) \{\rho^c(x), \rho^b(x')\} \\ &\quad +\frac{1}{2} K_{\mathfrak{B}}^P(x, x, y) f^{abc} f^{deh} U^{ae}(w') U^{lb}(z) U^{cd}(x) \{\rho^h(x), \rho^l(y)\} \\ &\quad +\frac{1}{2} K_{\mathfrak{B}}^P(x', x, x) f^{abc} f^{deh} U^{al}(w') U^{eb}(z) U^{cd}(x) \{\rho^h(x), \rho^l(x')\} \\ &\quad +\frac{1}{2} K_{\mathfrak{B}}^P(x, x, y) f^{abc} f^{cde} U^{bh}(z) U^{ld}(x) U^{hm}(y) U^{la}(w) \{\rho^e(x), \rho^m(y)\} \\ &\quad +\frac{1}{2} K_{\mathfrak{B}}^P(x', x, x) f^{abc} f^{cde} U^{bh}(z) U^{lm}(x') U^{hd}(x) U^{la}(w) \{\rho^e(x), \rho^m(x')\} \\ &\quad -\frac{1}{2} N_c K_{\mathfrak{B}}^P(x, x, y) U^{ab}(y) U^{ac}(x) \{\rho^c(x), \rho^b(y)\} \\ &\quad \left. +\frac{1}{2} N_c K_{\mathfrak{B}}^P(x', x, x) U^{ab}(x') U^{ac}(x) \{\rho^c(x), \rho^b(x')\} \right\} \end{aligned} \quad (6.38)$$

The kernel at the coincident points:

$$\begin{aligned} K_{\mathfrak{B}}^P(x, x, y) &= \left[\mathcal{K}(x-w') \cdot \mathcal{K}(z-w) \mathcal{K}(y-z) \cdot \mathcal{K}(x-z) - \frac{1}{2} \mathcal{K}(x-w') \cdot \mathcal{K}(x-w) \mathcal{K}(x-z) \cdot \mathcal{K}(y-z) \right] \\ K_{\mathfrak{B}}^P(x', x, x) &= \frac{1}{(x-z)^2} \left[\mathcal{K}(x'-w') \cdot \mathcal{K}(z-w) - \frac{1}{2} \mathcal{K}(x'-w') \cdot \mathcal{K}(x-w) \right] \end{aligned}$$

6.4 Group II, Row I, second term (region P)

The three- ρ term.

$$\begin{aligned} P \Big|_{\text{GH.II.2.P}} &= i\mathcal{N}_P \int e^{-ik \cdot (w'-w)} \bar{K}_{\mathfrak{B}}^P(x', x, y) \left\{ \right. \\ &\quad + f^{c'bd'} U^{ac'}(w') U^{ac}(w) \rho^{d'}(x) \{\rho^c(x'), \rho^b(y)\} \\ &\quad - f^{c'bd'} U^{ac'}(w') U^{bc}(z) U^{ad}(x') U^{ce}(y) \rho^{d'}(x) \{\rho^d(x'), \rho^e(y)\} \\ &\quad - f^{ac'd'} U^{bc'}(z) U^{d'e'}(x) U^{ac}(w) \rho^{e'}(x) \{\rho^c(x'), \rho^b(y)\} \\ &\quad \left. + f^{ac'd'} U^{bc'}(z) U^{d'e'}(x) U^{bc}(z) U^{ad}(x') U^{ce}(y) \rho^{e'}(x) \{\rho^d(x'), \rho^e(y)\} \right\} \end{aligned} \quad (6.39)$$

$$\text{with } \bar{K}_{\mathfrak{B}}^P(x', x, y) = \left[\mathcal{K}(x'-w) \cdot \mathcal{K}(z-w') \mathcal{K}(y-z) \cdot \mathcal{K}(x-z) - \frac{1}{2} \mathcal{K}(x'-w) \cdot \mathcal{K}(x-w') \mathcal{K}(x-z) \cdot \mathcal{K}(y-z) \right].$$

Reordering. In every sub-term (one product of the two brackets) the charges appear in the order shown, so we apply Eq. (2.4) with A, B, C the charges at x, x', y . The sub-terms are numbered 1–4 in the order above. For each sub-term we list the commutators, and each resulting two- ρ piece before and after the colour algebra.

Sub-term 1: $+ f^{c'bd'} U^{ac'}(w') U^{ac}(w) \rho^{d'}(x) \{\rho^c(x'), \rho^b(y)\}.$

$$\begin{aligned} [A, B] : [\rho^{d'}(x), \rho^c(x')] &= i f^{d'ch} \rho^h(x) \delta^{(2)}(x - x'), \quad x' \rightarrow x \\ \frac{1}{2} \{[A, B], C\} &= \frac{1}{2} i \bar{K}_{\mathfrak{B}}^P(x, x, y) f^{c'bd'} U^{ac'}(w') U^{ac}(w) f^{d'ch} \{\rho^h(x), \rho^b(y)\} \\ &= \frac{1}{2} i \bar{K}_{\mathfrak{B}}^P(x, x, y) f^{abc} f^{cde} U^{ha}(w') U^{hd}(w) \{\rho^e(x), \rho^b(y)\} \end{aligned} \quad (6.40)$$

$$\begin{aligned} [A, C] : [\rho^{d'}(x), \rho^b(y)] &= i f^{d'bh} \rho^h(x) \delta^{(2)}(x - y), \quad y \rightarrow x \\ \frac{1}{2} \{[A, C], B\} &= \frac{1}{2} i \bar{K}_{\mathfrak{B}}^P(x', x, x) f^{c'bd'} U^{ac'}(w') U^{ac}(w) f^{d'bh} \{\rho^h(x), \rho^c(x')\} \\ &= -\frac{1}{2} i N_c \bar{K}_{\mathfrak{B}}^P(x', x, x) U^{ab}(w') U^{ac}(w) \{\rho^b(x), \rho^c(x')\} \quad (\text{F}) \end{aligned} \quad (6.41)$$

Sub-term 2: $- f^{c'bd'} U^{ac'}(w') U^{bc}(z) U^{ad}(x') U^{ce}(y) \rho^{d'}(x) \{\rho^d(x'), \rho^e(y)\}.$

$$\begin{aligned} [A, B] : [\rho^{d'}(x), \rho^d(x')] &= i f^{d'dh} \rho^h(x) \delta^{(2)}(x - x'), \quad x' \rightarrow x \\ \frac{1}{2} \{[A, B], C\} &= -\frac{1}{2} i \bar{K}_{\mathfrak{B}}^P(x, x, y) f^{c'bd'} U^{ac'}(w') U^{bc}(z) U^{ad}(x') U^{ce}(y) f^{d'dh} \{\rho^h(x), \rho^e(y)\} \\ &= -\frac{1}{2} i \bar{K}_{\mathfrak{B}}^P(x, x, y) f^{abc} f^{cde} U^{ha}(w') U^{bl}(z) U^{hd}(x) U^{lm}(y) \{\rho^e(x), \rho^m(y)\} \end{aligned} \quad (6.42)$$

$$\begin{aligned} [A, C] : [\rho^{d'}(x), \rho^e(y)] &= i f^{d'eh} \rho^h(x) \delta^{(2)}(x - y), \quad y \rightarrow x \\ \frac{1}{2} \{[A, C], B\} &= -\frac{1}{2} i \bar{K}_{\mathfrak{B}}^P(x', x, x) f^{c'bd'} U^{ac'}(w') U^{bc}(z) U^{ad}(x') U^{ce}(x) f^{d'eh} \{\rho^h(x), \rho^d(x')\} \\ &= -\frac{1}{2} i \bar{K}_{\mathfrak{B}}^P(x', x, x) f^{abc} f^{cde} U^{ha}(w') U^{bl}(z) U^{hm}(x') U^{ld}(x) \{\rho^e(x), \rho^m(x')\} \end{aligned} \quad (6.43)$$

Sub-term 3: $- f^{ac'd'} U^{bc'}(z) U^{d'e'}(x) U^{ac}(w) \rho^{e'}(x) \{\rho^c(x'), \rho^b(y)\}.$

$$\begin{aligned} [A, B] : [\rho^{e'}(x), \rho^c(x')] &= i f^{e'ch} \rho^h(x) \delta^{(2)}(x - x'), \quad x' \rightarrow x \\ \frac{1}{2} \{[A, B], C\} &= -\frac{1}{2} i \bar{K}_{\mathfrak{B}}^P(x, x, y) f^{ac'd'} U^{bc'}(z) U^{d'e'}(x) U^{ac}(w) f^{e'ch} \{\rho^h(x), \rho^b(y)\} \\ &= -\frac{1}{2} i \bar{K}_{\mathfrak{B}}^P(x, x, y) f^{abc} f^{deh} U^{lb}(z) U^{cd}(x) U^{ae}(w) \{\rho^h(x), \rho^l(y)\} \end{aligned} \quad (6.44)$$

$$\begin{aligned} [A, C] : [\rho^{e'}(x), \rho^b(y)] &= i f^{e'bh} \rho^h(x) \delta^{(2)}(x - y), \quad y \rightarrow x \\ \frac{1}{2} \{[A, C], B\} &= -\frac{1}{2} i \bar{K}_{\mathfrak{B}}^P(x', x, x) f^{ac'd'} U^{bc'}(z) U^{d'e'}(x) U^{ac}(w) f^{e'bh} \{\rho^h(x), \rho^c(x')\} \\ &= -\frac{1}{2} i \bar{K}_{\mathfrak{B}}^P(x', x, x) f^{abc} f^{deh} U^{eb}(z) U^{cd}(x) U^{al}(w) \{\rho^h(x), \rho^l(x')\} \end{aligned} \quad (6.45)$$

Sub-term 4: $+ f^{ac'd'} U^{bc'}(z) U^{d'e'}(x) U^{bc}(z) U^{ad}(x') U^{ce}(y) \rho^{e'}(x) \{\rho^d(x'), \rho^e(y)\}.$

$$\begin{aligned} [A, B] : [\rho^{e'}(x), \rho^d(x')] &= i f^{e'dh} \rho^h(x) \delta^{(2)}(x - x'), \quad x' \rightarrow x \\ \frac{1}{2} \{[A, B], C\} &= \frac{1}{2} i \bar{K}_{\mathfrak{B}}^P(x, x, y) f^{ac'd'} U^{bc'}(z) U^{d'e'}(x) U^{bc}(z) U^{ad}(x') U^{ce}(y) f^{e'dh} \{\rho^h(x), \rho^e(y)\} \\ &= \frac{1}{2} i N_c \bar{K}_{\mathfrak{B}}^P(x, x, y) U^{ab}(y) U^{ac}(x) \{\rho^c(x), \rho^b(y)\} \quad (\text{U1}), (\text{U2}), (\text{F}) \end{aligned} \quad (6.46)$$

$$\begin{aligned} [A, C] : [\rho^{e'}(x), \rho^e(y)] &= i f^{e'eh} \rho^h(x) \delta^{(2)}(x - y), \quad y \rightarrow x \\ \frac{1}{2} \{[A, C], B\} &= \frac{1}{2} i \bar{K}_{\mathfrak{B}}^P(x', x, x) f^{ac'd'} U^{bc'}(z) U^{d'e'}(x) U^{bc}(z) U^{ad}(x') U^{ce}(x) f^{e'eh} \{\rho^h(x), \rho^d(x')\} \\ &= -\frac{1}{2} i N_c \bar{K}_{\mathfrak{B}}^P(x', x, x) U^{ab}(x') U^{ac}(x) \{\rho^c(x), \rho^b(x')\} \quad (\text{U1}), (\text{U2}), (\text{F}) \end{aligned} \quad (6.47)$$

Two- ρ terms of this source. Multiplying by the prefactor $i\mathcal{N}_P$ and collecting equal terms (dummy colour indices renamed),

$$\begin{aligned} P^{(2\rho)} \Big|_{\text{GII.I2.P}} &= \mathcal{N}_P \int e^{-ik \cdot (w' - w)} \left\{ \right. \\ &\quad - \frac{1}{2} \bar{K}_{\mathfrak{B}}^P(x, x, y) f^{abc} f^{cde} U^{ha}(w') U^{hd}(w) \{\rho^e(x), \rho^b(y)\} \\ &\quad + \frac{1}{2} N_c \bar{K}_{\mathfrak{B}}^P(x', x, x) U^{ab}(w') U^{ac}(w) \{\rho^b(x), \rho^c(x')\} \\ &\quad + \frac{1}{2} \bar{K}_{\mathfrak{B}}^P(x, x, y) f^{abc} f^{cde} U^{ha}(w') U^{bl}(z) U^{hd}(x) U^{lm}(y) \{\rho^e(x), \rho^m(y)\} \\ &\quad + \frac{1}{2} \bar{K}_{\mathfrak{B}}^P(x', x, x) f^{abc} f^{cde} U^{ha}(w') U^{bl}(z) U^{hm}(x') U^{ld}(x) \{\rho^e(x), \rho^m(x')\} \\ &\quad + \frac{1}{2} \bar{K}_{\mathfrak{B}}^P(x, x, y) f^{abc} f^{deh} U^{lb}(z) U^{cd}(x) U^{ae}(w) \{\rho^h(x), \rho^l(y)\} \\ &\quad + \frac{1}{2} \bar{K}_{\mathfrak{B}}^P(x', x, x) f^{abc} f^{deh} U^{eb}(z) U^{cd}(x) U^{al}(w) \{\rho^h(x), \rho^l(x')\} \\ &\quad - \frac{1}{2} N_c \bar{K}_{\mathfrak{B}}^P(x, x, y) U^{ab}(y) U^{ac}(x) \{\rho^c(x), \rho^b(y)\} \\ &\quad \left. + \frac{1}{2} N_c \bar{K}_{\mathfrak{B}}^P(x', x, x) U^{ab}(x') U^{ac}(x) \{\rho^c(x), \rho^b(x')\} \right\} \end{aligned} \quad (6.48)$$

The kernel at the coincident points:

$$\begin{aligned}\bar{K}_{\mathfrak{B}}^P(x, x, y) &= \left[\mathcal{K}(x - w) \cdot \mathcal{K}(z - w') \mathcal{K}(y - z) \cdot \mathcal{K}(x - z) - \frac{1}{2} \mathcal{K}(x - w) \cdot \mathcal{K}(x - w') \mathcal{K}(x - z) \cdot \mathcal{K}(y - z) \right] \\ \bar{K}_{\mathfrak{B}}^P(x', x, x) &= \frac{1}{(x - z)^2} \left[\mathcal{K}(x' - w) \cdot \mathcal{K}(z - w') - \frac{1}{2} \mathcal{K}(x' - w) \cdot \mathcal{K}(x - w') \right]\end{aligned}$$

6.5 Group III, Row I (region P)

The three- ρ term.

$$\begin{aligned}P \Big|_{\text{GIII.I.P}} &= -2i \mathcal{N}_P \int e^{-ik \cdot (w' - w)} K_{\mathfrak{B}}^P(x', x, y) \left\{ \right. \\ &\quad + U^{ab'}(x') U^{c'd'}(y) U^{c'b}(z) f^{cbd} U^{ac}(w) \rho^{b'}(x') \rho^{d'}(y) \rho^d(x) \\ &\quad - U^{ab'}(x') U^{c'd'}(y) U^{c'b}(z) f^{acd} U^{bc}(z) U^{de}(x) \rho^{b'}(x') \rho^{d'}(y) \rho^e(x) \\ &\quad - U^{bd'}(z) U^{d'e'}(y) U^{ac'}(w') f^{cbd} U^{ac}(w) \rho^{c'}(x') \rho^{e'}(y) \rho^d(x) \\ &\quad \left. + U^{bd'}(z) U^{d'e'}(y) U^{ac'}(w') f^{acd} U^{bc}(z) U^{de}(x) \rho^{c'}(x') \rho^{e'}(y) \rho^e(x) \right\} + \text{c.c.}\end{aligned}\quad (6.49)$$

$$\text{with } K_{\mathfrak{B}}^P(x', x, y) = \left[\mathcal{K}(x' - w') \cdot \mathcal{K}(z - w) \mathcal{K}(y - z) \cdot \mathcal{K}(x - z) - \frac{1}{2} \mathcal{K}(x' - w') \cdot \mathcal{K}(x - w) \mathcal{K}(x - z) \cdot \mathcal{K}(y - z) \right].$$

Reordering. In every sub-term (one product of the two brackets) the charges appear in the order shown, so we apply Eq. (2.3) with A, B, C the charges at x', y, x . The sub-terms are numbered 1–4 in the order above. For each sub-term we list the commutators, and each resulting two- ρ piece before and after the colour algebra.

Sub-term 1: $+ U^{ab'}(x') U^{c'd'}(y) U^{c'b}(z) f^{cbd} U^{ac}(w) \rho^{b'}(x') \rho^{d'}(y) \rho^d(x)$.

$$\begin{aligned}[A, B] : [\rho^{b'}(x'), \rho^{d'}(y)] &= i f^{b'd'h} \rho^h(x') \delta^{(2)}(x' - y), \quad y \rightarrow x' \\ \frac{1}{4} \{[A, B], C\} &= \frac{1}{4} i K_{\mathfrak{B}}^P(x', x, x') U^{ab'}(x') U^{c'd'}(x') U^{c'b}(z) f^{cbd} U^{ac}(w) f^{b'd'h} \{\rho^h(x'), \rho^d(x)\} \\ &= \frac{1}{4} i K_{\mathfrak{B}}^P(x', x, x') f^{abc} f^{deh} U^{eb}(z) U^{da}(w) U^{hl}(x') \{\rho^l(x'), \rho^c(x)\} \quad (\text{U2})\end{aligned}\quad (6.50)$$

$$\begin{aligned}[A, C] : [\rho^{b'}(x'), \rho^d(x)] &= i f^{b'dh} \rho^h(x') \delta^{(2)}(x' - x), \quad x \rightarrow x' \\ \frac{1}{4} \{[A, C], B\} &= \frac{1}{4} i K_{\mathfrak{B}}^P(x', x', y) U^{ab'}(x') U^{c'd'}(y) U^{c'b}(z) f^{cbd} U^{ac}(w) f^{b'dh} \{\rho^h(x'), \rho^{d'}(y)\} \\ &= \frac{1}{4} i K_{\mathfrak{B}}^P(x', x', y) f^{abc} f^{dce} U^{hd}(x') U^{lm}(y) U^{lb}(z) U^{ha}(w) \{\rho^e(x'), \rho^m(y)\}\end{aligned}\quad (6.51)$$

$$\begin{aligned}[B, C] : [\rho^{d'}(y), \rho^d(x)] &= i f^{d'dh} \rho^h(y) \delta^{(2)}(y - x), \quad x \rightarrow y \\ \frac{1}{4} \{[B, C], A\} &= \frac{1}{4} i K_{\mathfrak{B}}^P(x', y, y) U^{ab'}(x') U^{c'd'}(y) U^{c'b}(z) f^{cbd} U^{ac}(w) f^{d'dh} \{\rho^h(y), \rho^{b'}(x')\} \\ &= \frac{1}{4} i K_{\mathfrak{B}}^P(x', y, y) f^{abc} f^{dce} U^{hl}(x') U^{md}(y) U^{mb}(z) U^{ha}(w) \{\rho^e(y), \rho^l(x')\}\end{aligned}\quad (6.52)$$

Sub-term 2: $- U^{ab'}(x') U^{c'd'}(y) U^{c'b}(z) f^{acd} U^{bc}(z) U^{de}(x) \rho^{b'}(x') \rho^{d'}(y) \rho^e(x)$.

$$\begin{aligned}[A, B] : [\rho^{b'}(x'), \rho^{d'}(y)] &= i f^{b'd'h} \rho^h(x') \delta^{(2)}(x' - y), \quad y \rightarrow x' \\ \frac{1}{4} \{[A, B], C\} &= -\frac{1}{4} i K_{\mathfrak{B}}^P(x', x, x') U^{ab'}(x') U^{c'd'}(x') U^{c'b}(z) f^{acd} U^{bc}(z) U^{de}(x) f^{b'd'h} \{\rho^h(x'), \rho^e(x)\} \\ &= -\frac{1}{4} i N_c K_{\mathfrak{B}}^P(x', x, x') U^{ab}(x) U^{ac}(x') \{\rho^c(x'), \rho^b(x)\} \quad (\text{U1}^*), (\text{U2}), (\text{F})\end{aligned}\quad (6.53)$$

$$\begin{aligned}[A, C] : [\rho^{b'}(x'), \rho^e(x)] &= i f^{b'eh} \rho^h(x') \delta^{(2)}(x' - x), \quad x \rightarrow x' \\ \frac{1}{4} \{[A, C], B\} &= -\frac{1}{4} i K_{\mathfrak{B}}^P(x', x', y) U^{ab'}(x') U^{c'd'}(y) U^{c'b}(z) f^{acd} U^{bc}(z) U^{de}(x') f^{b'eh} \{\rho^h(x'), \rho^{d'}(y)\} \\ &= \frac{1}{4} i N_c K_{\mathfrak{B}}^P(x', x', y) U^{ab}(y) U^{ac}(x') \{\rho^c(x'), \rho^b(y)\} \quad (\text{U1}^*), (\text{U2}), (\text{F})\end{aligned}\quad (6.54)$$

$$\begin{aligned}[B, C] : [\rho^{d'}(y), \rho^e(x)] &= i f^{d'eh} \rho^h(y) \delta^{(2)}(y - x), \quad x \rightarrow y \\ \frac{1}{4} \{[B, C], A\} &= -\frac{1}{4} i K_{\mathfrak{B}}^P(x', y, y) U^{ab'}(x') U^{c'd'}(y) U^{c'b}(z) f^{acd} U^{bc}(z) U^{de}(y) f^{d'eh} \{\rho^h(y), \rho^{b'}(x')\} \\ &= -\frac{1}{4} i N_c K_{\mathfrak{B}}^P(x', y, y) U^{ab}(x') U^{ac}(y) \{\rho^c(y), \rho^b(x')\} \quad (\text{U1}^*), (\text{U2}), (\text{F})\end{aligned}\quad (6.55)$$

Sub-term 3: $- U^{bd'}(z) U^{d'e'}(y) U^{ac'}(w') f^{cbd} U^{ac}(w) \rho^{c'}(x') \rho^{e'}(y) \rho^d(x)$.

$$\begin{aligned}[A, B] : [\rho^{c'}(x'), \rho^{e'}(y)] &= i f^{c'e'h} \rho^h(x') \delta^{(2)}(x' - y), \quad y \rightarrow x' \\ \frac{1}{4} \{[A, B], C\} &= -\frac{1}{4} i K_{\mathfrak{B}}^P(x', x, x') U^{bd'}(z) U^{d'e'}(x') U^{ac'}(w') f^{cbd} U^{ac}(w) f^{c'e'h} \{\rho^h(x'), \rho^d(x)\} \\ &= -\frac{1}{4} i K_{\mathfrak{B}}^P(x', x, x') f^{abc} f^{deh} U^{bl}(z) U^{le}(x') U^{md}(w') U^{ma}(w) \{\rho^h(x'), \rho^c(x)\}\end{aligned}\quad (6.56)$$

$$\begin{aligned}
[A, C] : [\rho^{c'}(x'), \rho^d(x)] &= i f^{c'dh} \rho^h(x') \delta^{(2)}(x' - x), \quad x \rightarrow x' \\
\frac{1}{4} \{[A, C], B\} &= -\frac{1}{4} i K_{\mathfrak{B}}^P(x', x', y) U^{bd'}(z) U^{d'e'}(y) U^{ac'}(w') f^{cbd} U^{ac}(w) f^{c'dh} \{\rho^h(x'), \rho^{e'}(y)\} \\
&= -\frac{1}{4} i K_{\mathfrak{B}}^P(x', x', y) f^{abc} f^{dce} U^{bh}(z) U^{hl}(y) U^{md}(w') U^{ma}(w) \{\rho^e(x'), \rho^l(y)\} \quad (6.57)
\end{aligned}$$

$$\begin{aligned}
[B, C] : [\rho^{e'}(y), \rho^d(x)] &= i f^{e'dh} \rho^h(y) \delta^{(2)}(y - x), \quad x \rightarrow y \\
\frac{1}{4} \{[B, C], A\} &= -\frac{1}{4} i K_{\mathfrak{B}}^P(x', y, y) U^{bd'}(z) U^{d'e'}(y) U^{ac'}(w') f^{cbd} U^{ac}(w) f^{e'dh} \{\rho^h(y), \rho^{c'}(x')\} \\
&= -\frac{1}{4} i K_{\mathfrak{B}}^P(x', y, y) f^{abc} f^{dce} U^{bh}(z) U^{hd}(y) U^{lm}(w') U^{la}(w) \{\rho^e(y), \rho^m(x')\} \quad (6.58)
\end{aligned}$$

Sub-term 4: $+ U^{bd'}(z) U^{d'e'}(y) U^{ac'}(w') f^{acd} U^{bc}(z) U^{de}(x) \rho^{c'}(x') \rho^{e'}(y) \rho^e(x)$.

$$\begin{aligned}
[A, B] : [\rho^{c'}(x'), \rho^{e'}(y)] &= i f^{c'e'h} \rho^h(x') \delta^{(2)}(x' - y), \quad y \rightarrow x' \\
\frac{1}{4} \{[A, B], C\} &= \frac{1}{4} i K_{\mathfrak{B}}^P(x', x, x') U^{bd'}(z) U^{d'e'}(x') U^{ac'}(w') f^{acd} U^{bc}(z) U^{de}(x) f^{c'e'h} \{\rho^h(x'), \rho^e(x)\} \\
&= \frac{1}{4} i K_{\mathfrak{B}}^P(x', x, x') f^{abc} f^{deh} U^{be}(x') U^{ad}(w') U^{cl}(x) \{\rho^h(x'), \rho^l(x)\} \quad (U1) \quad (6.59)
\end{aligned}$$

$$\begin{aligned}
[A, C] : [\rho^{c'}(x'), \rho^e(x)] &= i f^{c'eh} \rho^h(x') \delta^{(2)}(x' - x), \quad x \rightarrow x' \\
\frac{1}{4} \{[A, C], B\} &= \frac{1}{4} i K_{\mathfrak{B}}^P(x', x', y) U^{bd'}(z) U^{d'e'}(y) U^{ac'}(w') f^{acd} U^{bc}(z) U^{de}(x') f^{c'eh} \{\rho^h(x'), \rho^{e'}(y)\} \\
&= \frac{1}{4} i K_{\mathfrak{B}}^P(x', x', y) f^{abc} f^{deh} U^{bl}(y) U^{ad}(w') U^{ce}(x') \{\rho^h(x'), \rho^l(y)\} \quad (U1) \quad (6.60)
\end{aligned}$$

$$\begin{aligned}
[B, C] : [\rho^{e'}(y), \rho^e(x)] &= i f^{e'eh} \rho^h(y) \delta^{(2)}(y - x), \quad x \rightarrow y \\
\frac{1}{4} \{[B, C], A\} &= \frac{1}{4} i K_{\mathfrak{B}}^P(x', y, y) U^{bd'}(z) U^{d'e'}(y) U^{ac'}(w') f^{acd} U^{bc}(z) U^{de}(y) f^{e'eh} \{\rho^h(y), \rho^{c'}(x')\} \\
&= \frac{1}{4} i N_c K_{\mathfrak{B}}^P(x', y, y) U^{ab}(w') U^{ac}(y) \{\rho^c(y), \rho^b(x')\} \quad (U1), (U2), (F) \quad (6.61)
\end{aligned}$$

Two- ρ terms of this source. Multiplying by the prefactor $-2i\mathcal{N}_P$ and collecting equal terms (dummy colour indices renamed),

$$\begin{aligned}
P^{(2\rho)} \Big|_{\text{GIII.I.P}} &= \mathcal{N}_P \int e^{-ik \cdot (w' - w)} \left\{ \right. \\
&+ \frac{1}{2} K_{\mathfrak{B}}^P(x', x, x') f^{abc} f^{deh} U^{eb}(z) U^{da}(w) U^{hl}(x') \{\rho^l(x'), \rho^c(x)\} \\
&+ \frac{1}{2} K_{\mathfrak{B}}^P(x', x', y) f^{abc} f^{dce} U^{hd}(x') U^{lm}(y) U^{lb}(z) U^{ha}(w) \{\rho^e(x'), \rho^m(y)\} \\
&+ \frac{1}{2} K_{\mathfrak{B}}^P(x', y, y) f^{abc} f^{dce} U^{hl}(x') U^{md}(y) U^{mb}(z) U^{ha}(w) \{\rho^e(y), \rho^l(x')\} \\
&- \frac{1}{2} N_c K_{\mathfrak{B}}^P(x', x, x') U^{ab}(x) U^{ac}(x') \{\rho^c(x'), \rho^b(x)\} \\
&+ \frac{1}{2} N_c K_{\mathfrak{B}}^P(x', x', y) U^{ab}(y) U^{ac}(x') \{\rho^c(x'), \rho^b(y)\} \\
&- \frac{1}{2} N_c K_{\mathfrak{B}}^P(x', y, y) U^{ab}(x') U^{ac}(y) \{\rho^c(y), \rho^b(x')\} \\
&- \frac{1}{2} K_{\mathfrak{B}}^P(x', x, x') f^{abc} f^{deh} U^{bl}(z) U^{le}(x') U^{md}(w') U^{ma}(w) \{\rho^h(x'), \rho^e(x)\} \\
&- \frac{1}{2} K_{\mathfrak{B}}^P(x', x', y) f^{abc} f^{dce} U^{bh}(z) U^{hl}(y) U^{md}(w') U^{ma}(w) \{\rho^e(x'), \rho^l(y)\} \\
&- \frac{1}{2} K_{\mathfrak{B}}^P(x', y, y) f^{abc} f^{dce} U^{bh}(z) U^{hd}(y) U^{lm}(w') U^{la}(w) \{\rho^e(y), \rho^m(x')\} \\
&+ \frac{1}{2} K_{\mathfrak{B}}^P(x', x, x') f^{abc} f^{deh} U^{be}(x') U^{ad}(w') U^{cl}(x) \{\rho^h(x'), \rho^l(x)\} \\
&+ \frac{1}{2} K_{\mathfrak{B}}^P(x', x', y) f^{abc} f^{deh} U^{bl}(y) U^{ad}(w') U^{ce}(x') \{\rho^h(x'), \rho^l(y)\} \\
&+ \frac{1}{2} N_c K_{\mathfrak{B}}^P(x', y, y) U^{ab}(w') U^{ac}(y) \{\rho^c(y), \rho^b(x')\} \left. \right\} + (w' \leftrightarrow w) \quad (6.62)
\end{aligned}$$

The kernel at the coincident points:

$$\begin{aligned}
K_{\mathfrak{B}}^P(x', x, x') &= \left[\mathcal{K}(x' - w') \cdot \mathcal{K}(z - w) \mathcal{K}(x' - z) \cdot \mathcal{K}(x - z) - \frac{1}{2} \mathcal{K}(x' - w') \cdot \mathcal{K}(x - w) \mathcal{K}(x - z) \cdot \mathcal{K}(x' - z) \right] \\
K_{\mathfrak{B}}^P(x', x', y) &= \left[\mathcal{K}(x' - w') \cdot \mathcal{K}(z - w) \mathcal{K}(y - z) \cdot \mathcal{K}(x' - z) - \frac{1}{2} \mathcal{K}(x' - w') \cdot \mathcal{K}(x' - w) \mathcal{K}(x' - z) \cdot \mathcal{K}(y - z) \right] \\
K_{\mathfrak{B}}^P(x', y, y) &= \frac{1}{(y - z)^2} \left[\mathcal{K}(x' - w') \cdot \mathcal{K}(z - w) - \frac{1}{2} \mathcal{K}(x' - w') \cdot \mathcal{K}(y - w) \right]
\end{aligned}$$

6.6 Group III, Row II (region P)

The three- ρ term.

$$\begin{aligned}
P \Big|_{\text{GIII.II.P}} &= -2i\mathcal{N}_P \int e^{-ik \cdot (w' - w)} K_{\mathfrak{B}}^P(x', x, y) \left\{ \right. \\
&+ U^{c'd'}(y) U^{ab'}(x') U^{bc'}(z) f^{cbd} U^{ac}(w) \rho^{d'}(y) \rho^{b'}(x') \rho^d(x)
\end{aligned}$$

$$\begin{aligned}
& - U^{c'd'}(y) U^{ab'}(x') U^{bc'}(z) f^{acd} U^{bc}(z) U^{de}(x) \rho^{d'}(y) \rho^{b'}(x') \rho^e(x) \\
& - U^{ab'}(x') f^{cbd} U^{ac}(w) \rho^b(y) \rho^{b'}(x') \rho^d(x) \\
& + U^{ab'}(x') f^{acd} U^{bc}(z) U^{de}(x) \rho^b(y) \rho^{b'}(x') \rho^e(x) \} + \text{c.c.}
\end{aligned} \tag{6.63}$$

with $K_{\mathfrak{B}}^P(x', x, y) = \left[\mathcal{K}(x' - w') \cdot \mathcal{K}(z - w) \mathcal{K}(y - z) \cdot \mathcal{K}(x - z) - \frac{1}{2} \mathcal{K}(x' - w') \cdot \mathcal{K}(x - w) \mathcal{K}(x - z) \cdot \mathcal{K}(y - z) \right]$.

Reordering. In every sub-term (one product of the two brackets) the charges appear in the order shown, so we apply Eq. (2.3) with A, B, C the charges at y, x', x . The sub-terms are numbered 1–4 in the order above. For each sub-term we list the commutators, and each resulting two- ρ piece before and after the colour algebra.

Sub-term 1: $+ U^{c'd'}(y) U^{ab'}(x') U^{bc'}(z) f^{cbd} U^{ac}(w) \rho^{d'}(y) \rho^{b'}(x') \rho^d(x)$.

$$\begin{aligned}
[A, B] : [\rho^{d'}(y), \rho^{b'}(x')] &= i f^{d'b'h} \rho^h(y) \delta^{(2)}(y - x'), \quad x' \rightarrow y \\
\frac{1}{4} \{ [A, B], C \} &= \frac{1}{4} i K_{\mathfrak{B}}^P(y, x, y) U^{c'd'}(y) U^{ab'}(y) U^{bc'}(z) f^{cbd} U^{ac}(w) f^{d'b'h} \{ \rho^h(y), \rho^d(x) \} \\
&= \frac{1}{4} i K_{\mathfrak{B}}^P(y, x, y) f^{abc} f^{deh} U^{bd}(z) U^{ea}(w) U^{hl}(y) \{ \rho^l(y), \rho^e(x) \} \quad (\text{U2})
\end{aligned} \tag{6.64}$$

$$\begin{aligned}
[A, C] : [\rho^{d'}(y), \rho^d(x)] &= i f^{d'dh} \rho^h(y) \delta^{(2)}(y - x), \quad x \rightarrow y \\
\frac{1}{4} \{ [A, C], B \} &= \frac{1}{4} i K_{\mathfrak{B}}^P(x', y, y) U^{c'd'}(y) U^{ab'}(x') U^{bc'}(z) f^{cbd} U^{ac}(w) f^{d'dh} \{ \rho^h(y), \rho^{b'}(x') \} \\
&= \frac{1}{4} i K_{\mathfrak{B}}^P(x', y, y) f^{abc} f^{dce} U^{hd}(y) U^{lm}(x') U^{bh}(z) U^{la}(w) \{ \rho^e(y), \rho^m(x') \}
\end{aligned} \tag{6.65}$$

$$\begin{aligned}
[B, C] : [\rho^{b'}(x'), \rho^d(x)] &= i f^{b'dh} \rho^h(x') \delta^{(2)}(x' - x), \quad x \rightarrow x' \\
\frac{1}{4} \{ [B, C], A \} &= \frac{1}{4} i K_{\mathfrak{B}}^P(x', x', y) U^{c'd'}(y) U^{ab'}(x') U^{bc'}(z) f^{cbd} U^{ac}(w) f^{b'dh} \{ \rho^h(x'), \rho^{d'}(y) \} \\
&= \frac{1}{4} i K_{\mathfrak{B}}^P(x', x', y) f^{abc} f^{dce} U^{hl}(y) U^{md}(x') U^{bh}(z) U^{ma}(w) \{ \rho^e(x'), \rho^l(y) \}
\end{aligned} \tag{6.66}$$

Sub-term 2: $- U^{c'd'}(y) U^{ab'}(x') U^{bc'}(z) f^{acd} U^{bc}(z) U^{de}(x) \rho^{d'}(y) \rho^{b'}(x') \rho^e(x)$.

$$\begin{aligned}
[A, B] : [\rho^{d'}(y), \rho^{b'}(x')] &= i f^{d'b'h} \rho^h(y) \delta^{(2)}(y - x'), \quad x' \rightarrow y \\
\frac{1}{4} \{ [A, B], C \} &= -\frac{1}{4} i K_{\mathfrak{B}}^P(y, x, y) U^{c'd'}(y) U^{ab'}(y) U^{bc'}(z) f^{acd} U^{bc}(z) U^{de}(x) f^{d'b'h} \{ \rho^h(y), \rho^e(x) \} \\
&= \frac{1}{4} i N_c K_{\mathfrak{B}}^P(y, x, y) U^{ab}(x) U^{ac}(y) \{ \rho^c(y), \rho^b(x) \} \quad (\text{U1}), (\text{U2}), (\text{F})
\end{aligned} \tag{6.67}$$

$$\begin{aligned}
[A, C] : [\rho^{d'}(y), \rho^e(x)] &= i f^{d'eh} \rho^h(y) \delta^{(2)}(y - x), \quad x \rightarrow y \\
\frac{1}{4} \{ [A, C], B \} &= -\frac{1}{4} i K_{\mathfrak{B}}^P(x', y, y) U^{c'd'}(y) U^{ab'}(x') U^{bc'}(z) f^{acd} U^{bc}(z) U^{de}(y) f^{d'eh} \{ \rho^h(y), \rho^{b'}(x') \} \\
&= -\frac{1}{4} i N_c K_{\mathfrak{B}}^P(x', y, y) U^{ab}(x') U^{ac}(y) \{ \rho^c(y), \rho^b(x') \} \quad (\text{U1}), (\text{U2}), (\text{F})
\end{aligned} \tag{6.68}$$

$$\begin{aligned}
[B, C] : [\rho^{b'}(x'), \rho^e(x)] &= i f^{b'eh} \rho^h(x') \delta^{(2)}(x' - x), \quad x \rightarrow x' \\
\frac{1}{4} \{ [B, C], A \} &= -\frac{1}{4} i K_{\mathfrak{B}}^P(x', x', y) U^{c'd'}(y) U^{ab'}(x') U^{bc'}(z) f^{acd} U^{bc}(z) U^{de}(x') f^{b'eh} \{ \rho^h(x'), \rho^{d'}(y) \} \\
&= \frac{1}{4} i N_c K_{\mathfrak{B}}^P(x', x', y) U^{ab}(y) U^{ac}(x') \{ \rho^c(x'), \rho^b(y) \} \quad (\text{U1}), (\text{U2}), (\text{F})
\end{aligned} \tag{6.69}$$

Sub-term 3: $- U^{ab'}(x') f^{cbd} U^{ac}(w) \rho^b(y) \rho^{b'}(x') \rho^d(x)$.

$$\begin{aligned}
[A, B] : [\rho^b(y), \rho^{b'}(x')] &= i f^{bb'h} \rho^h(y) \delta^{(2)}(y - x'), \quad x' \rightarrow y \\
\frac{1}{4} \{ [A, B], C \} &= -\frac{1}{4} i K_{\mathfrak{B}}^P(y, x, y) U^{ab'}(y) f^{cbd} U^{ac}(w) f^{bb'h} \{ \rho^h(y), \rho^d(x) \} \\
&= -\frac{1}{4} i K_{\mathfrak{B}}^P(y, x, y) f^{abc} f^{bde} U^{hd}(y) U^{ha}(w) \{ \rho^e(y), \rho^c(x) \}
\end{aligned} \tag{6.70}$$

$$\begin{aligned}
[A, C] : [\rho^b(y), \rho^d(x)] &= i f^{bdh} \rho^h(y) \delta^{(2)}(y - x), \quad x \rightarrow y \\
\frac{1}{4} \{ [A, C], B \} &= -\frac{1}{4} i K_{\mathfrak{B}}^P(x', y, y) U^{ab'}(x') f^{cbd} U^{ac}(w) f^{bdh} \{ \rho^h(y), \rho^{b'}(x') \} \\
&= -\frac{1}{4} i N_c K_{\mathfrak{B}}^P(x', y, y) U^{ab}(x') U^{ac}(w) \{ \rho^c(y), \rho^b(x') \} \quad (\text{F})
\end{aligned} \tag{6.71}$$

$$\begin{aligned}
[B, C] : [\rho^{b'}(x'), \rho^d(x)] &= i f^{b'dh} \rho^h(x') \delta^{(2)}(x' - x), \quad x \rightarrow x' \\
\frac{1}{4} \{ [B, C], A \} &= -\frac{1}{4} i K_{\mathfrak{B}}^P(x', x', y) U^{ab'}(x') f^{cbd} U^{ac}(w) f^{b'dh} \{ \rho^h(x'), \rho^b(y) \} \\
&= -\frac{1}{4} i K_{\mathfrak{B}}^P(x', x', y) f^{abc} f^{dce} U^{hd}(x') U^{ha}(w) \{ \rho^e(x'), \rho^b(y) \}
\end{aligned} \tag{6.72}$$

Sub-term 4: $+ U^{ab'}(x') f^{acd} U^{bc}(z) U^{de}(x) \rho^b(y) \rho^{b'}(x') \rho^e(x)$.

$$\begin{aligned}
[A, B] : [\rho^b(y), \rho^{b'}(x')] &= i f^{bb'h} \rho^h(y) \delta^{(2)}(y - x'), \quad x' \rightarrow y \\
\frac{1}{4} \{ [A, B], C \} &= \frac{1}{4} i K_{\mathfrak{B}}^P(y, x, y) U^{ab'}(y) f^{acd} U^{bc}(z) U^{de}(x) f^{bb'h} \{ \rho^h(y), \rho^e(x) \}
\end{aligned}$$

$$= \frac{1}{4} i K_{\mathfrak{B}}^P(y, x, y) f^{abc} f^{deh} U^{ae}(y) U^{db}(z) U^{cl}(x) \{\rho^h(y), \rho^l(x)\} \quad (6.73)$$

$$\begin{aligned} [A, C] : [\rho^b(y), \rho^e(x)] &= i f^{beh} \rho^h(y) \delta^{(2)}(y - x), \quad x \rightarrow y \\ \frac{1}{4} \{[A, C], B\} &= \frac{1}{4} i K_{\mathfrak{B}}^P(x', y, y) U^{ab'}(x') f^{acd} U^{bc}(z) U^{de}(y) f^{beh} \{\rho^h(y), \rho^{b'}(x')\} \\ &= \frac{1}{4} i K_{\mathfrak{B}}^P(x', y, y) f^{abc} f^{deh} U^{al}(x') U^{db}(z) U^{ce}(y) \{\rho^h(y), \rho^l(x')\} \end{aligned} \quad (6.74)$$

$$\begin{aligned} [B, C] : [\rho^{b'}(x'), \rho^e(x)] &= i f^{b'eh} \rho^h(x') \delta^{(2)}(x' - x), \quad x \rightarrow x' \\ \frac{1}{4} \{[B, C], A\} &= \frac{1}{4} i K_{\mathfrak{B}}^P(x', x', y) U^{ab'}(x') f^{acd} U^{bc}(z) U^{de}(x') f^{b'eh} \{\rho^h(x'), \rho^b(y)\} \\ &= -\frac{1}{4} i N_c K_{\mathfrak{B}}^P(x', x', y) U^{ab}(z) U^{bc}(x') \{\rho^c(x'), \rho^a(y)\} \quad (\text{U2}), \quad (\text{F}) \end{aligned} \quad (6.75)$$

Two- ρ terms of this source. Multiplying by the prefactor $-2i \mathcal{N}_P$ and collecting equal terms (dummy colour indices renamed),

$$\begin{aligned} P^{(2\rho)} \Big|_{\text{GIII.II.P}} &= \mathcal{N}_P \int e^{-ik \cdot (w' - w)} \left\{ \right. \\ &\quad + \frac{1}{2} K_{\mathfrak{B}}^P(y, x, y) f^{abc} f^{deh} U^{bd}(z) U^{ea}(w) U^{hl}(y) \{\rho^l(y), \rho^c(x)\} \\ &\quad + \frac{1}{2} K_{\mathfrak{B}}^P(x', y, y) f^{abc} f^{dce} U^{hd}(y) U^{lm}(x') U^{bh}(z) U^{la}(w) \{\rho^e(y), \rho^m(x')\} \\ &\quad + \frac{1}{2} K_{\mathfrak{B}}^P(x', x', y) f^{abc} f^{dce} U^{hl}(y) U^{md}(x') U^{bh}(z) U^{ma}(w) \{\rho^e(x'), \rho^l(y)\} \\ &\quad + \frac{1}{2} N_c K_{\mathfrak{B}}^P(y, x, y) U^{ab}(x) U^{ac}(y) \{\rho^c(y), \rho^b(x)\} \\ &\quad - \frac{1}{2} N_c K_{\mathfrak{B}}^P(x', y, y) U^{ab}(x') U^{ac}(y) \{\rho^c(y), \rho^b(x')\} \\ &\quad + \frac{1}{2} N_c K_{\mathfrak{B}}^P(x', x', y) U^{ab}(y) U^{ac}(x') \{\rho^c(x'), \rho^b(y)\} \\ &\quad - \frac{1}{2} K_{\mathfrak{B}}^P(y, x, y) f^{abc} f^{bde} U^{hd}(y) U^{ha}(w) \{\rho^e(y), \rho^c(x)\} \\ &\quad - \frac{1}{2} N_c K_{\mathfrak{B}}^P(x', y, y) U^{ab}(x') U^{ac}(w) \{\rho^c(y), \rho^b(x')\} \\ &\quad - \frac{1}{2} K_{\mathfrak{B}}^P(x', x', y) f^{abc} f^{dce} U^{hd}(x') U^{ha}(w) \{\rho^e(x'), \rho^b(y)\} \\ &\quad + \frac{1}{2} K_{\mathfrak{B}}^P(y, x, y) f^{abc} f^{deh} U^{ae}(y) U^{db}(z) U^{cl}(x) \{\rho^h(y), \rho^l(x)\} \\ &\quad + \frac{1}{2} K_{\mathfrak{B}}^P(x', y, y) f^{abc} f^{deh} U^{al}(x') U^{db}(z) U^{ce}(y) \{\rho^h(y), \rho^l(x')\} \\ &\quad \left. - \frac{1}{2} N_c K_{\mathfrak{B}}^P(x', x', y) U^{ab}(z) U^{bc}(x') \{\rho^c(x'), \rho^a(y)\} \right\} + (w' \leftrightarrow w) \end{aligned} \quad (6.76)$$

The kernel at the coincident points:

$$\begin{aligned} K_{\mathfrak{B}}^P(y, x, y) &= \left[\mathcal{K}(y - w') \cdot \mathcal{K}(z - w) \mathcal{K}(y - z) \cdot \mathcal{K}(x - z) - \frac{1}{2} \mathcal{K}(y - w') \cdot \mathcal{K}(x - w) \mathcal{K}(x - z) \cdot \mathcal{K}(y - z) \right] \\ K_{\mathfrak{B}}^P(x', y, y) &= \frac{1}{(y - z)^2} \left[\mathcal{K}(x' - w') \cdot \mathcal{K}(z - w) - \frac{1}{2} \mathcal{K}(x' - w') \cdot \mathcal{K}(y - w) \right] \\ K_{\mathfrak{B}}^P(x', x', y) &= \left[\mathcal{K}(x' - w') \cdot \mathcal{K}(z - w) \mathcal{K}(y - z) \cdot \mathcal{K}(x' - z) - \frac{1}{2} \mathcal{K}(x' - w') \cdot \mathcal{K}(x' - w) \mathcal{K}(x' - z) \cdot \mathcal{K}(y - z) \right] \end{aligned}$$

6.7 Group III, Row III, first part (region P)

The three- ρ term.

$$\begin{aligned} P \Big|_{\text{GIII.III.P}} &= -i \mathcal{N}_P \int e^{-ik \cdot (w' - w)} K_{\mathfrak{C}}^P(x', x, y) \left\{ \right. \\ &\quad + f^{c'd'a} U^{e'c'}(z) U^{d'b}(z) U^{ac}(w) \rho^{e'}(x') \{\rho^c(y), \rho^b(x)\} \\ &\quad - f^{c'd'a} U^{e'c'}(z) U^{d'b}(z) U^{bc}(z) U^{ad}(y) U^{ce}(x) \rho^{e'}(x') \{\rho^d(y), \rho^e(x)\} \\ &\quad - f^{c'd'a} U^{bd'}(z) U^{c'e'}(x') U^{ac}(w) \rho^{e'}(x') \{\rho^c(y), \rho^b(x)\} \\ &\quad \left. + f^{c'd'a} U^{bd'}(z) U^{c'e'}(x') U^{bc}(z) U^{ad}(y) U^{ce}(x) \rho^{e'}(x') \{\rho^d(y), \rho^e(x)\} \right\} + \text{c.c.} \end{aligned} \quad (6.77)$$

with $K_{\mathfrak{C}}^P(x', x, y) = \mathcal{K}(w' - z) \cdot \mathcal{K}(y - w) \mathcal{K}(x' - z) \cdot \mathcal{K}(x - z)$.

Reordering. In every sub-term (one product of the two brackets) the charges appear in the order shown, so we apply Eq. (2.4) with A, B, C the charges at x', y, x . The sub-terms are numbered 1–4 in the order above. For each sub-term we list the commutators, and each resulting two- ρ piece before and after the colour algebra.

Sub-term 1: $+ f^{c'd'a} U^{e'c'}(z) U^{d'b}(z) U^{ac}(w) \rho^{e'}(x') \{\rho^c(y), \rho^b(x)\}$.

$$[A, B] : [\rho^{e'}(x'), \rho^c(y)] = i f^{e'ch} \rho^h(x') \delta^{(2)}(x' - y), \quad y \rightarrow x'$$

$$\begin{aligned}\frac{1}{2}\{[A, B], C\} &= \frac{1}{2} i K_{\mathcal{C}}^P(x', x, x') f^{c'd'a} U^{e'c'}(z) U^{d'b}(z) U^{ac}(w) f^{e'ch} \{\rho^h(x'), \rho^b(x)\} \\ &= \frac{1}{2} i K_{\mathcal{C}}^P(x', x, x') f^{abc} f^{ade} U^{hb}(w) U^{eh}(z) \{\rho^c(x'), \rho^d(x)\} \quad (\text{U2}^*)\end{aligned}\quad (6.78)$$

$$\begin{aligned}[A, C] : [\rho^{e'}(x'), \rho^b(x)] &= i f^{e'bh} \rho^h(x') \delta^{(2)}(x' - x), \quad x \rightarrow x' \\ \frac{1}{2}\{[A, C], B\} &= \frac{1}{2} i K_{\mathcal{C}}^P(x', x', y) f^{c'd'a} U^{e'c'}(z) U^{d'b}(z) U^{ac}(w) f^{e'bh} \{\rho^h(x'), \rho^c(y)\} \\ &= \frac{1}{2} i N_c K_{\mathcal{C}}^P(x', x', y) U^{ab}(w) U^{ca}(z) \{\rho^c(x'), \rho^b(y)\} \quad (\text{U2}^*), \quad (\text{F})\end{aligned}\quad (6.79)$$

Sub-term 2: $- f^{c'd'a} U^{e'c'}(z) U^{d'b}(z) U^{bc}(z) U^{ad}(y) U^{ce}(x) \rho^{e'}(x') \{\rho^d(y), \rho^e(x)\}.$

$$\begin{aligned}[A, B] : [\rho^{e'}(x'), \rho^d(y)] &= i f^{e'dh} \rho^h(x') \delta^{(2)}(x' - y), \quad y \rightarrow x' \\ \frac{1}{2}\{[A, B], C\} &= -\frac{1}{2} i K_{\mathcal{C}}^P(x', x, x') f^{c'd'a} U^{e'c'}(z) U^{d'b}(z) U^{bc}(z) U^{ad}(x') U^{ce}(x) f^{e'dh} \{\rho^h(x'), \rho^e(x)\} \\ &= -\frac{1}{2} i K_{\mathcal{C}}^P(x', x, x') f^{abc} f^{deh} U^{da}(z) U^{ce}(x') U^{bl}(x) \{\rho^h(x'), \rho^l(x)\} \quad (\text{U1}^*)\end{aligned}\quad (6.80)$$

$$\begin{aligned}[A, C] : [\rho^{e'}(x'), \rho^e(x)] &= i f^{e'eh} \rho^h(x') \delta^{(2)}(x' - x), \quad x \rightarrow x' \\ \frac{1}{2}\{[A, C], B\} &= -\frac{1}{2} i K_{\mathcal{C}}^P(x', x', y) f^{c'd'a} U^{e'c'}(z) U^{d'b}(z) U^{bc}(z) U^{ad}(y) U^{ce}(x') f^{e'eh} \{\rho^h(x'), \rho^d(y)\} \\ &= -\frac{1}{2} i K_{\mathcal{C}}^P(x', x', y) f^{abc} f^{deh} U^{da}(z) U^{cl}(y) U^{be}(x') \{\rho^h(x'), \rho^l(y)\} \quad (\text{U1}^*)\end{aligned}\quad (6.81)$$

Sub-term 3: $- f^{c'd'a} U^{bd'}(z) U^{c'e'}(x') U^{ac}(w) \rho^{e'}(x') \{\rho^c(y), \rho^b(x)\}.$

$$\begin{aligned}[A, B] : [\rho^{e'}(x'), \rho^c(y)] &= i f^{e'ch} \rho^h(x') \delta^{(2)}(x' - y), \quad y \rightarrow x' \\ \frac{1}{2}\{[A, B], C\} &= -\frac{1}{2} i K_{\mathcal{C}}^P(x', x, x') f^{c'd'a} U^{bd'}(z) U^{c'e'}(x') U^{ac}(w) f^{e'ch} \{\rho^h(x'), \rho^b(x)\} \\ &= -\frac{1}{2} i K_{\mathcal{C}}^P(x', x, x') f^{abc} f^{deh} U^{lb}(z) U^{ad}(x') U^{ce}(w) \{\rho^h(x'), \rho^l(x)\}\end{aligned}\quad (6.82)$$

$$\begin{aligned}[A, C] : [\rho^{e'}(x'), \rho^b(x)] &= i f^{e'bh} \rho^h(x') \delta^{(2)}(x' - x), \quad x \rightarrow x' \\ \frac{1}{2}\{[A, C], B\} &= -\frac{1}{2} i K_{\mathcal{C}}^P(x', x', y) f^{c'd'a} U^{bd'}(z) U^{c'e'}(x') U^{ac}(w) f^{e'bh} \{\rho^h(x'), \rho^c(y)\} \\ &= -\frac{1}{2} i K_{\mathcal{C}}^P(x', x', y) f^{abc} f^{deh} U^{eb}(z) U^{ad}(x') U^{cl}(w) \{\rho^h(x'), \rho^l(y)\}\end{aligned}\quad (6.83)$$

Sub-term 4: $+ f^{c'd'a} U^{bd'}(z) U^{c'e'}(x') U^{bc}(z) U^{ad}(y) U^{ce}(x) \rho^{e'}(x') \{\rho^d(y), \rho^e(x)\}.$

$$\begin{aligned}[A, B] : [\rho^{e'}(x'), \rho^d(y)] &= i f^{e'dh} \rho^h(x') \delta^{(2)}(x' - y), \quad y \rightarrow x' \\ \frac{1}{2}\{[A, B], C\} &= \frac{1}{2} i K_{\mathcal{C}}^P(x', x, x') f^{c'd'a} U^{bd'}(z) U^{c'e'}(x') U^{bc}(z) U^{ad}(x') U^{ce}(x) f^{e'dh} \{\rho^h(x'), \rho^e(x)\} \\ &= -\frac{1}{2} i N_c K_{\mathcal{C}}^P(x', x, x') U^{ab}(x) U^{ac}(x') \{\rho^c(x'), \rho^b(x)\} \quad (\text{U1}), \quad (\text{U2}), \quad (\text{F})\end{aligned}\quad (6.84)$$

$$\begin{aligned}[A, C] : [\rho^{e'}(x'), \rho^e(x)] &= i f^{e'eh} \rho^h(x') \delta^{(2)}(x' - x), \quad x \rightarrow x' \\ \frac{1}{2}\{[A, C], B\} &= \frac{1}{2} i K_{\mathcal{C}}^P(x', x', y) f^{c'd'a} U^{bd'}(z) U^{c'e'}(x') U^{bc}(z) U^{ad}(y) U^{ce}(x') f^{e'eh} \{\rho^h(x'), \rho^d(y)\} \\ &= \frac{1}{2} i N_c K_{\mathcal{C}}^P(x', x', y) U^{ab}(y) U^{ac}(x') \{\rho^c(x'), \rho^b(y)\} \quad (\text{U1}), \quad (\text{U2}), \quad (\text{F})\end{aligned}\quad (6.85)$$

Two- ρ terms of this source. Multiplying by the prefactor $-i\mathcal{N}_P$ and collecting equal terms (dummy colour indices renamed),

$$\begin{aligned}P^{(2\rho)} \Big|_{\text{GIII.III.P}} &= \mathcal{N}_P \int e^{-ik \cdot (w' - w)} \left\{ \right. \\ &\quad + \frac{1}{2} K_{\mathcal{C}}^P(x', x, x') f^{abc} f^{ade} U^{hb}(w) U^{eh}(z) \{\rho^c(x'), \rho^d(x)\} \\ &\quad + \frac{1}{2} N_c K_{\mathcal{C}}^P(x', x', y) U^{ab}(w) U^{ca}(z) \{\rho^c(x'), \rho^b(y)\} \\ &\quad - \frac{1}{2} K_{\mathcal{C}}^P(x', x, x') f^{abc} f^{deh} U^{da}(z) U^{ce}(x') U^{bl}(x) \{\rho^h(x'), \rho^l(x)\} \\ &\quad - \frac{1}{2} K_{\mathcal{C}}^P(x', x', y) f^{abc} f^{deh} U^{da}(z) U^{cl}(y) U^{be}(x') \{\rho^h(x'), \rho^l(y)\} \\ &\quad - \frac{1}{2} K_{\mathcal{C}}^P(x', x, x') f^{abc} f^{deh} U^{lb}(z) U^{ad}(x') U^{ce}(w) \{\rho^h(x'), \rho^l(x)\} \\ &\quad - \frac{1}{2} K_{\mathcal{C}}^P(x', x', y) f^{abc} f^{deh} U^{eb}(z) U^{ad}(x') U^{cl}(w) \{\rho^h(x'), \rho^l(y)\} \\ &\quad - \frac{1}{2} N_c K_{\mathcal{C}}^P(x', x, x') U^{ab}(x) U^{ac}(x') \{\rho^c(x'), \rho^b(x)\} \\ &\quad \left. + \frac{1}{2} N_c K_{\mathcal{C}}^P(x', x', y) U^{ab}(y) U^{ac}(x') \{\rho^c(x'), \rho^b(y)\} \right\} + (w' \leftrightarrow w)\end{aligned}\quad (6.86)$$

The kernel at the coincident points:

$$\begin{aligned}K_{\mathcal{C}}^P(x', x, x') &= \mathcal{K}(w' - z) \cdot \mathcal{K}(x' - w) \mathcal{K}(x' - z) \cdot \mathcal{K}(x - z) \\ K_{\mathcal{C}}^P(x', x', y) &= \mathcal{K}(w' - z) \cdot \mathcal{K}(y - w) \frac{1}{(x' - z)^2}\end{aligned}$$

6.8 Group III, Row V (region P)

The three- ρ term.

$$\begin{aligned}
 P \Big|_{\text{GIII.V.P}} = & -2i \mathcal{N}_P \int e^{-ik \cdot (w' - w)} K_{\mathcal{E}}^P(x', x, y) \{ \\
 & + f^{c'ca} U^{e'e'}(z) U^{ce}(x) U^{ad}(y) \rho^{e'}(x') \rho^e(x) \rho^d(y) \\
 & - f^{c'ca} U^{e'e'}(z) U^{ad}(w) U^{ce}(x) \rho^{e'}(x') \rho^e(x) \rho^d(y) \\
 & - f^{c'ca} U^{c'e'}(x') U^{ce}(x) U^{ad}(y) \rho^{e'}(x') \rho^e(x) \rho^d(y) \\
 & + f^{c'ca} U^{c'e'}(x') U^{ad}(w) U^{ce}(x) \rho^{e'}(x') \rho^e(x) \rho^d(y) \} + \text{c.c.}
 \end{aligned} \tag{6.87}$$

with $K_{\mathcal{E}}^P(x', x, y) = \mathcal{K}(w' - z) \cdot \mathcal{K}(y - w) \mathcal{K}(x' - z) \cdot \mathcal{K}(x - z)$.

Reordering. In every sub-term (one product of the two brackets) the charges appear in the order shown, so we apply Eq. (2.3) with A, B, C the charges at x', x, y . The sub-terms are numbered 1–4 in the order above. For each sub-term we list the commutators, and each resulting two- ρ piece before and after the colour algebra.

Sub-term 1: $+ f^{c'ca} U^{e'e'}(z) U^{ce}(x) U^{ad}(y) \rho^{e'}(x') \rho^e(x) \rho^d(y)$.

$$\begin{aligned}
 [A, B] : [\rho^{e'}(x'), \rho^e(x)] &= i f^{e'eh} \rho^h(x') \delta^{(2)}(x' - x), \quad x \rightarrow x' \\
 \frac{1}{4} \{[A, B], C\} &= \frac{1}{4} i K_{\mathcal{E}}^P(x', x', y) f^{c'ca} U^{e'e'}(z) U^{ce}(x') U^{ad}(y) f^{e'eh} \{\rho^h(x'), \rho^d(y)\} \\
 &= \frac{1}{4} i K_{\mathcal{E}}^P(x', x', y) f^{abc} f^{deh} U^{da}(z) U^{be}(x') U^{cl}(y) \{\rho^h(x'), \rho^l(y)\}
 \end{aligned} \tag{6.88}$$

$$\begin{aligned}
 [A, C] : [\rho^{e'}(x'), \rho^d(y)] &= i f^{e'dh} \rho^h(x') \delta^{(2)}(x' - y), \quad y \rightarrow x' \\
 \frac{1}{4} \{[A, C], B\} &= \frac{1}{4} i K_{\mathcal{E}}^P(x', x, x') f^{c'ca} U^{e'e'}(z) U^{ce}(x) U^{ad}(x') f^{e'dh} \{\rho^h(x'), \rho^e(x)\} \\
 &= \frac{1}{4} i K_{\mathcal{E}}^P(x', x, x') f^{abc} f^{deh} U^{da}(z) U^{bl}(x) U^{ce}(x') \{\rho^h(x'), \rho^l(x)\}
 \end{aligned} \tag{6.89}$$

$$\begin{aligned}
 [B, C] : [\rho^e(x), \rho^d(y)] &= i f^{edh} \rho^h(x) \delta^{(2)}(x - y), \quad y \rightarrow x \\
 \frac{1}{4} \{[B, C], A\} &= \frac{1}{4} i K_{\mathcal{E}}^P(x', x, x) f^{c'ca} U^{e'e'}(z) U^{ce}(x) U^{ad}(x) f^{edh} \{\rho^h(x), \rho^{e'}(x')\} \\
 &= \frac{1}{4} i N_c K_{\mathcal{E}}^P(x', x, x) U^{ab}(z) U^{bc}(x) \{\rho^c(x), \rho^a(x')\} \quad (\text{U2}), \quad (\text{F})
 \end{aligned} \tag{6.90}$$

Sub-term 2: $- f^{c'ca} U^{e'e'}(z) U^{ad}(w) U^{ce}(x) \rho^{e'}(x') \rho^e(x) \rho^d(y)$.

$$\begin{aligned}
 [A, B] : [\rho^{e'}(x'), \rho^e(x)] &= i f^{e'eh} \rho^h(x') \delta^{(2)}(x' - x), \quad x \rightarrow x' \\
 \frac{1}{4} \{[A, B], C\} &= -\frac{1}{4} i K_{\mathcal{E}}^P(x', x', y) f^{c'ca} U^{e'e'}(z) U^{ad}(w) U^{ce}(x') f^{e'eh} \{\rho^h(x'), \rho^d(y)\} \\
 &= -\frac{1}{4} i K_{\mathcal{E}}^P(x', x', y) f^{abc} f^{deh} U^{da}(z) U^{cl}(w) U^{be}(x') \{\rho^h(x'), \rho^l(y)\}
 \end{aligned} \tag{6.91}$$

$$\begin{aligned}
 [A, C] : [\rho^{e'}(x'), \rho^d(y)] &= i f^{e'dh} \rho^h(x') \delta^{(2)}(x' - y), \quad y \rightarrow x' \\
 \frac{1}{4} \{[A, C], B\} &= -\frac{1}{4} i K_{\mathcal{E}}^P(x', x, x') f^{c'ca} U^{e'e'}(z) U^{ad}(w) U^{ce}(x) f^{e'dh} \{\rho^h(x'), \rho^e(x)\} \\
 &= -\frac{1}{4} i K_{\mathcal{E}}^P(x', x, x') f^{abc} f^{deh} U^{da}(z) U^{ce}(w) U^{bl}(x) \{\rho^h(x'), \rho^l(x)\}
 \end{aligned} \tag{6.92}$$

$$\begin{aligned}
 [B, C] : [\rho^e(x), \rho^d(y)] &= i f^{edh} \rho^h(x) \delta^{(2)}(x - y), \quad y \rightarrow x \\
 \frac{1}{4} \{[B, C], A\} &= -\frac{1}{4} i K_{\mathcal{E}}^P(x', x, x) f^{c'ca} U^{e'e'}(z) U^{ad}(w) U^{ce}(x) f^{edh} \{\rho^h(x), \rho^{e'}(x')\} \\
 &= -\frac{1}{4} i K_{\mathcal{E}}^P(x', x, x) f^{abc} f^{deh} U^{la}(z) U^{ce}(w) U^{bd}(x) \{\rho^h(x), \rho^l(x')\}
 \end{aligned} \tag{6.93}$$

Sub-term 3: $- f^{c'ca} U^{c'e'}(x') U^{ce}(x) U^{ad}(y) \rho^{e'}(x') \rho^e(x) \rho^d(y)$.

$$\begin{aligned}
 [A, B] : [\rho^{e'}(x'), \rho^e(x)] &= i f^{e'eh} \rho^h(x') \delta^{(2)}(x' - x), \quad x \rightarrow x' \\
 \frac{1}{4} \{[A, B], C\} &= -\frac{1}{4} i K_{\mathcal{E}}^P(x', x', y) f^{c'ca} U^{c'e'}(x') U^{ce}(x') U^{ad}(y) f^{e'eh} \{\rho^h(x'), \rho^d(y)\} \\
 &= -\frac{1}{4} i N_c K_{\mathcal{E}}^P(x', x', y) U^{ab}(y) U^{ac}(x') \{\rho^c(x'), \rho^b(y)\} \quad (\text{U2}), \quad (\text{F})
 \end{aligned} \tag{6.94}$$

$$\begin{aligned}
 [A, C] : [\rho^{e'}(x'), \rho^d(y)] &= i f^{e'dh} \rho^h(x') \delta^{(2)}(x' - y), \quad y \rightarrow x' \\
 \frac{1}{4} \{[A, C], B\} &= -\frac{1}{4} i K_{\mathcal{E}}^P(x', x, x') f^{c'ca} U^{c'e'}(x') U^{ce}(x) U^{ad}(x') f^{e'dh} \{\rho^h(x'), \rho^e(x)\} \\
 &= \frac{1}{4} i N_c K_{\mathcal{E}}^P(x', x, x') U^{ab}(x) U^{ac}(x') \{\rho^c(x'), \rho^b(x)\} \quad (\text{U2}), \quad (\text{F})
 \end{aligned} \tag{6.95}$$

$$\begin{aligned}
 [B, C] : [\rho^e(x), \rho^d(y)] &= i f^{edh} \rho^h(x) \delta^{(2)}(x - y), \quad y \rightarrow x \\
 \frac{1}{4} \{[B, C], A\} &= -\frac{1}{4} i K_{\mathcal{E}}^P(x', x, x) f^{c'ca} U^{c'e'}(x') U^{ce}(x) U^{ad}(x) f^{edh} \{\rho^h(x), \rho^{e'}(x')\} \\
 &= -\frac{1}{4} i N_c K_{\mathcal{E}}^P(x', x, x) U^{ab}(x') U^{ac}(x) \{\rho^c(x), \rho^b(x')\} \quad (\text{U2}), \quad (\text{F})
 \end{aligned} \tag{6.96}$$

Sub-term 4: $+ f^{c'ca} U^{c'e'}(x') U^{ad}(w) U^{ce}(x) \rho^{e'}(x') \rho^e(x) \rho^d(y).$

$$\begin{aligned} [A, B] : [\rho^{e'}(x'), \rho^e(x)] &= i f^{e'eh} \rho^h(x') \delta^{(2)}(x' - x), \quad x \rightarrow x' \\ \frac{1}{4} \{[A, B], C\} &= \frac{1}{4} i K_{\mathfrak{C}}^P(x', x', y) f^{c'ca} U^{c'e'}(x') U^{ad}(w) U^{ce}(x') f^{e'eh} \{\rho^h(x'), \rho^d(y)\} \\ &= \frac{1}{4} i N_c K_{\mathfrak{C}}^P(x', x', y) U^{ab}(w) U^{ac}(x') \{\rho^c(x'), \rho^b(y)\} \quad (\text{U2}), \quad (\text{F}) \end{aligned} \quad (6.97)$$

$$\begin{aligned} [A, C] : [\rho^{e'}(x'), \rho^d(y)] &= i f^{e'dh} \rho^h(x') \delta^{(2)}(x' - y), \quad y \rightarrow x' \\ \frac{1}{4} \{[A, C], B\} &= \frac{1}{4} i K_{\mathfrak{C}}^P(x', x, x') f^{c'ca} U^{c'e'}(x') U^{ad}(w) U^{ce}(x) f^{e'dh} \{\rho^h(x'), \rho^e(x)\} \\ &= \frac{1}{4} i K_{\mathfrak{C}}^P(x', x, x') f^{abc} f^{deh} U^{ad}(x') U^{ce}(w) U^{bl}(x) \{\rho^h(x'), \rho^l(x)\} \end{aligned} \quad (6.98)$$

$$\begin{aligned} [B, C] : [\rho^e(x), \rho^d(y)] &= i f^{edh} \rho^h(x) \delta^{(2)}(x - y), \quad y \rightarrow x \\ \frac{1}{4} \{[B, C], A\} &= \frac{1}{4} i K_{\mathfrak{C}}^P(x', x, x) f^{c'ca} U^{c'e'}(x') U^{ad}(w) U^{ce}(x) f^{edh} \{\rho^h(x), \rho^{e'}(x')\} \\ &= \frac{1}{4} i K_{\mathfrak{C}}^P(x', x, x) f^{abc} f^{deh} U^{al}(x') U^{ce}(w) U^{bd}(x) \{\rho^h(x), \rho^l(x')\} \end{aligned} \quad (6.99)$$

Two- ρ terms of this source. Multiplying by the prefactor $-2i\mathcal{N}_P$ and collecting equal terms (dummy colour indices renamed),

$$\begin{aligned} P^{(2\rho)} \Big|_{\text{GIII.V.P}} &= \mathcal{N}_P \int e^{-ik \cdot (w' - w)} \left\{ \right. \\ &+ \frac{1}{2} K_{\mathfrak{C}}^P(x', x', y) f^{abc} f^{deh} U^{da}(z) U^{be}(x') U^{cl}(y) \{\rho^h(x'), \rho^l(y)\} \\ &+ \frac{1}{2} K_{\mathfrak{C}}^P(x', x, x') f^{abc} f^{deh} U^{da}(z) U^{bl}(x) U^{ce}(x') \{\rho^h(x'), \rho^l(x)\} \\ &+ \frac{1}{2} N_c K_{\mathfrak{C}}^P(x', x, x) U^{ab}(z) U^{bc}(x) \{\rho^c(x), \rho^a(x')\} \\ &- \frac{1}{2} K_{\mathfrak{C}}^P(x', x', y) f^{abc} f^{deh} U^{da}(z) U^{cl}(w) U^{be}(x') \{\rho^h(x'), \rho^l(y)\} \\ &- \frac{1}{2} K_{\mathfrak{C}}^P(x', x, x') f^{abc} f^{deh} U^{da}(z) U^{ce}(w) U^{bl}(x) \{\rho^h(x'), \rho^l(x)\} \\ &- \frac{1}{2} K_{\mathfrak{C}}^P(x', x, x) f^{abc} f^{deh} U^{la}(z) U^{ce}(w) U^{bd}(x) \{\rho^h(x), \rho^l(x')\} \\ &- \frac{1}{2} N_c K_{\mathfrak{C}}^P(x', x', y) U^{ab}(y) U^{ac}(x') \{\rho^c(x'), \rho^b(y)\} \\ &+ \frac{1}{2} N_c K_{\mathfrak{C}}^P(x', x, x') U^{ab}(x) U^{ac}(x') \{\rho^c(x'), \rho^b(x)\} \\ &- \frac{1}{2} N_c K_{\mathfrak{C}}^P(x', x, x) U^{ab}(x') U^{ac}(x) \{\rho^c(x), \rho^b(x')\} \\ &+ \frac{1}{2} N_c K_{\mathfrak{C}}^P(x', x', y) U^{ab}(w) U^{ac}(x') \{\rho^c(x'), \rho^b(y)\} \\ &+ \frac{1}{2} K_{\mathfrak{C}}^P(x', x, x') f^{abc} f^{deh} U^{ad}(x') U^{ce}(w) U^{bl}(x) \{\rho^h(x'), \rho^l(x)\} \\ &+ \frac{1}{2} K_{\mathfrak{C}}^P(x', x, x) f^{abc} f^{deh} U^{al}(x') U^{ce}(w) U^{bd}(x) \{\rho^h(x), \rho^l(x')\} \left. \right\} + (w' \leftrightarrow w) \end{aligned} \quad (6.100)$$

The kernel at the coincident points:

$$\begin{aligned} K_{\mathfrak{C}}^P(x', x', y) &= \mathcal{K}(w' - z) \cdot \mathcal{K}(y - w) \frac{1}{(x' - z)^2} \\ K_{\mathfrak{C}}^P(x', x, x') &= \mathcal{K}(w' - z) \cdot \mathcal{K}(x' - w) \mathcal{K}(x' - z) \cdot \mathcal{K}(x - z) \\ K_{\mathfrak{C}}^P(x', x, x) &= \mathcal{K}(w' - z) \cdot \mathcal{K}(x - w) \mathcal{K}(x' - z) \cdot \mathcal{K}(x - z) \end{aligned}$$

6.9 Group III, Row VI (region P)

The three- ρ term.

$$\begin{aligned} P \Big|_{\text{GIII.VI.P}} &= -2i\mathcal{N}_P \int e^{-ik \cdot (w' - w)} K_{\mathfrak{C}}^P(x', x, y) \left\{ \right. \\ &+ f^{c'd'a} U^{e'c'}(z) U^{d'b}(z) U^{bc}(z) U^{ad}(y) U^{ce}(x) \rho^{e'}(x') \rho^d(y) \rho^e(x) \\ &- f^{c'd'a} U^{e'c'}(z) U^{d'b}(z) U^{ad}(y) \rho^{e'}(x') \rho^d(y) \rho^b(x) \\ &- f^{c'd'a} U^{bd'}(z) U^{c'e'}(x') U^{bc}(z) U^{ad}(y) U^{ce}(x) \rho^{e'}(x') \rho^d(y) \rho^e(x) \\ &+ f^{c'd'a} U^{bd'}(z) U^{c'e'}(x') U^{ad}(y) \rho^{e'}(x') \rho^d(y) \rho^b(x) \left. \right\} + \text{c.c.} \end{aligned} \quad (6.101)$$

with $K_{\mathfrak{C}}^P(x', x, y) = \mathcal{K}(w' - z) \cdot \mathcal{K}(y - w) \mathcal{K}(x' - z) \cdot \mathcal{K}(x - z).$

Reordering. In every sub-term (one product of the two brackets) the charges appear in the order shown, so we apply Eq. (2.3) with A, B, C the charges at x', y, x . The sub-terms are numbered 1–4 in the order above. For each sub-term we list the commutators, and each resulting two- ρ piece before and after the colour algebra.

Sub-term 1: $+ f^{c'd'a} U^{e'e'}(z) U^{d'b}(z) U^{bc}(z) U^{ad}(y) U^{ce}(x) \rho^d(x') \rho^e(y) \rho^e(x).$

$$\begin{aligned} [A, B] : [\rho^{e'}(x'), \rho^d(y)] &= i f^{e'dh} \rho^h(x') \delta^{(2)}(x' - y), \quad y \rightarrow x' \\ \frac{1}{4} \{[A, B], C\} &= \frac{1}{4} i K_{\mathcal{E}}^P(x', x, x') f^{c'd'a} U^{e'e'}(z) U^{d'b}(z) U^{bc}(z) U^{ad}(x') U^{ce}(x) f^{e'dh} \{\rho^h(x'), \rho^e(x)\} \\ &= \frac{1}{4} i K_{\mathcal{E}}^P(x', x, x') f^{abc} f^{deh} U^{da}(z) U^{ce}(x') U^{bl}(x) \{\rho^h(x'), \rho^l(x)\} \quad (\text{U1}^*) \end{aligned} \quad (6.102)$$

$$\begin{aligned} [A, C] : [\rho^{e'}(x'), \rho^e(x)] &= i f^{e'eh} \rho^h(x') \delta^{(2)}(x' - x), \quad x \rightarrow x' \\ \frac{1}{4} \{[A, C], B\} &= \frac{1}{4} i K_{\mathcal{E}}^P(x', x', y) f^{c'd'a} U^{e'e'}(z) U^{d'b}(z) U^{bc}(z) U^{ad}(y) U^{ce}(x') f^{e'eh} \{\rho^h(x'), \rho^d(y)\} \\ &= \frac{1}{4} i K_{\mathcal{E}}^P(x', x', y) f^{abc} f^{deh} U^{da}(z) U^{cl}(y) U^{be}(x') \{\rho^h(x'), \rho^l(y)\} \quad (\text{U1}^*) \end{aligned} \quad (6.103)$$

$$\begin{aligned} [B, C] : [\rho^d(y), \rho^e(x)] &= i f^{deh} \rho^h(y) \delta^{(2)}(y - x), \quad x \rightarrow y \\ \frac{1}{4} \{[B, C], A\} &= \frac{1}{4} i K_{\mathcal{E}}^P(x', y, y) f^{c'd'a} U^{e'e'}(z) U^{d'b}(z) U^{bc}(z) U^{ad}(y) U^{ce}(y) f^{deh} \{\rho^h(y), \rho^{e'}(x')\} \\ &= -\frac{1}{4} i N_c K_{\mathcal{E}}^P(x', y, y) U^{ab}(z) U^{bc}(y) \{\rho^c(y), \rho^a(x')\} \quad (\text{U1}^*), (\text{U2}), (\text{F}) \end{aligned} \quad (6.104)$$

Sub-term 2: $- f^{c'd'a} U^{e'e'}(z) U^{d'b}(z) U^{ad}(y) \rho^{e'}(x') \rho^d(y) \rho^b(x).$

$$\begin{aligned} [A, B] : [\rho^{e'}(x'), \rho^d(y)] &= i f^{e'dh} \rho^h(x') \delta^{(2)}(x' - y), \quad y \rightarrow x' \\ \frac{1}{4} \{[A, B], C\} &= -\frac{1}{4} i K_{\mathcal{E}}^P(x', x, x') f^{c'd'a} U^{e'e'}(z) U^{d'b}(z) U^{ad}(x') f^{e'dh} \{\rho^h(x'), \rho^b(x)\} \\ &= -\frac{1}{4} i K_{\mathcal{E}}^P(x', x, x') f^{abc} f^{ade} U^{hb}(x') U^{eh}(z) \{\rho^c(x'), \rho^d(x)\} \quad (\text{U2}^*) \end{aligned} \quad (6.105)$$

$$\begin{aligned} [A, C] : [\rho^{e'}(x'), \rho^b(x)] &= i f^{e'bh} \rho^h(x') \delta^{(2)}(x' - x), \quad x \rightarrow x' \\ \frac{1}{4} \{[A, C], B\} &= -\frac{1}{4} i K_{\mathcal{E}}^P(x', x', y) f^{c'd'a} U^{e'e'}(z) U^{d'b}(z) U^{ad}(y) f^{e'bh} \{\rho^h(x'), \rho^d(y)\} \\ &= -\frac{1}{4} i N_c K_{\mathcal{E}}^P(x', x', y) U^{ab}(y) U^{ca}(z) \{\rho^c(x'), \rho^b(y)\} \quad (\text{U2}^*), (\text{F}) \end{aligned} \quad (6.106)$$

$$\begin{aligned} [B, C] : [\rho^d(y), \rho^b(x)] &= i f^{dbh} \rho^h(y) \delta^{(2)}(y - x), \quad x \rightarrow y \\ \frac{1}{4} \{[B, C], A\} &= -\frac{1}{4} i K_{\mathcal{E}}^P(x', y, y) f^{c'd'a} U^{e'e'}(z) U^{d'b}(z) U^{ad}(y) f^{dbh} \{\rho^h(y), \rho^{e'}(x')\} \\ &= -\frac{1}{4} i K_{\mathcal{E}}^P(x', y, y) f^{abc} f^{dbe} U^{ha}(y) U^{eh}(z) \{\rho^c(y), \rho^d(x')\} \quad (\text{U2}^*) \end{aligned} \quad (6.107)$$

Sub-term 3: $- f^{c'd'a} U^{bd'}(z) U^{c'e'}(x') U^{bc}(z) U^{ad}(y) U^{ce}(x) \rho^{e'}(x') \rho^d(y) \rho^e(x).$

$$\begin{aligned} [A, B] : [\rho^{e'}(x'), \rho^d(y)] &= i f^{e'dh} \rho^h(x') \delta^{(2)}(x' - y), \quad y \rightarrow x' \\ \frac{1}{4} \{[A, B], C\} &= -\frac{1}{4} i K_{\mathcal{E}}^P(x', x, x') f^{c'd'a} U^{bd'}(z) U^{c'e'}(x') U^{bc}(z) U^{ad}(x') U^{ce}(x) f^{e'dh} \{\rho^h(x'), \rho^e(x)\} \\ &= \frac{1}{4} i N_c K_{\mathcal{E}}^P(x', x, x') U^{ab}(x) U^{ac}(x') \{\rho^c(x'), \rho^b(x)\} \quad (\text{U1}), (\text{U2}), (\text{F}) \end{aligned} \quad (6.108)$$

$$\begin{aligned} [A, C] : [\rho^{e'}(x'), \rho^e(x)] &= i f^{e'eh} \rho^h(x') \delta^{(2)}(x' - x), \quad x \rightarrow x' \\ \frac{1}{4} \{[A, C], B\} &= -\frac{1}{4} i K_{\mathcal{E}}^P(x', x', y) f^{c'd'a} U^{bd'}(z) U^{c'e'}(x') U^{bc}(z) U^{ad}(y) U^{ce}(x') f^{e'eh} \{\rho^h(x'), \rho^d(y)\} \\ &= -\frac{1}{4} i N_c K_{\mathcal{E}}^P(x', x', y) U^{ab}(y) U^{ac}(x') \{\rho^c(x'), \rho^b(y)\} \quad (\text{U1}), (\text{U2}), (\text{F}) \end{aligned} \quad (6.109)$$

$$\begin{aligned} [B, C] : [\rho^d(y), \rho^e(x)] &= i f^{deh} \rho^h(y) \delta^{(2)}(y - x), \quad x \rightarrow y \\ \frac{1}{4} \{[B, C], A\} &= -\frac{1}{4} i K_{\mathcal{E}}^P(x', y, y) f^{c'd'a} U^{bd'}(z) U^{c'e'}(x') U^{bc}(z) U^{ad}(y) U^{ce}(y) f^{deh} \{\rho^h(y), \rho^{e'}(x')\} \\ &= \frac{1}{4} i N_c K_{\mathcal{E}}^P(x', y, y) U^{ab}(x') U^{ac}(y) \{\rho^c(y), \rho^b(x')\} \quad (\text{U1}), (\text{U2}), (\text{F}) \end{aligned} \quad (6.110)$$

Sub-term 4: $+ f^{c'd'a} U^{bd'}(z) U^{c'e'}(x') U^{ad}(y) \rho^{e'}(x') \rho^d(y) \rho^b(x).$

$$\begin{aligned} [A, B] : [\rho^{e'}(x'), \rho^d(y)] &= i f^{e'dh} \rho^h(x') \delta^{(2)}(x' - y), \quad y \rightarrow x' \\ \frac{1}{4} \{[A, B], C\} &= \frac{1}{4} i K_{\mathcal{E}}^P(x', x, x') f^{c'd'a} U^{bd'}(z) U^{c'e'}(x') U^{ad}(x') f^{e'dh} \{\rho^h(x'), \rho^b(x)\} \\ &= -\frac{1}{4} i N_c K_{\mathcal{E}}^P(x', x, x') U^{ab}(z) U^{bc}(x') \{\rho^c(x'), \rho^a(x)\} \quad (\text{U2}), (\text{F}) \end{aligned} \quad (6.111)$$

$$\begin{aligned} [A, C] : [\rho^{e'}(x'), \rho^b(x)] &= i f^{e'bh} \rho^h(x') \delta^{(2)}(x' - x), \quad x \rightarrow x' \\ \frac{1}{4} \{[A, C], B\} &= \frac{1}{4} i K_{\mathcal{E}}^P(x', x', y) f^{c'd'a} U^{bd'}(z) U^{c'e'}(x') U^{ad}(y) f^{e'bh} \{\rho^h(x'), \rho^d(y)\} \\ &= \frac{1}{4} i K_{\mathcal{E}}^P(x', x', y) f^{abc} f^{deh} U^{eb}(z) U^{ad}(x') U^{cl}(y) \{\rho^h(x'), \rho^l(y)\} \end{aligned} \quad (6.112)$$

$$\begin{aligned} [B, C] : [\rho^d(y), \rho^b(x)] &= i f^{dbh} \rho^h(y) \delta^{(2)}(y - x), \quad x \rightarrow y \\ \frac{1}{4} \{[B, C], A\} &= \frac{1}{4} i K_{\mathcal{E}}^P(x', y, y) f^{c'd'a} U^{bd'}(z) U^{c'e'}(x') U^{ad}(y) f^{dbh} \{\rho^h(y), \rho^{e'}(x')\} \\ &= \frac{1}{4} i K_{\mathcal{E}}^P(x', y, y) f^{abc} f^{deh} U^{eb}(z) U^{al}(x') U^{cd}(y) \{\rho^h(y), \rho^l(x')\} \end{aligned} \quad (6.113)$$

Two- ρ terms of this source. Multiplying by the prefactor $-2i\mathcal{N}_P$ and collecting equal terms (dummy colour indices renamed),

$$\begin{aligned}
P^{(2\rho)} \Big|_{\text{III.VI.P}} &= \mathcal{N}_P \int e^{-ik \cdot (w' - w)} \left\{ \right. \\
&+ \frac{1}{2} K_{\mathcal{E}}^P(x', x, x') f^{abc} f^{deh} U^{da}(z) U^{ce}(x') U^{bl}(x) \{\rho^h(x'), \rho^l(x)\} \\
&+ \frac{1}{2} K_{\mathcal{E}}^P(x', x', y) f^{abc} f^{deh} U^{da}(z) U^{cl}(y) U^{be}(x') \{\rho^h(x'), \rho^l(y)\} \\
&- \frac{1}{2} N_c K_{\mathcal{E}}^P(x', y, y) U^{ab}(z) U^{bc}(y) \{\rho^c(y), \rho^a(x')\} \\
&- \frac{1}{2} K_{\mathcal{E}}^P(x', x, x') f^{abc} f^{ade} U^{hb}(x') U^{eh}(z) \{\rho^c(x'), \rho^d(x)\} \\
&- \frac{1}{2} N_c K_{\mathcal{E}}^P(x', x', y) U^{ab}(y) U^{ca}(z) \{\rho^c(x'), \rho^b(y)\} \\
&- \frac{1}{2} K_{\mathcal{E}}^P(x', y, y) f^{abc} f^{dbe} U^{ha}(y) U^{eh}(z) \{\rho^c(y), \rho^d(x')\} \\
&+ \frac{1}{2} N_c K_{\mathcal{E}}^P(x', x, x') U^{ab}(x) U^{ac}(x') \{\rho^c(x'), \rho^b(x)\} \\
&- \frac{1}{2} N_c K_{\mathcal{E}}^P(x', x', y) U^{ab}(y) U^{ac}(x') \{\rho^c(x'), \rho^b(y)\} \\
&+ \frac{1}{2} N_c K_{\mathcal{E}}^P(x', y, y) U^{ab}(x') U^{ac}(y) \{\rho^c(y), \rho^b(x')\} \\
&- \frac{1}{2} N_c K_{\mathcal{E}}^P(x', x, x') U^{ab}(z) U^{bc}(x') \{\rho^c(x'), \rho^a(x)\} \\
&+ \frac{1}{2} K_{\mathcal{E}}^P(x', x', y) f^{abc} f^{deh} U^{eb}(z) U^{ad}(x') U^{cl}(y) \{\rho^h(x'), \rho^l(y)\} \\
&+ \frac{1}{2} K_{\mathcal{E}}^P(x', y, y) f^{abc} f^{deh} U^{eb}(z) U^{al}(x') U^{cd}(y) \{\rho^h(y), \rho^l(x')\} \left. \right\} + (w' \leftrightarrow w) \quad (6.114)
\end{aligned}$$

The kernel at the coincident points:

$$\begin{aligned}
K_{\mathcal{E}}^P(x', x, x') &= \mathcal{K}(w' - z) \cdot \mathcal{K}(x' - w) \mathcal{K}(x' - z) \cdot \mathcal{K}(x - z) \\
K_{\mathcal{E}}^P(x', x', y) &= \mathcal{K}(w' - z) \cdot \mathcal{K}(y - w) \frac{1}{(x' - z)^2} \\
K_{\mathcal{E}}^P(x', y, y) &= \mathcal{K}(w' - z) \cdot \mathcal{K}(y - w) \mathcal{K}(x' - z) \cdot \mathcal{K}(y - z)
\end{aligned}$$

7 The symmetrized three- ρ terms cancel

All symmetrized three- ρ terms proportional to $\log(\vee/k^+)$, after the fixes, sorted by kernel structure and colour network. Every entry is renamed so that the charge tied to w is x , the one tied to w' is x' , and the third is y . If several namings are equivalent, the entry is averaged over them. The symmetrized product $S(\rho\rho\rho)$ is classical, so the order of the charges no longer matters. Contributions are labelled by row: subscript P for the $\log \vee$ part, subscript Λ for minus the $\log(\vee/\Lambda)$ part. Coefficients are in units of $\mathcal{N}_P$ and include the c.c. where the row has one. In every table the sum of each row is zero.

Structure 1: $\mathcal{K}(x - w) \cdot \mathcal{K}(x' - w') K(y, w, z)$

colour structure	contributions (row: coefficient)	sum
$f^{abc} U^{de}(x') U^{da}(w) \rho^e(x') \rho^b(x) \rho^c(y)$	I.II $_{\Lambda}$: $2i$; III.II $_{\Lambda}$: $-2i$	0 ✓
$f^{abc} U^{de}(w') U^{da}(w) \rho^e(x') \rho^b(x) \rho^c(y)$	I.II $_{\Lambda}$: $-2i$; II.I $_{\Lambda}$: $2i$	0 ✓
$f^{abc} U^{ad}(x') U^{be}(x) U^{ch}(y) \rho^d(x') \rho^e(x) \rho^h(y)$	I.II $_{\Lambda}$: $-2i$; I.III $_{\Lambda}$: $-2i$; I.IV $_{\Lambda}$: $2i$; I.V $_{\Lambda}$: $2i$; II.I $_{\Lambda}$: $2i$; III.I $_{\Lambda}$: $-2i$; III.II $_{\Lambda}$: $-2i$; III.III $_{\Lambda}$: $-2i$; III.V $_{\Lambda}$: $2i$; III.VI $_{\Lambda}$: $2i$	0 ✓
$f^{abc} U^{ad}(w') U^{be}(x) U^{ch}(y) \rho^d(x') \rho^e(x) \rho^h(y)$	I.II $_{\Lambda}$: $2i$; I.III $_{\Lambda}$: $2i$; I.IV $_{\Lambda}$: $-2i$; I.V $_{\Lambda}$: $-2i$; III.I $_{\Lambda}$: $2i$; III.V $_{\Lambda}$: $-2i$	0 ✓
$f^{abc} U^{ad}(x') U^{ce}(y) U^{bh}(w) \rho^d(x') \rho^e(y) \rho^h(x)$	I.III $_{\Lambda}$: $2i$; III.III $_{\Lambda}$: $2i$; III.V $_{\Lambda}$: $-2i$; III.VI $_{\Lambda}$: $-2i$	0 ✓
$f^{abc} U^{ad}(w') U^{ce}(y) U^{bh}(w) \rho^d(x') \rho^e(y) \rho^h(x)$	I.III $_{\Lambda}$: $-2i$; III.V $_{\Lambda}$: $2i$	0 ✓
$f^{abc} U^{ad}(x') U^{be}(z) U^{ch}(w) \rho^d(x') \rho^h(x) \rho^e(y)$	I.V $_{\Lambda}$: $2i$; III.VI $_{\Lambda}$: $-2i$	0 ✓
$f^{abc} U^{ad}(w') U^{be}(z) U^{ch}(w) \rho^d(x') \rho^h(x) \rho^e(y)$	I.V $_{\Lambda}$: $-2i$; III.III $_{\Lambda}$: $2i$	0 ✓
$f^{abc} U^{ad}(w') U^{eb}(z) U^{ch}(x) \rho^d(x') \rho^e(y) \rho^h(x)$	II.I $_{\Lambda}$: $2i$; III.III $_{\Lambda}$: $-2i$	0 ✓
$f^{abc} U^{ad}(x') U^{eb}(z) U^{ch}(x) \rho^e(y) \rho^d(x') \rho^h(x)$	III.II $_{\Lambda}$: $-2i$; III.VI $_{\Lambda}$: $2i$	0 ✓
$f^{abc} U^{de}(x') U^{da}(w) U^{hb}(z) U^{hl}(y) \rho^e(x') \rho^l(y)$	I.IV $_{\Lambda}$: $2i$; II.I $_{\Lambda}$: $2i$; III.I $_{\Lambda}$: $-2i$; III.II $_{\Lambda}$: $-2i$	0 ✓
$\rho^c(x)$		
$f^{abc} U^{de}(w') U^{da}(w) U^{hb}(z) U^{hl}(y) \rho^e(x') \rho^l(y)$	I.IV $_{\Lambda}$: $-2i$; III.I $_{\Lambda}$: $2i$	0 ✓
$\rho^c(x)$		

Structure 2: $\mathcal{K}(x - w) \cdot \mathcal{K}(x' - w') K(y, w', z)$

colour structure	contributions (row: coefficient)	sum
$f^{abc} U^{de}(x) U^{da}(w') \rho^e(y) \rho^b(x') \rho^e(x)$	I.II $_{\Lambda}$: $-2i$; III.II $_{\Lambda}$: $2i$	0 ✓
$f^{abc} U^{de}(w) U^{da}(w') \rho^e(y) \rho^b(x') \rho^e(x)$	I.II $_{\Lambda}$: $2i$; II.I $_{\Lambda}$: $-2i$	0 ✓
$f^{abc} U^{ad}(x) U^{be}(x') U^{ch}(y) \rho^h(y) \rho^e(x') \rho^d(x)$	I.II $_{\Lambda}$: $2i$; I.III $_{\Lambda}$: $2i$; I.IV $_{\Lambda}$: $-2i$; I.V $_{\Lambda}$: $-2i$; II.I $_{\Lambda}$: $-2i$; III.I $_{\Lambda}$: $2i$; III.II $_{\Lambda}$: $2i$; III.III $_{\Lambda}$: $2i$; III.V $_{\Lambda}$: $-2i$; III.VI $_{\Lambda}$: $-2i$	0 ✓
$f^{abc} U^{ad}(w) U^{be}(x') U^{ch}(y) \rho^h(y) \rho^e(x') \rho^d(x)$	I.II $_{\Lambda}$: $-2i$; I.III $_{\Lambda}$: $-2i$; I.IV $_{\Lambda}$: $2i$; I.V $_{\Lambda}$: $2i$; III.I $_{\Lambda}$: $-2i$; III.V $_{\Lambda}$: $2i$	0 ✓
$f^{abc} U^{ad}(x) U^{ce}(y) U^{bh}(w') \rho^h(x') \rho^e(y) \rho^d(x)$	I.III $_{\Lambda}$: $-2i$; III.III $_{\Lambda}$: $-2i$; III.V $_{\Lambda}$: $2i$; III.VI $_{\Lambda}$: $2i$	0 ✓
$f^{abc} U^{ad}(w) U^{ce}(y) U^{bh}(w') \rho^h(x') \rho^e(y) \rho^d(x)$	I.III $_{\Lambda}$: $2i$; III.V $_{\Lambda}$: $-2i$	0 ✓
$f^{abc} U^{ad}(x) U^{be}(z) U^{ch}(w') \rho^e(y) \rho^h(x') \rho^d(x)$	I.V $_{\Lambda}$: $-2i$; III.VI $_{\Lambda}$: $2i$	0 ✓
$f^{abc} U^{ad}(w) U^{be}(z) U^{ch}(w') \rho^e(y) \rho^h(x') \rho^d(x)$	I.V $_{\Lambda}$: $2i$; III.III $_{\Lambda}$: $-2i$	0 ✓
$f^{abc} U^{db}(z) U^{ce}(x') U^{ah}(w) \rho^e(x') \rho^h(x) \rho^d(y)$	II.I $_{\Lambda}$: $-2i$; III.III $_{\Lambda}$: $2i$	0 ✓
$f^{abc} U^{ad}(x) U^{eb}(z) U^{ch}(x') \rho^h(x') \rho^d(x) \rho^e(y)$	III.II $_{\Lambda}$: $2i$; III.VI $_{\Lambda}$: $-2i$	0 ✓
$f^{abc} U^{de}(x) U^{da}(w') U^{hb}(z) U^{hl}(y) \rho^c(x') \rho^l(y)$	I.IV $_{\Lambda}$: $-2i$; II.I $_{\Lambda}$: $-2i$; III.I $_{\Lambda}$: $2i$; III.II $_{\Lambda}$: $2i$	0 ✓
$\rho^e(x)$		
$f^{abc} U^{de}(w) U^{da}(w') U^{hb}(z) U^{hl}(y) \rho^c(x') \rho^l(y)$	I.IV $_{\Lambda}$: $2i$; III.I $_{\Lambda}$: $-2i$	0 ✓
$\rho^e(x)$		

Structure 3: $\mathcal{K}(x - w) \cdot \mathcal{K}(x' - w') K(x, y, z)$

colour structure	contributions (row: coefficient)	sum
$f^{abc} U^{de}(x') U^{da}(w) \rho^e(x') \rho^b(x) \rho^c(y)$	I.II $_{\Lambda}$: $-2i$; III.II $_P$: i ; III.II $_{\Lambda}$: i	0 ✓
$f^{abc} U^{de}(w') U^{da}(w) \rho^e(x') \rho^b(x) \rho^c(y)$	I.II $_{\Lambda}$: $2i$; II.I $_P$: $-i$; II.I $_{\Lambda}$: $-i$	0 ✓
$f^{abc} U^{ad}(x') U^{be}(x) U^{ch}(y) \rho^d(x') \rho^e(x) \rho^h(y)$	I.II $_{\Lambda}$: $2i$; I.III $_P$: $-i$; I.III $_{\Lambda}$: $-i$; I.IV $_P$: $-i$; I.IV $_{\Lambda}$: $-i$; II.I $_P$: $-i$; II.I $_{\Lambda}$: $-i$; III.I $_P$: i ; III.I $_{\Lambda}$: i ; III.II $_P$: i ; III.II $_{\Lambda}$: i	0 ✓
$f^{abc} U^{ad}(w') U^{be}(x) U^{ch}(y) \rho^d(x') \rho^e(x) \rho^h(y)$	I.II $_{\Lambda}$: $-2i$; I.III $_P$: i ; I.III $_{\Lambda}$: i ; I.IV $_P$: i ; I.IV $_{\Lambda}$: i ; III.I $_P$: $-i$; III.I $_{\Lambda}$: $-i$	0 ✓
$f^{abc} U^{ad}(x') U^{ce}(x) U^{bh}(z) \rho^d(x') \rho^e(x) \rho^h(y)$	I.III $_P$: $-i$; I.III $_{\Lambda}$: $-i$; III.II $_P$: i ; III.II $_{\Lambda}$: i	0 ✓
$f^{abc} U^{ad}(w') U^{ce}(x) U^{bh}(z) \rho^d(x') \rho^e(x) \rho^h(y)$	I.III $_P$: i ; I.III $_{\Lambda}$: i ; II.I $_P$: $-i$; II.I $_{\Lambda}$: $-i$	0 ✓
$f^{abc} U^{de}(x') U^{da}(w) U^{hb}(z) U^{hl}(y) \rho^e(x') \rho^l(y)$	I.IV $_P$: $-i$; I.IV $_{\Lambda}$: $-i$; II.I $_P$: $-i$; II.I $_{\Lambda}$: $-i$; III.I $_P$: i ; III.I $_{\Lambda}$: i ; III.II $_P$: i ; III.II $_{\Lambda}$: i	0 ✓
$\rho^c(x)$		
$f^{abc} U^{de}(w') U^{da}(w) U^{hb}(z) U^{hl}(y) \rho^e(x') \rho^l(y)$	I.IV $_P$: i ; I.IV $_{\Lambda}$: i ; III.I $_P$: $-i$; III.I $_{\Lambda}$: $-i$	0 ✓
$\rho^c(x)$		

Structure 4: $\mathcal{K}(x - w) \cdot \mathcal{K}(x' - w') K(x', y, z)$

colour structure	contributions (row: coefficient)	sum
$f^{abc} U^{de}(x) U^{da}(w') \rho^e(y) \rho^b(x') \rho^e(x)$	I.II $_{\Lambda}$: $2i$; III.II $_P$: $-i$; III.II $_{\Lambda}$: $-i$	0 ✓
$f^{abc} U^{de}(w) U^{da}(w') \rho^e(y) \rho^b(x') \rho^e(x)$	I.II $_{\Lambda}$: $-2i$; II.I $_P$: i ; II.I $_{\Lambda}$: i	0 ✓
$f^{abc} U^{ad}(x) U^{be}(x') U^{ch}(y) \rho^h(y) \rho^e(x') \rho^d(x)$	I.II $_{\Lambda}$: $-2i$; I.III $_P$: i ; I.III $_{\Lambda}$: i ; I.IV $_P$: i ; I.IV $_{\Lambda}$: i ; II.I $_P$: i ; II.I $_{\Lambda}$: i ; III.I $_P$: $-i$; III.I $_{\Lambda}$: $-i$; III.II $_P$: $-i$; III.II $_{\Lambda}$: $-i$	0 ✓
$f^{abc} U^{ad}(w) U^{be}(x') U^{ch}(y) \rho^h(y) \rho^e(x') \rho^d(x)$	I.II $_{\Lambda}$: $2i$; I.III $_P$: $-i$; I.III $_{\Lambda}$: $-i$; I.IV $_P$: $-i$; I.IV $_{\Lambda}$: $-i$; III.I $_P$: i ; III.I $_{\Lambda}$: i	0 ✓
$f^{abc} U^{ad}(x) U^{ce}(x') U^{bh}(z) \rho^h(y) \rho^e(x') \rho^d(x)$	I.III $_P$: i ; I.III $_{\Lambda}$: i ; III.II $_P$: $-i$; III.II $_{\Lambda}$: $-i$	0 ✓
$f^{abc} U^{ad}(w) U^{ce}(x') U^{bh}(z) \rho^h(y) \rho^e(x') \rho^d(x)$	I.III $_P$: $-i$; I.III $_{\Lambda}$: $-i$; II.I $_P$: i ; II.I $_{\Lambda}$: i	0 ✓
$f^{abc} U^{de}(x) U^{da}(w') U^{hb}(z) U^{hl}(y) \rho^c(x') \rho^l(y)$	I.IV $_P$: i ; I.IV $_{\Lambda}$: i ; II.I $_P$: i ; II.I $_{\Lambda}$: i ; III.I $_P$: $-i$; III.I $_{\Lambda}$: $-i$; III.II $_P$: $-i$; III.II $_{\Lambda}$: $-i$	0 ✓
$\rho^e(x)$		
$f^{abc} U^{de}(w) U^{da}(w') U^{hb}(z) U^{hl}(y) \rho^c(x') \rho^l(y)$	I.IV $_P$: $-i$; I.IV $_{\Lambda}$: $-i$; III.I $_P$: i ; III.I $_{\Lambda}$: i	0 ✓
$\rho^e(x)$		

Structure 5: $\mathcal{K}(x' - w') \cdot \mathcal{K}(w - z) K(x, y, z)$

colour structure	contributions (row: coefficient)	sum
$f^{abc} U^{ad}(x') U^{ce}(x) U^{bh}(z) \rho^d(x') \rho^e(x) \rho^h(y)$	I.III $_P$: $-i$; III.II $_P$: i ; III.III $_P$: $-i$; III.V $_P$: i	0 ✓
$f^{abc} U^{ad}(x') U^{ce}(y) U^{bh}(z) \rho^d(x') \rho^e(y) \rho^h(x)$	I.III $_P$: $-i$; III.II $_P$: i ; III.III $_P$: $-i$; III.V $_P$: i	0 ✓

colour structure	contributions (row: coefficient)	sum
$f^{abc} U^{ad}(w') U^{ce}(x) U^{bh}(z) \rho^d(x') \rho^e(x) \rho^h(y)$	I.III _P : i ; II.I _P : $-i$; III.III _P : i ; III.V _P : $-i$	0 ✓
$f^{abc} U^{ad}(w') U^{ce}(y) U^{bh}(z) \rho^d(x') \rho^e(y) \rho^h(x)$	I.III _P : i ; II.I _P : $-i$; III.III _P : i ; III.V _P : $-i$	0 ✓
$f^{abc} U^{de}(x') U^{da}(w) U^{hb}(z) U^{hl}(x) \rho^e(x') \rho^l(x)$	I.IV _P : $-i$; II.I _P : $-i$; III.I _P : i ; III.II _P : i	0 ✓
$\rho^c(y)$ $f^{abc} U^{de}(x') U^{da}(w) U^{hb}(z) U^{hl}(y) \rho^e(x') \rho^l(y)$	I.IV _P : $-i$; II.I _P : $-i$; III.I _P : i ; III.II _P : i	0 ✓
$\rho^c(x)$ $f^{abc} U^{de}(w') U^{da}(w) U^{hb}(z) U^{hl}(x) \rho^e(x') \rho^l(x)$	I.IV _P : i ; III.I _P : $-i$	0 ✓
$\rho^c(y)$ $f^{abc} U^{de}(w') U^{da}(w) U^{hb}(z) U^{hl}(y) \rho^e(x') \rho^l(y)$	I.IV _P : i ; III.I _P : $-i$	0 ✓
$\rho^c(x)$		

Structure 6: $\mathcal{K}(x - w) \cdot \mathcal{K}(w' - z) K(x', y, z)$

colour structure	contributions (row: coefficient)	sum
$f^{abc} U^{ad}(x) U^{ce}(y) U^{bh}(z) \rho^h(x') \rho^e(y) \rho^d(x)$	I.III _P : i ; III.II _P : $-i$; III.III _P : i ; III.V _P : $-i$	0 ✓
$f^{abc} U^{ad}(x) U^{ce}(x') U^{bh}(z) \rho^h(y) \rho^e(x') \rho^d(x)$	I.III _P : i ; III.II _P : $-i$; III.III _P : i ; III.V _P : $-i$	0 ✓
$f^{abc} U^{ad}(w) U^{ce}(y) U^{bh}(z) \rho^h(x') \rho^e(y) \rho^d(x)$	I.III _P : $-i$; II.I _P : i ; III.III _P : $-i$; III.V _P : i	0 ✓
$f^{abc} U^{ad}(w) U^{ce}(x') U^{bh}(z) \rho^h(y) \rho^e(x') \rho^d(x)$	I.III _P : $-i$; II.I _P : i ; III.III _P : $-i$; III.V _P : i	0 ✓
$f^{abc} U^{de}(x) U^{da}(w') U^{hb}(z) U^{hl}(y) \rho^c(x') \rho^l(y)$	I.IV _P : i ; II.I _P : i ; III.I _P : $-i$; III.II _P : $-i$	0 ✓
$\rho^e(x)$ $f^{abc} U^{de}(x) U^{da}(w') U^{hb}(z) U^{hl}(x') \rho^c(y) \rho^l(x')$	I.IV _P : i ; II.I _P : i ; III.I _P : $-i$; III.II _P : $-i$	0 ✓
$\rho^e(x)$ $f^{abc} U^{de}(w) U^{da}(w') U^{hb}(z) U^{hl}(y) \rho^c(x') \rho^l(y)$	I.IV _P : $-i$; III.I _P : i	0 ✓
$\rho^e(x)$ $f^{abc} U^{de}(w) U^{da}(w') U^{hb}(z) U^{hl}(x') \rho^c(y) \rho^l(x')$	I.IV _P : $-i$; III.I _P : i	0 ✓
$\rho^e(x)$		

8 The two- ρ terms: comparison with BFKL $\otimes$ LO–JIMWLK $\otimes$ LO

All two- ρ terms proportional to $\log(\sqrt{s}/k^+)$, after the fixes:

- the genuine two- ρ terms of Sec. 4 (Group I Rows II, V; Group II Rows I, IV; Group III Row III part 2);
- the two- ρ terms of the three- ρ terms (Sec. 6).

Grouped by kernel structure and colour network. For each structure the colour networks are reduced to an independent basis, with coefficients exact polynomials in N_c (checked with SU(3) and SU(4); the relations are in App. A). The last two columns are BFKL $\otimes$ LO and JIMWLK $\otimes$ LO in the same basis. A check mark means sum = BFKL–JIMWLK. The labels are as in Sec. 7. The two integrated genuine terms are written as z integrands, as in the previous note: Group I Row II as $\frac{1}{\pi} [K(x, w, z) - K(w, w, z)]$, Group II Row IV as $K(w, w', z) - \frac{1}{2} K(w, w, z) - \frac{1}{2} K(w', w', z)$.

Structure 1: $\mathcal{K}(x - w) \cdot \mathcal{K}(x' - w') K(x, w, z)$

Neither BFKL $\otimes$ LO nor JIMWLK $\otimes$ LO has a term with this kernel: the $\log(\sqrt{s}/k^+)$ terms cancel among themselves.

colour structure	contributions (row: coefficient)	sum	BFKL	JIMWLK
$U^{ab}(x') U^{ac}(w) \rho^b(x') \rho^c(x)$	I.II _A : N_c ; III.II _A : $-N_c$	0	0	0 ✓
$U^{ab}(x') U^{ac}(x) \rho^b(x') \rho^c(x)$	I.II _A : $-N_c$; I.III _A : $-N_c$; I.IV _A : N_c ; I.V _A : N_c ; II.I _A : N_c ; III.I _A : $-N_c$; III.II _A : $-N_c$; III.III _A : $-N_c$; III.V _A : N_c ; III.VI _A : N_c	0	0	0 ✓
$U^{ab}(w') U^{ac}(w) \rho^b(x') \rho^c(x)$	I.II _A : $-N_c$; II.I _A : N_c	0	0	0 ✓
$U^{ab}(w') U^{ac}(x) \rho^b(x') \rho^c(x)$	I.II _A : N_c ; I.III _A : N_c ; I.IV _A : $-N_c$; I.V _A : $-N_c$; III.I _A : N_c ; III.V _A : $-N_c$	0	0	0 ✓
$f^{abc} f^{deh} U^{al}(x') U^{ce}(x) U^{bd}(w) \rho^l(x')$	I.III _A : 1; III.III _A : 1; III.V _A : -1 ; III.VI _A : -1	0	0	0 ✓
$\rho^h(x)$ $f^{abc} f^{deh} U^{al}(w') U^{ce}(x) U^{bd}(w) \rho^l(x')$	I.III _A : -1 ; III.V _A : 1	0	0	0 ✓
$\rho^h(x)$ $f^{abc} f^{deh} U^{al}(x') U^{bd}(z) U^{ce}(w) \rho^l(x')$	I.V _A : -1 ; III.VI _A : 1	0	0	0 ✓
$\rho^h(x)$ $f^{abc} f^{deh} U^{al}(w') U^{bd}(z) U^{ce}(w) \rho^l(x')$	I.V _A : 1; III.III _A : -1	0	0	0 ✓
$\rho^h(x)$				

colour structure	contributions (row: coefficient)	sum	BFKL	JIMWLK
$f^{abc} f^{deh} U^{al}(w') U^{eb}(z) U^{cd}(x) \rho^l(x')$ $\rho^h(x)$	II.I _A : 1; III.III _A : -1	0	0	0 ✓
$f^{abc} f^{deh} U^{al}(x') U^{eb}(z) U^{cd}(x) \rho^l(x')$ $\rho^h(x)$	III.II _A : -1; III.VI _A : 1	0	0	0 ✓
$f^{abc} f^{cde} U^{hl}(x') U^{ha}(w) U^{mb}(z)$ $U^{md}(x) \rho^l(x') \rho^e(x)$	I.IV _A : 1; II.I _A : 1; III.I _A : -1; III.II _A : -1	0	0	0 ✓
$f^{abc} f^{cde} U^{hl}(w') U^{ha}(w) U^{mb}(z)$ $U^{md}(x) \rho^l(x') \rho^e(x)$	I.IV _A : -1; III.I _A : 1	0	0	0 ✓

Structure 2: $\mathcal{K}(x' - w) \cdot \mathcal{K}(x' - w') K(x, w, z)$

Neither BFKL $\otimes$ LO nor JIMWLK $\otimes$ LO has a term with this kernel: the $\log(\sqrt{s}/k^+)$ terms cancel among themselves.

colour structure	contributions (row: coefficient)	sum	BFKL	JIMWLK
$U^{ab}(x) U^{ac}(x') \rho^c(x') \rho^b(x)$	I.II _A : N_c ; I.III _A : N_c ; I.IV _A : $-N_c$; I.V _A : $-N_c$; II.I _A : $-N_c$; III.I _A : N_c ; III.II _A : N_c ; III.III _A : N_c ; III.V _A : $-N_c$; III.VI _A : $-N_c$	0	0	0 ✓
$U^{ab}(z) U^{bc}(x') \rho^a(x) \rho^c(x')$	III.II _A : $-N_c$; III.VI _A : N_c	0	0	0 ✓
$f^{abc} f^{bde} U^{hd}(x') U^{ha}(w) \rho^e(x') \rho^c(x)$	I.II _A : 1; III.II _A : -1	0	0	0 ✓
$f^{abc} f^{bde} U^{hd}(w') U^{ha}(w) \rho^e(x') \rho^c(x)$	I.II _A : -1; II.I _A : 1	0	0	0 ✓
$f^{abc} f^{deh} U^{ae}(w') U^{bd}(x') U^{cl}(x) \rho^h(x')$ $\rho^l(x)$	I.II _A : 1; I.III _A : 1; I.IV _A : -1; I.V _A : -1; III.I _A : 1; III.V _A : -1	0	0	0 ✓
$f^{abc} f^{deh} U^{ae}(x') U^{cl}(x) U^{bd}(w) \rho^l(x)$ $\rho^h(x')$	I.III _A : 1; III.III _A : 1; III.V _A : -1; III.VI _A : -1	0	0	0 ✓
$f^{abc} f^{deh} U^{ae}(w') U^{cl}(x) U^{bd}(w) \rho^l(x)$ $\rho^h(x')$	I.III _A : -1; III.V _A : 1	0	0	0 ✓
$f^{abc} f^{deh} U^{ae}(x') U^{bl}(z) U^{cd}(w) \rho^h(x')$ $\rho^l(x)$	I.V _A : 1; III.VI _A : -1	0	0	0 ✓
$f^{abc} f^{deh} U^{ae}(w') U^{bl}(z) U^{cd}(w) \rho^h(x')$ $\rho^l(x)$	I.V _A : -1; III.III _A : 1	0	0	0 ✓
$f^{abc} f^{deh} U^{ae}(w') U^{lb}(z) U^{cd}(x') \rho^l(x)$ $\rho^h(x')$	II.I _A : 1; III.III _A : -1	0	0	0 ✓
$f^{abc} f^{cde} U^{hd}(x') U^{ha}(w) U^{lb}(z)$ $U^{lm}(x) \rho^m(x) \rho^e(x')$	I.IV _A : 1; II.I _A : 1; III.I _A : -1; III.II _A : -1	0	0	0 ✓
$f^{abc} f^{cde} U^{hd}(w') U^{ha}(w) U^{lb}(z)$ $U^{lm}(x) \rho^m(x) \rho^e(x')$	I.IV _A : -1; III.I _A : 1	0	0	0 ✓

Structure 3: $\mathcal{K}(x - w) \cdot \mathcal{K}(x' - w') K(x', w, z)$

colour structure	contributions (row: coefficient)	sum	BFKL	JIMWLK
$U^{ab}(w) U^{ac}(x') \rho^c(x') \rho^b(x)$	I.III _A : N_c ; III.III _A : N_c ; III.V _A : $-N_c$; III.VI _A : N_c	$2N_c$	0	$-2N_c$ ✓
$f^{abc} f^{cde} U^{hd}(x') U^{ha}(w) \rho^b(x) \rho^e(x')$	I.II _A : 1; III.II _A : 1	2	0	-2 ✓
$f^{abc} f^{deh} U^{ea}(w) U^{db}(z) U^{hl}(x') \rho^l(x')$ $\rho^c(x)$	I.IV _A : 1; II.I _A : 1; III.I _A : -1; III.II _A : 1	2	0	-2 ✓
$f^{abc} f^{deh} U^{ae}(x') U^{bd}(z) U^{cl}(w) \rho^l(x)$ $\rho^h(x')$	I.V _A : 1; III.VI _A : 1	2	0	-2 ✓
$U^{ab}(x) U^{ac}(x') \rho^b(x) \rho^c(x')$	I.II _A : $-N_c$; I.III _A : $-N_c$; I.IV _A : N_c ; I.V _A : N_c ; II.I _A : N_c ; III.I _A : $-N_c$; III.II _A : N_c ; III.III _A : $-N_c$; III.V _A : N_c ; III.VI _A : $-N_c$	0	0	0 ✓
$f^{abc} f^{cde} U^{hd}(w') U^{ha}(w) \rho^b(x) \rho^e(x')$	I.II _A : -1; II.I _A : 1	0	0	0 ✓
$f^{abc} f^{deh} U^{ae}(w') U^{bl}(x) U^{cd}(x') \rho^l(x)$ $\rho^h(x')$	I.II _A : 1; I.III _A : 1; I.IV _A : -1; I.V _A : -1; III.I _A : 1; III.V _A : -1	0	0	0 ✓
$f^{abc} f^{deh} U^{ae}(w') U^{cd}(x') U^{bl}(w) \rho^h(x')$ $\rho^l(x)$	I.III _A : -1; III.V _A : 1	0	0	0 ✓
$f^{abc} f^{deh} U^{ae}(w') U^{bd}(z) U^{cl}(w) \rho^l(x)$ $\rho^h(x')$	I.V _A : -1; III.III _A : 1	0	0	0 ✓
$f^{abc} f^{deh} U^{ae}(w') U^{db}(z) U^{cl}(x) \rho^h(x')$ $\rho^l(x)$	II.I _A : 1; III.III _A : -1	0	0	0 ✓
$f^{abc} f^{deh} U^{ad}(x') U^{eb}(z) U^{cl}(x) \rho^h(x')$ $\rho^l(x)$	III.II _A : -1; III.VI _A : 1	0	0	0 ✓

colour structure	contributions (row: coefficient)	sum	BFKL	JIMWLK
$f^{abc} f^{deh} U^{le}(w') U^{la}(w) U^{mb}(z)$ $U^{md}(x') \rho^h(x') \rho^e(x)$	I.IV $_{\Lambda}$: -1; III.I $_{\Lambda}$: 1	0	0	0 ✓

Structure 4: $\mathcal{K}(x' - w) \cdot \mathcal{K}(x' - w') K(x, w', z)$

Neither BFKL $\otimes$ LO nor JIMWLK $\otimes$ LO has a term with this kernel: the $\log(\sqrt{s}/k^+)$ terms cancel among themselves.

colour structure	contributions (row: coefficient)	sum	BFKL	JIMWLK
$U^{ab}(x) U^{ac}(x') \rho^b(x) \rho^c(x')$	I.II $_{\Lambda}$: N_c ; I.III $_{\Lambda}$: N_c ; I.IV $_{\Lambda}$: $-N_c$; I.V $_{\Lambda}$: $-N_c$; II.I $_{\Lambda}$: $-N_c$; III.I $_{\Lambda}$: N_c ; III.II $_{\Lambda}$: N_c ; III.III $_{\Lambda}$: N_c ; III.V $_{\Lambda}$: $-N_c$; III.VI $_{\Lambda}$: $-N_c$	0	0	0 ✓
$U^{ab}(z) U^{bc}(x') \rho^c(x') \rho^a(x)$	III.II $_{\Lambda}$: $-N_c$; III.VI $_{\Lambda}$: N_c	0	0	0 ✓
$f^{abc} f^{dbe} U^{hd}(x') U^{ha}(w') \rho^c(x) \rho^e(x')$	I.II $_{\Lambda}$: -1; III.II $_{\Lambda}$: 1	0	0	0 ✓
$f^{abc} f^{dbe} U^{hd}(w) U^{ha}(w') \rho^c(x) \rho^e(x')$	I.II $_{\Lambda}$: 1; II.I $_{\Lambda}$: -1	0	0	0 ✓
$f^{abc} f^{deh} U^{ad}(w) U^{be}(x') U^{cl}(x) \rho^l(x)$ $\rho^h(x')$	I.II $_{\Lambda}$: -1; I.III $_{\Lambda}$: -1; I.IV $_{\Lambda}$: 1; I.V $_{\Lambda}$: 1; III.I $_{\Lambda}$: -1; III.V $_{\Lambda}$: 1	0	0	0 ✓
$f^{abc} f^{deh} U^{ad}(x') U^{cl}(x) U^{be}(w') \rho^l(x)$ $\rho^h(x')$	I.III $_{\Lambda}$: -1; III.III $_{\Lambda}$: -1; III.V $_{\Lambda}$: 1; III.VI $_{\Lambda}$: 1	0	0	0 ✓
$f^{abc} f^{deh} U^{ad}(w) U^{cl}(x) U^{be}(w') \rho^l(x)$ $\rho^h(x')$	I.III $_{\Lambda}$: 1; III.V $_{\Lambda}$: -1	0	0	0 ✓
$f^{abc} f^{deh} U^{ad}(x') U^{bl}(z) U^{ce}(w') \rho^l(x)$ $\rho^h(x')$	I.V $_{\Lambda}$: -1; III.VI $_{\Lambda}$: 1	0	0	0 ✓
$f^{abc} f^{deh} U^{ad}(w) U^{bl}(z) U^{ce}(w') \rho^l(x)$ $\rho^h(x')$	I.V $_{\Lambda}$: 1; III.III $_{\Lambda}$: -1	0	0	0 ✓
$f^{abc} f^{deh} U^{lb}(z) U^{ce}(x') U^{ad}(w) \rho^h(x')$ $\rho^l(x)$	II.I $_{\Lambda}$: -1; III.III $_{\Lambda}$: 1	0	0	0 ✓
$f^{abc} f^{dce} U^{hd}(x') U^{ha}(w') U^{lb}(z)$	I.IV $_{\Lambda}$: -1; II.I $_{\Lambda}$: -1; III.I $_{\Lambda}$: 1; III.II $_{\Lambda}$: 1	0	0	0 ✓
$U^{lm}(x) \rho^m(x) \rho^e(x')$ $f^{abc} f^{dce} U^{hd}(w) U^{ha}(w') U^{lb}(z)$	I.IV $_{\Lambda}$: 1; III.I $_{\Lambda}$: -1	0	0	0 ✓
$U^{lm}(x) \rho^m(x) \rho^e(x')$				

Structure 5: $\mathcal{K}(x - w) \cdot \mathcal{K}(x' - w') K(x', w', z)$

Neither BFKL $\otimes$ LO nor JIMWLK $\otimes$ LO has a term with this kernel: the $\log(\sqrt{s}/k^+)$ terms cancel among themselves.

colour structure	contributions (row: coefficient)	sum	BFKL	JIMWLK
$U^{ab}(x) U^{ac}(w') \rho^c(x') \rho^b(x)$	I.II $_{\Lambda}$: N_c ; III.II $_{\Lambda}$: $-N_c$	0	0	0 ✓
$U^{ab}(x) U^{ac}(x') \rho^c(x') \rho^b(x)$	I.II $_{\Lambda}$: $-N_c$; I.III $_{\Lambda}$: $-N_c$; I.IV $_{\Lambda}$: N_c ; I.V $_{\Lambda}$: N_c ; II.I $_{\Lambda}$: N_c ; III.I $_{\Lambda}$: $-N_c$; III.II $_{\Lambda}$: $-N_c$; III.III $_{\Lambda}$: $-N_c$; III.V $_{\Lambda}$: N_c ; III.VI $_{\Lambda}$: N_c	0	0	0 ✓
$U^{ab}(w) U^{ac}(w') \rho^c(x') \rho^b(x)$	I.II $_{\Lambda}$: $-N_c$; II.I $_{\Lambda}$: N_c	0	0	0 ✓
$U^{ab}(w) U^{ac}(x') \rho^c(x') \rho^b(x)$	I.II $_{\Lambda}$: N_c ; I.III $_{\Lambda}$: N_c ; I.IV $_{\Lambda}$: $-N_c$; I.V $_{\Lambda}$: $-N_c$; III.I $_{\Lambda}$: N_c ; III.V $_{\Lambda}$: $-N_c$	0	0	0 ✓
$f^{abc} f^{deh} U^{al}(x) U^{cd}(x') U^{be}(w') \rho^h(x')$ $\rho^l(x)$	I.III $_{\Lambda}$: -1; III.III $_{\Lambda}$: -1; III.V $_{\Lambda}$: 1; III.VI $_{\Lambda}$: 1	0	0	0 ✓
$f^{abc} f^{deh} U^{al}(w) U^{cd}(x') U^{be}(w') \rho^h(x')$ $\rho^l(x)$	I.III $_{\Lambda}$: 1; III.V $_{\Lambda}$: -1	0	0	0 ✓
$f^{abc} f^{deh} U^{al}(x) U^{be}(z) U^{cd}(w') \rho^h(x')$ $\rho^l(x)$	I.V $_{\Lambda}$: 1; III.VI $_{\Lambda}$: -1	0	0	0 ✓
$f^{abc} f^{deh} U^{al}(w) U^{be}(z) U^{cd}(w') \rho^h(x')$ $\rho^l(x)$	I.V $_{\Lambda}$: -1; III.III $_{\Lambda}$: 1	0	0	0 ✓
$f^{abc} f^{deh} U^{db}(z) U^{ce}(x') U^{al}(w) \rho^l(x)$ $\rho^h(x')$	II.I $_{\Lambda}$: -1; III.III $_{\Lambda}$: 1	0	0	0 ✓
$f^{abc} f^{deh} U^{al}(x) U^{db}(z) U^{ce}(x') \rho^l(x)$ $\rho^h(x')$	III.II $_{\Lambda}$: 1; III.VI $_{\Lambda}$: -1	0	0	0 ✓
$f^{abc} f^{dce} U^{hl}(x) U^{ha}(w') U^{mb}(z)$	I.IV $_{\Lambda}$: -1; II.I $_{\Lambda}$: -1; III.I $_{\Lambda}$: 1; III.II $_{\Lambda}$: 1	0	0	0 ✓
$U^{md}(x') \rho^e(x') \rho^l(x)$ $f^{abc} f^{dce} U^{hl}(w) U^{ha}(w') U^{mb}(z)$	I.IV $_{\Lambda}$: 1; III.I $_{\Lambda}$: -1	0	0	0 ✓
$U^{md}(x') \rho^e(x') \rho^l(x)$				

Structure 6: $\mathcal{K}(x - w) \cdot \mathcal{K}(x' - w') K(x, w', z)$

colour structure	contributions (row: coefficient)	sum	BFKL	JIMWLK
$U^{ab}(w') U^{ac}(x) \rho^b(x') \rho^c(x)$	I.III $_{\Lambda}$: N_c ; III.III $_{\Lambda}$: N_c ; III.V $_{\Lambda}$: $-N_c$; III.VI $_{\Lambda}$: N_c	$2N_c$	0	$-2N_c$ ✓
$f^{abc} f^{dce} U^{hd}(x) U^{ha}(w') \rho^b(x') \rho^e(x)$	I.II $_{\Lambda}$: -1 ; III.II $_{\Lambda}$: -1	-2	0	2 ✓
$f^{abc} f^{deh} U^{da}(w') U^{eb}(z) U^{hl}(x) \rho^c(x')$	I.IV $_{\Lambda}$: -1 ; II.I $_{\Lambda}$: -1 ; III.I $_{\Lambda}$: 1 ; III.II $_{\Lambda}$: -1	-2	0	2 ✓
$\rho^l(x)$				
$f^{abc} f^{deh} U^{ad}(x) U^{be}(z) U^{cl}(w') \rho^l(x')$	I.V $_{\Lambda}$: -1 ; III.VI $_{\Lambda}$: -1	-2	0	2 ✓
$\rho^h(x)$				
$U^{ab}(x') U^{ac}(x) \rho^b(x') \rho^c(x)$	I.II $_{\Lambda}$: $-N_c$; I.III $_{\Lambda}$: $-N_c$; I.IV $_{\Lambda}$: N_c ; I.V $_{\Lambda}$: N_c ; II.I $_{\Lambda}$: N_c ; III.I $_{\Lambda}$: $-N_c$; III.II $_{\Lambda}$: N_c ; III.III $_{\Lambda}$: $-N_c$; III.V $_{\Lambda}$: N_c ; III.VI $_{\Lambda}$: $-N_c$	0	0	0 ✓
$f^{abc} f^{dce} U^{hd}(w) U^{ha}(w') \rho^b(x') \rho^e(x)$	I.II $_{\Lambda}$: 1 ; II.I $_{\Lambda}$: -1	0	0	0 ✓
$f^{abc} f^{deh} U^{ad}(w) U^{bl}(x') U^{ce}(x) \rho^l(x')$	I.II $_{\Lambda}$: -1 ; I.III $_{\Lambda}$: -1 ; I.IV $_{\Lambda}$: 1 ; I.V $_{\Lambda}$: 1 ; III.I $_{\Lambda}$: -1 ; III.V $_{\Lambda}$: 1	0	0	0 ✓
$\rho^h(x)$				
$f^{abc} f^{deh} U^{ad}(w) U^{ce}(x) U^{bl}(w') \rho^l(x')$	I.III $_{\Lambda}$: 1 ; III.V $_{\Lambda}$: -1	0	0	0 ✓
$\rho^h(x)$				
$f^{abc} f^{deh} U^{ad}(w) U^{be}(z) U^{cl}(w') \rho^l(x')$	I.V $_{\Lambda}$: 1 ; III.III $_{\Lambda}$: -1	0	0	0 ✓
$\rho^h(x)$				
$f^{abc} f^{deh} U^{db}(z) U^{cl}(x') U^{ae}(w) \rho^l(x')$	II.I $_{\Lambda}$: 1 ; III.III $_{\Lambda}$: -1	0	0	0 ✓
$\rho^h(x)$				
$f^{abc} f^{deh} U^{ae}(x) U^{db}(z) U^{cl}(x') \rho^l(x')$	III.II $_{\Lambda}$: 1 ; III.VI $_{\Lambda}$: -1	0	0	0 ✓
$\rho^h(x)$				
$f^{abc} f^{deh} U^{ld}(w) U^{la}(w') U^{mb}(z)$	I.IV $_{\Lambda}$: 1 ; III.I $_{\Lambda}$: -1	0	0	0 ✓
$U^{me}(x) \rho^c(x') \rho^h(x)$				

Structure 7: $\mathcal{K}(x - w) \cdot \mathcal{K}(x' - w') K(x, x, z)$

colour structure	contributions (row: coefficient)	sum	BFKL	JIMWLK
$U^{ab}(x') U^{ac}(w) \rho^b(x') \rho^c(x)$	III.II $_P$: $\frac{N_c}{2}$; III.II $_{\Lambda}$: $\frac{N_c}{2}$	N_c	N_c	0 ✓
$U^{ab}(x') U^{ac}(x) \rho^b(x') \rho^c(x)$	I.III $_P$: $\frac{N_c}{2}$; I.III $_{\Lambda}$: $\frac{N_c}{2}$; I.IV $_P$: $-\frac{N_c}{2}$; I.IV $_{\Lambda}$: $-\frac{N_c}{2}$; II.I $_P$: $-\frac{N_c}{2}$; II.I $_{\Lambda}$: $-\frac{N_c}{2}$; III.I $_P$: $\frac{N_c}{2}$; III.I $_{\Lambda}$: $\frac{N_c}{2}$; III.II $_P$: $\frac{N_c}{2}$; III.II $_{\Lambda}$: $\frac{N_c}{2}$	N_c	$-N_c$	$-2N_c$ ✓
$U^{ab}(w') U^{ac}(w) \rho^b(x') \rho^c(x)$	II.I $_P$: $-\frac{N_c}{2}$; II.I $_{\Lambda}$: $-\frac{N_c}{2}$	$-N_c$	$-N_c$	0 ✓
$U^{ab}(w') U^{ac}(x) \rho^b(x') \rho^c(x)$	I.III $_P$: $-\frac{N_c}{2}$; I.III $_{\Lambda}$: $-\frac{N_c}{2}$; I.IV $_P$: $\frac{N_c}{2}$; I.IV $_{\Lambda}$: $\frac{N_c}{2}$; III.I $_P$: $-\frac{N_c}{2}$; III.I $_{\Lambda}$: $-\frac{N_c}{2}$	$-N_c$	N_c	$2N_c$ ✓
$f^{abc} f^{deh} U^{al}(x') U^{ce}(x) U^{bd}(z) \rho^l(x')$	I.III $_P$: $-\frac{1}{2}$; I.III $_{\Lambda}$: $-\frac{1}{2}$; III.II $_P$: $-\frac{1}{2}$; III.II $_{\Lambda}$: $-\frac{1}{2}$	-2	0	2 ✓
$\rho^h(x)$				
$f^{abc} f^{deh} U^{al}(w') U^{ce}(x) U^{bd}(z) \rho^l(x')$	I.III $_P$: $\frac{1}{2}$; I.III $_{\Lambda}$: $\frac{1}{2}$; II.I $_P$: $\frac{1}{2}$; II.I $_{\Lambda}$: $\frac{1}{2}$	2	0	-2 ✓
$\rho^h(x)$				
$f^{abc} f^{cde} U^{hl}(x') U^{ha}(w) U^{mb}(z)$	I.IV $_P$: $-\frac{1}{2}$; I.IV $_{\Lambda}$: $-\frac{1}{2}$; II.I $_P$: $-\frac{1}{2}$; II.I $_{\Lambda}$: $-\frac{1}{2}$;	0	0	0 ✓
$U^{md}(x) \rho^l(x') \rho^e(x)$	III.I $_P$: $\frac{1}{2}$; III.I $_{\Lambda}$: $\frac{1}{2}$; III.II $_P$: $\frac{1}{2}$; III.II $_{\Lambda}$: $\frac{1}{2}$			
$f^{abc} f^{cde} U^{hl}(w') U^{ha}(w) U^{mb}(z)$	I.IV $_P$: $\frac{1}{2}$; I.IV $_{\Lambda}$: $\frac{1}{2}$; III.I $_P$: $-\frac{1}{2}$; III.I $_{\Lambda}$: $-\frac{1}{2}$	0	0	0 ✓
$U^{md}(x) \rho^l(x') \rho^e(x)$				

Structure 8: $\mathcal{K}(x' - w) \cdot \mathcal{K}(x' - w') K(x, x', z)$

Neither BFKL $\otimes$ LO nor JIMWLK $\otimes$ LO has a term with this kernel: the $\log(\sqrt{v}/k^+)$ terms cancel among themselves.

colour structure	contributions (row: coefficient)	sum	BFKL	JIMWLK
$U^{ab}(x) U^{ac}(x') \rho^c(x') \rho^b(x)$	I.II $_{\Lambda}$: $-2N_c$; I.III $_P$: N_c ; I.III $_{\Lambda}$: N_c ; I.IV $_P$: N_c ; I.IV $_{\Lambda}$: N_c ; II.I $_P$: N_c ; II.I $_{\Lambda}$: N_c ; III.I $_P$: $-N_c$; III.I $_{\Lambda}$: $-N_c$; III.II $_P$: $-N_c$; III.II $_{\Lambda}$: $-N_c$	0	0	0 ✓
$U^{ab}(z) U^{ac}(x') \rho^c(x') \rho^b(x)$	I.III $_P$: $-N_c$; I.III $_{\Lambda}$: $-N_c$; III.II $_P$: N_c ; III.II $_{\Lambda}$: N_c	0	0	0 ✓
$f^{abc} f^{bde} U^{hd}(x') U^{ha}(w) \rho^e(x') \rho^c(x)$	I.II $_{\Lambda}$: -1 ; III.II $_P$: $\frac{1}{2}$; III.II $_{\Lambda}$: $\frac{1}{2}$	0	0	0 ✓
$f^{abc} f^{dbe} U^{hd}(x') U^{ha}(w') \rho^c(x) \rho^e(x')$	I.II $_{\Lambda}$: 1 ; III.II $_P$: $-\frac{1}{2}$; III.II $_{\Lambda}$: $-\frac{1}{2}$	0	0	0 ✓
$f^{abc} f^{bde} U^{hd}(w') U^{ha}(w) \rho^e(x') \rho^c(x)$	I.II $_{\Lambda}$: 1 ; II.I $_P$: $-\frac{1}{2}$; II.I $_{\Lambda}$: $-\frac{1}{2}$	0	0	0 ✓
$f^{abc} f^{dbe} U^{hd}(w) U^{ha}(w') \rho^c(x) \rho^e(x')$	I.II $_{\Lambda}$: -1 ; II.I $_P$: $\frac{1}{2}$; II.I $_{\Lambda}$: $\frac{1}{2}$	0	0	0 ✓
$f^{abc} f^{deh} U^{ae}(w') U^{bd}(x') U^{cl}(x) \rho^h(x')$	I.II $_{\Lambda}$: -1 ; I.III $_P$: $\frac{1}{2}$; I.III $_{\Lambda}$: $\frac{1}{2}$; I.IV $_P$: $\frac{1}{2}$; I.IV $_{\Lambda}$: $\frac{1}{2}$; III.I $_P$: $-\frac{1}{2}$; III.I $_{\Lambda}$: $-\frac{1}{2}$	0	0	0 ✓
$\rho^l(x)$				
$f^{abc} f^{deh} U^{ad}(w) U^{be}(x') U^{cl}(x) \rho^l(x)$	I.II $_{\Lambda}$: 1 ; I.III $_P$: $-\frac{1}{2}$; I.III $_{\Lambda}$: $-\frac{1}{2}$; I.IV $_P$: $-\frac{1}{2}$; I.IV $_{\Lambda}$: $-\frac{1}{2}$; III.I $_P$: $\frac{1}{2}$; III.I $_{\Lambda}$: $\frac{1}{2}$	0	0	0 ✓
$\rho^h(x')$				

colour structure	contributions (row: coefficient)	sum	BFKL	JIMWLK
$f^{abc} f^{deh} U^{ae}(w') U^{cd}(x') U^{bl}(z) \rho^h(x')$	I.III _P : $\frac{1}{2}$; I.III _{Λ} : $\frac{1}{2}$; II.I _P : $-\frac{1}{2}$; II.I _{Λ} : $-\frac{1}{2}$	0	0	0 ✓
$\rho^l(x)$				
$f^{abc} f^{deh} U^{ad}(w) U^{ce}(x') U^{bl}(z) \rho^l(x)$	I.III _P : $-\frac{1}{2}$; I.III _{Λ} : $-\frac{1}{2}$; II.I _P : $\frac{1}{2}$; II.I _{Λ} : $\frac{1}{2}$	0	0	0 ✓
$\rho^h(x')$				
$f^{abc} f^{cde} U^{hd}(x') U^{ha}(w) U^{lb}(z)$	I.IV _P : $-\frac{1}{2}$; I.IV _{Λ} : $-\frac{1}{2}$; II.I _P : $-\frac{1}{2}$; II.I _{Λ} : $-\frac{1}{2}$;	0	0	0 ✓
$U^{lm}(x) \rho^m(x) \rho^e(x')$	III.I _P : $\frac{1}{2}$; III.I _{Λ} : $\frac{1}{2}$; III.II _P : $\frac{1}{2}$; III.II _{Λ} : $\frac{1}{2}$			
$f^{abc} f^{dce} U^{hd}(x') U^{ha}(w') U^{lb}(z)$	I.IV _P : $\frac{1}{2}$; I.IV _{Λ} : $\frac{1}{2}$; II.I _P : $\frac{1}{2}$; II.I _{Λ} : $\frac{1}{2}$;	0	0	0 ✓
$U^{lm}(x) \rho^m(x) \rho^e(x')$	III.I _P : $-\frac{1}{2}$; III.I _{Λ} : $-\frac{1}{2}$; III.II _P : $-\frac{1}{2}$; III.II _{Λ} : $-\frac{1}{2}$			
$f^{abc} f^{cde} U^{hd}(w') U^{ha}(w) U^{lb}(z)$	I.IV _P : $\frac{1}{2}$; I.IV _{Λ} : $\frac{1}{2}$; III.I _P : $-\frac{1}{2}$; III.I _{Λ} : $-\frac{1}{2}$	0	0	0 ✓
$U^{lm}(x) \rho^m(x) \rho^e(x')$				
$f^{abc} f^{dce} U^{hd}(w) U^{ha}(w') U^{lb}(z)$	I.IV _P : $-\frac{1}{2}$; I.IV _{Λ} : $-\frac{1}{2}$; III.I _P : $\frac{1}{2}$; III.I _{Λ} : $\frac{1}{2}$	0	0	0 ✓
$U^{lm}(x) \rho^m(x) \rho^e(x')$				

Structure 9: $\mathcal{K}(x - w) \cdot \mathcal{K}(x' - w') K(x, x', z)$

colour structure	contributions (row: coefficient)	sum	BFKL	JIMWLK
$U^{ab}(x) U^{ac}(x') \rho^b(x) \rho^c(x')$	I.II _{Λ} : $2N_c$; I.III _P : $-N_c$; I.III _{Λ} : $-N_c$; I.IV _P : $-N_c$; I.IV _{Λ} : $-N_c$; II.I _P : $\frac{N_c}{2}$; II.I _{Λ} : $-\frac{N_c}{2}$; III.I _P : N_c ; III.I _{Λ} : N_c ; III.II _P : $-N_c$; III.II _{Λ} : $-N_c$	$-2N_c$	0	$2N_c$ ✓
$f^{abc} f^{cde} U^{hd}(x') U^{ha}(w) \rho^b(x) \rho^e(x')$	I.II _{Λ} : -1 ; III.II _P : $-\frac{1}{2}$; III.II _{Λ} : $-\frac{1}{2}$	-2	-2	0 ✓
$f^{abc} f^{dce} U^{hd}(x) U^{ha}(w') \rho^b(x') \rho^e(x)$	I.II _{Λ} : 1; III.II _P : $\frac{1}{2}$; III.II _{Λ} : $\frac{1}{2}$	2	2	0 ✓
$f^{abc} f^{cde} U^{hd}(w') U^{ha}(w) \rho^b(x) \rho^e(x')$	I.II _{Λ} : 2; II.I _P : $\frac{1}{2}$; II.I _{Λ} : $-\frac{1}{2}$	2	2	0 ✓
$f^{abc} f^{dbe} U^{ha}(x') U^{hd}(x) \rho^c(x') \rho^e(x)$	-	0	2	2 ✓
$f^{abc} f^{deh} U^{ae}(x') U^{cl}(x) U^{bd}(z) \rho^l(x)$	I.III _P : $-\frac{1}{2}$; I.III _{Λ} : $-\frac{1}{2}$; III.II _P : $-\frac{1}{2}$; III.II _{Λ} : $-\frac{1}{2}$	-2	0	2 ✓
$\rho^h(x')$				
$f^{abc} f^{deh} U^{ad}(x) U^{cl}(x') U^{be}(z) \rho^l(x')$	I.III _P : $\frac{1}{2}$; I.III _{Λ} : $\frac{1}{2}$; III.II _P : $\frac{1}{2}$; III.II _{Λ} : $\frac{1}{2}$	2	0	-2 ✓
$\rho^h(x)$				
$f^{abc} f^{deh} U^{ae}(w') U^{bl}(x) U^{cd}(x') \rho^l(x)$	I.II _{Λ} : -1 ; I.III _P : $\frac{1}{2}$; I.III _{Λ} : $\frac{1}{2}$; I.IV _P : $\frac{1}{2}$; I.IV _{Λ} : $\frac{1}{2}$; III.I _P : $-\frac{1}{2}$; III.I _{Λ} : $-\frac{1}{2}$	0	0	0 ✓
$\rho^h(x')$				
$f^{abc} f^{deh} U^{ad}(w) U^{bl}(x') U^{ce}(x) \rho^l(x')$	I.II _{Λ} : 1; I.III _P : $-\frac{1}{2}$; I.III _{Λ} : $-\frac{1}{2}$; I.IV _P : $-\frac{1}{2}$; I.IV _{Λ} : $-\frac{1}{2}$; III.I _P : $\frac{1}{2}$; III.I _{Λ} : $\frac{1}{2}$	0	0	0 ✓
$\rho^h(x)$				
$f^{abc} f^{deh} U^{ae}(w') U^{cl}(x) U^{bd}(z) \rho^l(x)$	I.III _P : $\frac{1}{2}$; I.III _{Λ} : $\frac{1}{2}$; I.IV _P : $-\frac{1}{2}$; I.IV _{Λ} : $-\frac{1}{2}$; II.I _P : $\frac{1}{2}$; II.I _{Λ} : $-\frac{1}{2}$; III.I _P : $\frac{1}{2}$; III.I _{Λ} : $\frac{1}{2}$; III.II _P : $-\frac{1}{2}$; III.II _{Λ} : $-\frac{1}{2}$	0	0	0 ✓
$\rho^h(x')$				
$f^{abc} f^{deh} U^{ad}(w) U^{cl}(x') U^{be}(z) \rho^l(x')$	I.III _P : $-\frac{1}{2}$; I.III _{Λ} : $-\frac{1}{2}$; I.IV _P : $\frac{1}{2}$; I.IV _{Λ} : $\frac{1}{2}$; II.I _P : $-\frac{1}{2}$; II.I _{Λ} : $\frac{1}{2}$; III.I _P : $-\frac{1}{2}$; III.I _{Λ} : $-\frac{1}{2}$; III.II _P : $\frac{1}{2}$; III.II _{Λ} : $\frac{1}{2}$	0	0	0 ✓
$\rho^h(x)$				
$f^{abc} f^{deh} U^{le}(w') U^{la}(w) U^{mb}(z)$	I.IV _P : $\frac{1}{2}$; I.IV _{Λ} : $\frac{1}{2}$; III.I _P : $-\frac{1}{2}$; III.I _{Λ} : $-\frac{1}{2}$	0	0	0 ✓
$U^{md}(x') \rho^h(x') \rho^c(x)$				
$f^{abc} f^{deh} U^{ld}(w) U^{la}(w') U^{mb}(z)$	I.IV _P : $-\frac{1}{2}$; I.IV _{Λ} : $-\frac{1}{2}$; III.I _P : $\frac{1}{2}$; III.I _{Λ} : $\frac{1}{2}$	0	0	0 ✓
$U^{me}(x) \rho^c(x') \rho^h(x)$				

Structure 10: $\mathcal{K}(x - w) \cdot \mathcal{K}(x' - w') K(x', x', z)$

colour structure	contributions (row: coefficient)	sum	BFKL	JIMWLK
$U^{ab}(x) U^{ac}(w') \rho^c(x') \rho^b(x)$	III.II _P : $\frac{N_c}{2}$; III.II _{Λ} : $\frac{N_c}{2}$	N_c	N_c	0 ✓
$U^{ab}(x) U^{ac}(x') \rho^c(x') \rho^b(x)$	I.III _P : $\frac{N_c}{2}$; I.III _{Λ} : $\frac{N_c}{2}$; I.IV _P : $-\frac{N_c}{2}$; I.IV _{Λ} : $-\frac{N_c}{2}$; II.I _P : $-\frac{N_c}{2}$; II.I _{Λ} : $-\frac{N_c}{2}$; III.I _P : $\frac{N_c}{2}$; III.I _{Λ} : $\frac{N_c}{2}$; III.II _P : $\frac{N_c}{2}$; III.II _{Λ} : $\frac{N_c}{2}$	N_c	$-N_c$	$-2N_c$ ✓
$U^{ab}(w) U^{ac}(w') \rho^c(x') \rho^b(x)$	II.I _P : $-\frac{N_c}{2}$; II.I _{Λ} : $-\frac{N_c}{2}$	$-N_c$	$-N_c$	0 ✓
$U^{ab}(w) U^{ac}(x') \rho^c(x') \rho^b(x)$	I.III _P : $-\frac{N_c}{2}$; I.III _{Λ} : $-\frac{N_c}{2}$; I.IV _P : $\frac{N_c}{2}$; I.IV _{Λ} : $\frac{N_c}{2}$; III.I _P : $-\frac{N_c}{2}$; III.I _{Λ} : $-\frac{N_c}{2}$	$-N_c$	N_c	$2N_c$ ✓
$f^{abc} f^{deh} U^{al}(x) U^{cd}(x') U^{be}(z) \rho^h(x')$	I.III _P : $\frac{1}{2}$; I.III _{Λ} : $\frac{1}{2}$; III.II _P : $\frac{1}{2}$; III.II _{Λ} : $\frac{1}{2}$	2	0	-2 ✓
$\rho^l(x)$				
$f^{abc} f^{deh} U^{al}(w) U^{cd}(x') U^{be}(z) \rho^h(x')$	I.III _P : $-\frac{1}{2}$; I.III _{Λ} : $-\frac{1}{2}$; II.I _P : $-\frac{1}{2}$; II.I _{Λ} : $-\frac{1}{2}$	-2	0	2 ✓
$\rho^l(x)$				
$f^{abc} f^{dce} U^{hl}(x) U^{ha}(w') U^{mb}(z)$	I.IV _P : $\frac{1}{2}$; I.IV _{Λ} : $\frac{1}{2}$; II.I _P : $\frac{1}{2}$; II.I _{Λ} : $\frac{1}{2}$;	0	0	0 ✓
$U^{md}(x') \rho^e(x') \rho^l(x)$	III.I _P : $-\frac{1}{2}$; III.I _{Λ} : $-\frac{1}{2}$; III.II _P : $-\frac{1}{2}$; III.II _{Λ} : $-\frac{1}{2}$			

colour structure	contributions (row: coefficient)	sum	BFKL	JIMWLK
$f^{abc} f^{dce} U^{hl}(w) U^{ha}(w') U^{mb}(z)$ $U^{md}(x') \rho^e(x') \rho^l(x)$	I.IV _P : $-\frac{1}{2}$; I.IV _{Λ} : $-\frac{1}{2}$; III.I _P : $\frac{1}{2}$; III.I _{Λ} : $\frac{1}{2}$	0	0	0 ✓

Structure 11: $\mathcal{K}(x-w) \cdot \mathcal{K}(x'-w') K(w, w, z)$

colour structure	contributions (row: coefficient)	sum	BFKL	JIMWLK
$U^{ab}(x') U^{ac}(w) \rho^b(x') \rho^c(x)$	I.II _{Λ} : $-N_c$; II.IV _{Λ} : $-N_c$	$-2N_c$	0	$2N_c$ ✓
$U^{ab}(w') U^{ac}(w) \rho^b(x') \rho^c(x)$	I.II _{Λ} : N_c ; II.IV _{Λ} : N_c	$2N_c$	0	$-2N_c$ ✓
$f^{abc} f^{deh} U^{al}(x') U^{be}(z) U^{ch}(w) \rho^l(x')$ $\rho^d(x)$	I.V _{Λ} : 2	2	0	-2 ✓
$f^{abc} f^{deh} U^{al}(w') U^{be}(z) U^{ch}(w) \rho^l(x')$ $\rho^d(x)$	I.V _{Λ} : -2	-2	0	2 ✓
$U^{ab}(x') U^{ac}(x) \rho^b(x') \rho^c(x)$	I.II _{Λ} : N_c ; I.V _{Λ} : $-2N_c$; II.IV _{Λ} : N_c	0	0	0 ✓
$U^{ab}(w') U^{ac}(x) \rho^b(x') \rho^c(x)$	I.II _{Λ} : $-N_c$; I.V _{Λ} : $2N_c$; II.IV _{Λ} : $-N_c$	0	0	0 ✓

Structure 12: $\mathcal{K}(x-w) \cdot \mathcal{K}(x'-w') K(w', w', z)$

colour structure	contributions (row: coefficient)	sum	BFKL	JIMWLK
$U^{ab}(x) U^{ac}(w') \rho^c(x') \rho^b(x)$	I.II _{Λ} : $-N_c$; II.IV _{Λ} : $-N_c$	$-2N_c$	0	$2N_c$ ✓
$U^{ab}(w) U^{ac}(w') \rho^c(x') \rho^b(x)$	I.II _{Λ} : N_c ; II.IV _{Λ} : N_c	$2N_c$	0	$-2N_c$ ✓
$f^{abc} f^{deh} U^{al}(x) U^{be}(z) U^{ch}(w') \rho^d(x')$ $\rho^l(x)$	I.V _{Λ} : 2	2	0	-2 ✓
$f^{abc} f^{deh} U^{al}(w) U^{be}(z) U^{ch}(w') \rho^d(x')$ $\rho^l(x)$	I.V _{Λ} : -2	-2	0	2 ✓
$U^{ab}(x) U^{ac}(x') \rho^c(x') \rho^b(x)$	I.II _{Λ} : N_c ; I.V _{Λ} : $-2N_c$; II.IV _{Λ} : N_c	0	0	0 ✓
$U^{ab}(w) U^{ac}(x') \rho^c(x') \rho^b(x)$	I.II _{Λ} : $-N_c$; I.V _{Λ} : $2N_c$; II.IV _{Λ} : $-N_c$	0	0	0 ✓

Structure 13: $\mathcal{K}(x'-w') \cdot \mathcal{K}(w-z) K(x, x, z)$

colour structure	contributions (row: coefficient)	sum	BFKL	JIMWLK
$U^{ab}(w') U^{ac}(w) \rho^b(x') \rho^c(x)$	II.I _P : $-N_c$	$-N_c$	$-N_c$	0 ✓
$U^{ab}(x') U^{ac}(w) \rho^b(x') \rho^c(x)$	III.II _P : N_c	N_c	N_c	0 ✓
$U^{ab}(w') U^{ca}(z) \rho^b(x') \rho^c(x)$	III.III _P : N_c	N_c	N_c	0 ✓
$U^{ab}(x') U^{ca}(z) \rho^b(x') \rho^c(x)$	III.VI _P : $-N_c$	$-N_c$	$-N_c$	0 ✓
$U^{ab}(x') U^{ac}(x) \rho^b(x') \rho^c(x)$	I.III _P : N_c ; I.IV _P : $-N_c$; II.I _P : $-N_c$; III.I _P : N_c ; III.II _P : N_c ; III.III _P : N_c ; III.V _P : $-N_c$; III.VI _P : $-N_c$	0	0	0 ✓
$U^{ab}(w') U^{ac}(x) \rho^b(x') \rho^c(x)$	I.III _P : $-N_c$; I.IV _P : N_c ; III.I _P : $-N_c$; III.V _P : N_c	0	0	0 ✓
$f^{abc} f^{deh} U^{al}(x') U^{ce}(x) U^{bd}(z) \rho^l(x')$ $\rho^h(x)$	I.III _P : -1 ; III.II _P : -1 ; III.III _P : -1 ; III.V _P : 1; III.VI _P : 2	0	0	0 ✓
$f^{abc} f^{deh} U^{al}(w') U^{ce}(x) U^{bd}(z) \rho^l(x')$ $\rho^h(x)$	I.III _P : 1; II.I _P : 1; III.III _P : -1 ; III.V _P : -1	0	0	0 ✓
$f^{abc} f^{cde} U^{hl}(x') U^{ha}(w) U^{mb}(z)$ $U^{md}(x) \rho^l(x') \rho^e(x)$	I.IV _P : -1 ; II.I _P : -1 ; III.I _P : 1; III.II _P : 1	0	0	0 ✓
$f^{abc} f^{cde} U^{hl}(w') U^{ha}(w) U^{mb}(z)$ $U^{md}(x) \rho^l(x') \rho^e(x)$	I.IV _P : 1; III.I _P : -1	0	0	0 ✓

Structure 14: $\mathcal{K}(x'-w') \cdot \mathcal{K}(w-z) K(x, x', z)$

colour structure	contributions (row: coefficient)	sum	BFKL	JIMWLK
$f^{abc} f^{bde} U^{hd}(w') U^{ha}(w) \rho^e(x') \rho^c(x)$	II.I _P : -2	-2	-2	0 ✓
$f^{abc} f^{cde} U^{hd}(x') U^{ha}(w) \rho^b(x) \rho^e(x')$	III.II _P : -2	-2	-2	0 ✓
$f^{abc} f^{ade} U^{hb}(w') U^{eh}(z) \rho^d(x) \rho^c(x')$	III.III _P : 2	2	2	0 ✓
$f^{abc} f^{ade} U^{hb}(x') U^{eh}(z) \rho^d(x) \rho^c(x')$	III.VI _P : -2	-2	-2	0 ✓
$U^{ab}(z) U^{ac}(x') \rho^c(x') \rho^b(x)$	I.III _P : $-N_c$; III.II _P : N_c ; III.III _P : N_c ; III.V _P : N_c ; III.VI _P : $-2N_c$	0	0	0 ✓

colour structure	contributions (row: coefficient)	sum	BFKL	JIMWLK
$U^{ab}(x) U^{ac}(x') \rho^c(x') \rho^b(x)$	II.I _P : $2N_c$; III.II _P : $-2N_c$; III.III _P : $-2N_c$; III.VI _P : $2N_c$	0	0	0 ✓
$f^{abc} f^{deh} U^{ae}(x') U^{cl}(x) U^{bd}(z) \rho^l(x)$ $\rho^h(x')$	I.III _P : -1 ; III.II _P : -1 ; III.III _P : -1 ; III.V _P : 1 ; III.VI _P : 2	0	0	0 ✓
$f^{abc} f^{deh} U^{ae}(w') U^{cd}(x') U^{bl}(z) \rho^h(x')$ $\rho^l(x)$	I.III _P : 1 ; II.I _P : -1 ; III.III _P : 1 ; III.V _P : -1	0	0	0 ✓
$f^{abc} f^{deh} U^{ae}(w') U^{cl}(x) U^{bd}(z) \rho^l(x)$ $\rho^h(x')$	I.III _P : 1 ; II.I _P : 1 ; III.III _P : -1 ; III.V _P : -1	0	0	0 ✓
$f^{abc} f^{deh} U^{ea}(w) U^{db}(z) U^{hl}(x') \rho^l(x')$ $\rho^c(x)$	I.IV _P : -1 ; II.I _P : 1 ; III.I _P : 1 ; III.II _P : -1	0	0	0 ✓
$f^{abc} f^{cde} U^{hd}(x') U^{ha}(w) U^{lb}(z)$ $U^{lm}(x) \rho^m(x) \rho^e(x')$	I.IV _P : -1 ; II.I _P : -1 ; III.I _P : 1 ; III.II _P : 1	0	0	0 ✓
$f^{abc} f^{deh} U^{le}(w') U^{la}(w) U^{mb}(z)$ $U^{md}(x') \rho^h(x') \rho^c(x)$	I.IV _P : 1 ; III.I _P : -1	0	0	0 ✓
$f^{abc} f^{cde} U^{hd}(w') U^{ha}(w) U^{lb}(z)$ $U^{lm}(x) \rho^m(x) \rho^e(x')$	I.IV _P : 1 ; III.I _P : -1	0	0	0 ✓

Structure 15: $\mathcal{K}(x - w) \cdot \mathcal{K}(w' - z) K(x, x', z)$

colour structure	contributions (row: coefficient)	sum	BFKL	JIMWLK
$f^{abc} f^{bde} U^{ha}(w') U^{hd}(w) \rho^c(x') \rho^e(x)$	II.I _P : -2	-2	-2	0 ✓
$f^{abc} f^{bde} U^{hd}(x) U^{ha}(w') \rho^c(x') \rho^e(x)$	III.II _P : 2	2	2	0 ✓
$f^{abc} f^{dae} U^{hb}(w) U^{eh}(z) \rho^d(x') \rho^c(x)$	III.III _P : -2	-2	-2	0 ✓
$f^{abc} f^{dae} U^{hb}(x) U^{eh}(z) \rho^d(x') \rho^c(x)$	III.VI _P : 2	2	2	0 ✓
$U^{ab}(z) U^{ac}(x) \rho^b(x') \rho^c(x)$	I.III _P : $-N_c$; III.II _P : N_c ; III.III _P : N_c ; III.V _P : N_c ; III.VI _P : $-2N_c$	0	0	0 ✓
$U^{ab}(x') U^{ac}(x) \rho^b(x') \rho^c(x)$	II.I _P : $2N_c$; III.II _P : $-2N_c$; III.III _P : $-2N_c$; III.VI _P : $2N_c$	0	0	0 ✓
$f^{abc} f^{deh} U^{ad}(x) U^{cl}(x') U^{be}(z) \rho^l(x')$ $\rho^h(x)$	I.III _P : 1 ; III.II _P : 1 ; III.III _P : 1 ; III.V _P : -1 ; III.VI _P : -2	0	0	0 ✓
$f^{abc} f^{deh} U^{ad}(w) U^{ce}(x) U^{bl}(z) \rho^l(x')$ $\rho^h(x)$	I.III _P : -1 ; II.I _P : 1 ; III.III _P : -1 ; III.V _P : 1	0	0	0 ✓
$f^{abc} f^{deh} U^{ad}(w) U^{cl}(x') U^{be}(z) \rho^l(x')$ $\rho^h(x)$	I.III _P : -1 ; II.I _P : -1 ; III.III _P : 1 ; III.V _P : 1	0	0	0 ✓
$f^{abc} f^{deh} U^{da}(w') U^{eb}(z) U^{hl}(x) \rho^c(x')$ $\rho^l(x)$	I.IV _P : 1 ; II.I _P : -1 ; III.I _P : -1 ; III.II _P : 1	0	0	0 ✓
$f^{abc} f^{dce} U^{hd}(x) U^{ha}(w') U^{lb}(z)$ $U^{lm}(x') \rho^m(x') \rho^e(x)$	I.IV _P : 1 ; II.I _P : 1 ; III.I _P : -1 ; III.II _P : -1	0	0	0 ✓
$f^{abc} f^{deh} U^{ld}(w) U^{la}(w') U^{mb}(z)$ $U^{me}(x) \rho^c(x') \rho^h(x)$	I.IV _P : -1 ; III.I _P : 1	0	0	0 ✓
$f^{abc} f^{dce} U^{hd}(w) U^{ha}(w') U^{lb}(z)$ $U^{lm}(x') \rho^m(x') \rho^e(x)$	I.IV _P : -1 ; III.I _P : 1	0	0	0 ✓

Structure 16: $\mathcal{K}(x - w) \cdot \mathcal{K}(w' - z) K(x', x', z)$

colour structure	contributions (row: coefficient)	sum	BFKL	JIMWLK
$U^{ab}(w') U^{ac}(w) \rho^c(x) \rho^b(x')$	II.I _P : $-N_c$	$-N_c$	$-N_c$	0 ✓
$U^{ab}(x) U^{ac}(w') \rho^b(x) \rho^c(x')$	III.II _P : N_c	N_c	N_c	0 ✓
$U^{ab}(w) U^{ca}(z) \rho^b(x) \rho^c(x')$	III.III _P : N_c	N_c	N_c	0 ✓
$U^{ab}(x) U^{ca}(z) \rho^b(x) \rho^c(x')$	III.VI _P : $-N_c$	$-N_c$	$-N_c$	0 ✓
$U^{ab}(x) U^{ac}(x') \rho^c(x') \rho^b(x)$	I.III _P : N_c ; I.IV _P : $-N_c$; II.I _P : $-N_c$; III.I _P : N_c ; III.II _P : N_c ; III.III _P : N_c ; III.V _P : $-N_c$; III.VI _P : $-N_c$	0	0	0 ✓
$U^{ab}(w) U^{ac}(x') \rho^c(x') \rho^b(x)$	I.III _P : $-N_c$; I.IV _P : N_c ; III.I _P : $-N_c$; III.V _P : N_c	0	0	0 ✓
$f^{abc} f^{deh} U^{al}(x) U^{cd}(x') U^{be}(z) \rho^h(x')$ $\rho^l(x)$	I.III _P : 1 ; III.II _P : 1 ; III.III _P : 1 ; III.V _P : -1 ; III.VI _P : -2	0	0	0 ✓
$f^{abc} f^{deh} U^{al}(w) U^{cd}(x') U^{be}(z) \rho^h(x')$ $\rho^l(x)$	I.III _P : -1 ; II.I _P : -1 ; III.III _P : 1 ; III.V _P : 1	0	0	0 ✓

colour structure	contributions (row: coefficient)	sum	BFKL	JIMWLK
$f^{abc} f^{dce} U^{hl}(x) U^{ha}(w') U^{mb}(z)$ $U^{md}(x') \rho^e(x') \rho^l(x)$	I.IV _P : 1; II.I _P : 1; III.I _P : -1; III.II _P : -1	0	0	0 ✓
$f^{abc} f^{dce} U^{hl}(w) U^{ha}(w') U^{mb}(z)$ $U^{md}(x') \rho^e(x') \rho^l(x)$	I.IV _P : -1; III.I _P : 1	0	0	0 ✓

Structure 17: $\mathcal{K}(x-w) \cdot \mathcal{K}(x'-w') K(w, w', z)$

colour structure	contributions (row: coefficient)	sum	BFKL	JIMWLK
$U^{ab}(w') U^{cb}(w) \rho^a(x') \rho^c(x)$	II.IV _Λ : $-2N_c$	$-2N_c$	0	$2N_c$ ✓
$f^{abc} f^{dbe} U^{ha}(w') U^{hd}(w) \rho^c(x') \rho^e(x)$	II.I _Λ : -2	-2	0	2 ✓
$f^{abc} f^{deh} U^{la}(w') U^{be}(z) U^{cd}(w) \rho^l(x')$ $\rho^h(x)$	III.III _Λ : -2	-2	0	2 ✓
$f^{abc} f^{deh} U^{la}(w) U^{be}(z) U^{cd}(w') \rho^h(x')$ $\rho^l(x)$	III.III _Λ : -2	-2	0	2 ✓
$U^{ab}(x') U^{bc}(x) \rho^a(x') \rho^c(x)$	II.I _Λ : $-2N_c$; II.IV _Λ : $-2N_c$; III.III _Λ : $4N_c$	0	0	0 ✓
$U^{ab}(w') U^{bc}(x) \rho^a(x') \rho^c(x)$	II.IV _Λ : $2N_c$; III.III _Λ : $-2N_c$	0	0	0 ✓
$U^{ab}(w) U^{bc}(x') \rho^c(x') \rho^a(x)$	II.IV _Λ : $2N_c$; III.III _Λ : $-2N_c$	0	0	0 ✓
$f^{abc} f^{deh} U^{da}(w') U^{be}(z) U^{hl}(x) \rho^c(x')$ $\rho^l(x)$	II.I _Λ : 2; III.III _Λ : -2	0	0	0 ✓
$f^{abc} f^{deh} U^{ad}(w) U^{eb}(z) U^{lc}(x') \rho^l(x')$ $\rho^h(x)$	II.I _Λ : 2; III.III _Λ : -2	0	0	0 ✓

Structure 18: $K(w, w', z) K(x, x', z)$

colour structure	contributions (row: coefficient)	sum	BFKL	JIMWLK
$\rho^a(x) \rho^a(x')$	II.IV _P : $2N_c$	$2N_c$	$2N_c$	0 ✓
$f^{abc} f^{dbe} U^{ha}(w') U^{hd}(w) \rho^c(x) \rho^e(x')$	II.I _P : 1	1	1	0 ✓
$f^{abc} f^{dbe} U^{ha}(w') U^{hd}(w) \rho^c(x') \rho^e(x)$	II.I _P : 1	1	1	0 ✓
$f^{abc} f^{dbe} U^{ha}(w) U^{eh}(z) \rho^d(x) \rho^c(x')$	III.III _P : 1	1	1	0 ✓
$f^{abc} f^{dbe} U^{ha}(w) U^{eh}(z) \rho^d(x') \rho^c(x)$	III.III _P : 1	1	1	0 ✓
$f^{abc} f^{dbe} U^{ha}(w') U^{eh}(z) \rho^c(x) \rho^d(x')$	III.III _P : 1	1	1	0 ✓
$f^{abc} f^{dbe} U^{ha}(w') U^{eh}(z) \rho^c(x') \rho^d(x)$	III.III _P : 1	1	1	0 ✓
$U^{ab}(x) U^{bc}(x') \rho^a(x) \rho^c(x')$	II.I _P : $2N_c$; II.IV _P : $2N_c$; III.III _P : $-4N_c$	0	0	0 ✓
$U^{ab}(z) U^{bc}(x') \rho^a(x) \rho^c(x')$	II.IV _P : $-2N_c$; III.III _P : $2N_c$	0	0	0 ✓
$U^{ab}(z) U^{bc}(x) \rho^a(x') \rho^c(x)$	II.IV _P : $-2N_c$; III.III _P : $2N_c$	0	0	0 ✓
$f^{abc} f^{deh} U^{da}(w') U^{be}(z) U^{hl}(x') \rho^c(x)$ $\rho^l(x')$	II.I _P : -1; III.III _P : 1	0	0	0 ✓
$f^{abc} f^{deh} U^{da}(w') U^{be}(z) U^{hl}(x) \rho^c(x')$ $\rho^l(x)$	II.I _P : -1; III.III _P : 1	0	0	0 ✓
$f^{abc} f^{deh} U^{ad}(w) U^{eb}(z) U^{lc}(x) \rho^l(x)$ $\rho^h(x')$	II.I _P : -1; III.III _P : 1	0	0	0 ✓
$f^{abc} f^{deh} U^{ad}(w) U^{eb}(z) U^{lc}(x') \rho^l(x')$ $\rho^h(x)$	II.I _P : -1; III.III _P : 1	0	0	0 ✓

9 The BFKL terms

Operator form. Let $O(x', x) = \rho^{b'}(x') M^{b'b}(x', x) \rho^b(x)$ with $M^{b'b}(p', p) = [U(w') - U(p')]^{ab'} [U(w) - U(p)]^{ab}$. The leading-order cross section is $\frac{d^3 N_{LO}}{d^3 k} \propto \int \mathcal{K}(x' - w') \cdot \mathcal{K}(x - w) \langle O(x', x) \rangle$. For $p^+ \gg k^+$ the gluon at z is added to the projectile: its charge $j^a(z)$ adds to ρ^a , and the valence charges acquire the virtual corrections. At order α_s the result is

$$\begin{aligned}
\text{BFKL} \otimes \text{LO} = 2\mathcal{N}_P \int_{x, x', u, v, z} e^{-ik \cdot (w' - w)} K(u, v, z) \Big\{ \\
& - \frac{1}{2} \mathcal{K}(x' - w') \cdot \mathcal{K}(x - w) \left[\rho^h(u) \rho^h(v) O(x', x) - 2\rho^h(u) O(x', x) \rho^h(v) + O(x', x) \rho^h(u) \rho^h(v) \right] \\
& + \mathcal{K}(z - w') \cdot \mathcal{K}(z - w) \rho^{e'}(u) \rho^e(v) (T^{b'} T^b)_{e'e} M^{b'b}(z, z) \\
& + \mathcal{K}(x' - w') \cdot \mathcal{K}(z - w) \rho^{e'}(u) \rho^{b'}(x') \rho^e(v) (T^b)_{e'e} M^{b'b}(x', z) \\
& + \mathcal{K}(z - w') \cdot \mathcal{K}(x - w) \rho^{e'}(u) \rho^b(x) \rho^e(v) (T^{b'})_{e'e} M^{b'b}(z, x) \Big\}, \tag{9.1}
\end{aligned}$$

with $(T^a)_{bc} = -if^{abc}$. The first line is the virtual correction (the double commutator $-\frac{1}{2}[\rho^h(u), [\rho^h(v), O]]$). The second line has both measured gluons emitted by the gluon at z . The last two lines have one measured gluon emitted by a valence charge and the other by the gluon at z .

Fully written out. After reordering the charges (Eq. (2.2)) and simplifying the colour algebra, Eq. (9.1) is a sum of two- ρ terms. Its symmetrized three- ρ and four- ρ parts vanish. In the basis of Sec. 8:

$$\begin{aligned}
\text{BFKL} \otimes \text{LO} = \mathcal{N}_P \int e^{-ik \cdot (w' - w)} \int_{x, x', z} \Big\{ & \\
& + \mathcal{K}(x - w) \cdot \mathcal{K}(x' - w') K(x, x, z) \Big[\\
& \quad + N_c U^{ab}(x') U^{ac}(w) \rho^b(x') \rho^c(x) \\
& \quad - N_c U^{ab}(x') U^{ac}(x) \rho^b(x') \rho^c(x) \\
& \quad - N_c U^{ab}(w') U^{ac}(w) \rho^b(x') \rho^c(x) \\
& \quad + N_c U^{ab}(w') U^{ac}(x) \rho^b(x') \rho^c(x) \Big] \\
& + \mathcal{K}(x - w) \cdot \mathcal{K}(x' - w') K(x, x', z) \Big[\\
& \quad - 2 f^{abc} f^{cde} U^{hd}(x') U^{ha}(w) \rho^b(x) \rho^e(x') \\
& \quad + 2 f^{abc} f^{dce} U^{hd}(x) U^{ha}(w') \rho^b(x') \rho^e(x) \\
& \quad + 2 f^{abc} f^{cde} U^{hd}(w') U^{ha}(w) \rho^b(x) \rho^e(x') \\
& \quad + 2 f^{abc} f^{dbe} U^{ha}(x') U^{hd}(x) \rho^c(x') \rho^e(x) \Big] \\
& + \mathcal{K}(x - w) \cdot \mathcal{K}(x' - w') K(x', x', z) \Big[\\
& \quad + N_c U^{ab}(x) U^{ac}(w') \rho^c(x') \rho^b(x) \\
& \quad - N_c U^{ab}(x) U^{ac}(x') \rho^c(x') \rho^b(x) \\
& \quad - N_c U^{ab}(w) U^{ac}(w') \rho^c(x') \rho^b(x) \\
& \quad + N_c U^{ab}(w) U^{ac}(x') \rho^c(x') \rho^b(x) \Big] \\
& + \mathcal{K}(x' - w') \cdot \mathcal{K}(w - z) K(x, x, z) \Big[\\
& \quad - N_c U^{ab}(w') U^{ac}(w) \rho^b(x') \rho^c(x) \\
& \quad + N_c U^{ab}(x') U^{ac}(w) \rho^b(x') \rho^c(x) \\
& \quad + N_c U^{ab}(w') U^{ca}(z) \rho^b(x') \rho^c(x) \\
& \quad - N_c U^{ab}(x') U^{ca}(z) \rho^b(x') \rho^c(x) \Big] \\
& + \mathcal{K}(x' - w') \cdot \mathcal{K}(w - z) K(x, x', z) \Big[\\
& \quad - 2 f^{abc} f^{bde} U^{hd}(w') U^{ha}(w) \rho^e(x') \rho^c(x) \\
& \quad - 2 f^{abc} f^{cde} U^{hd}(x') U^{ha}(w) \rho^b(x) \rho^e(x') \\
& \quad + 2 f^{abc} f^{ade} U^{hb}(w') U^{eh}(z) \rho^d(x) \rho^c(x') \\
& \quad - 2 f^{abc} f^{ade} U^{hb}(x') U^{eh}(z) \rho^d(x) \rho^c(x') \Big] \\
& + \mathcal{K}(x - w) \cdot \mathcal{K}(w' - z) K(x, x', z) \Big[\\
& \quad - 2 f^{abc} f^{bde} U^{ha}(w') U^{hd}(w) \rho^c(x') \rho^e(x) \\
& \quad + 2 f^{abc} f^{bde} U^{hd}(x) U^{ha}(w') \rho^c(x') \rho^e(x) \\
& \quad - 2 f^{abc} f^{dae} U^{hb}(w) U^{eh}(z) \rho^d(x') \rho^c(x) \\
& \quad + 2 f^{abc} f^{dae} U^{hb}(x) U^{eh}(z) \rho^d(x') \rho^c(x) \Big] \\
& + \mathcal{K}(x - w) \cdot \mathcal{K}(w' - z) K(x', x', z) \Big[\\
& \quad - N_c U^{ab}(w') U^{ac}(w) \rho^c(x) \rho^b(x') \\
& \quad + N_c U^{ab}(x) U^{ac}(w') \rho^b(x) \rho^c(x') \\
& \quad + N_c U^{ab}(w) U^{ca}(z) \rho^b(x) \rho^c(x') \\
& \quad - N_c U^{ab}(x) U^{ca}(z) \rho^b(x) \rho^c(x') \Big]
\end{aligned}$$

$$\begin{aligned}
& + K(w, w', z) K(x, x', z) \Big[\\
& \quad + 2N_c \rho^a(x) \rho^a(x') \\
& \quad + f^{abc} f^{dbe} U^{ha}(w') U^{hd}(w) \rho^c(x) \rho^e(x') \\
& \quad + f^{abc} f^{dbe} U^{ha}(w') U^{hd}(w) \rho^c(x') \rho^e(x) \\
& \quad + f^{abc} f^{dbe} U^{ha}(w) U^{eh}(z) \rho^d(x) \rho^c(x') \\
& \quad + f^{abc} f^{dbe} U^{ha}(w) U^{eh}(z) \rho^d(x') \rho^c(x) \\
& \quad + f^{abc} f^{dbe} U^{ha}(w') U^{eh}(z) \rho^c(x) \rho^d(x') \\
& \quad + f^{abc} f^{dbe} U^{ha}(w') U^{eh}(z) \rho^c(x') \rho^d(x) \Big] \Big\} \tag{9.2}
\end{aligned}$$

Colour-singlet projection: Eq. (LOBFKL). For $\langle \rho^a(p) \rho^b(q) \rangle = \delta^{ab} \mu(p, q) / (N_c^2 - 1)$, Eq. (9.1) reduces to Eq. (LOBFKL) of your Appendix E:

$$\begin{aligned}
& - \frac{\alpha_s^2 N_c \delta Y}{8\pi^6 (N_c^2 - 1)} \int_{x, y, z} \mu(x, y) \Big[\mathcal{K}(x - w) \cdot \mathcal{K}(y - w') K_D(x, y, z) \text{Tr}\{(U^\dagger(x) - U^\dagger(w))(U(y) - U(w'))\} \\
& \quad - 2 \mathcal{K}(z - w) \cdot \mathcal{K}(z - w') K(x, y, z) \text{Tr}\{(U^\dagger(z) - U^\dagger(w))(U(z) - U(w'))\} \\
& \quad + \mathcal{K}(z - w) \cdot \mathcal{K}(y - w') (2K(x, y, z) - K(x, x, z)) \text{Tr}\{(U^\dagger(z) - U^\dagger(w))(U(y) - U(w'))\} \\
& \quad + \mathcal{K}(x - w) \cdot \mathcal{K}(z - w') (2K(x, y, z) - K(y, y, z)) \text{Tr}\{(U^\dagger(x) - U^\dagger(w))(U(z) - U(w'))\} \Big], \tag{9.3}
\end{aligned}$$

with $K_D(x, y, z) = K(x, x, z) - 2K(x, y, z) + K(y, y, z)$. Its lines match those of Eq. (9.1): the first is the virtual (dipole) term, the second the term with both measured gluons from z , and the last two are the mixed terms. Table 2 compares the two numerically at random points, for symmetric and for general adjoint U . They agree to machine precision, so **Appendix E is correct as written** (unlike Appendix F; see the previous note).

U	seed	Eq. (9.1), singlet projection	Eq. (LOBFKL)
symmetric	1	79.30732887	79.30732887
symmetric	2	-107.56552336	-107.56552336
symmetric	3	0.08995397	0.08995397
general	1	95.24753301	95.24753301
general	2	91.64627107	91.64627107
general	3	0.13321489	0.13321489

Table 2: BFKL $\otimes$ LO at random points (x, y, w, w', z) , in units of $P\mu$ with $P = \alpha_s^2 \delta Y / (8\pi^6 (N_c^2 - 1))$.

The coefficient of $\log \vee$. Adding the coefficients of $\log(\vee/\Lambda)$ (JIMWLK $\otimes$ LO, previous note) and of $\log(\vee/k^+)$ (this note) gives BFKL $\otimes$ LO, Eq. (9.1): the projectile evolves with $\log \vee$, and the target with $\log(k^+/\Lambda)$.

10 Numerical checks

Row by row, region P . The coefficient of $\log \vee$ of each row of the notes, P_{row} , compared with an independent leading-log computation of the region $p^+ \gg k^+$. That computation is the one of `BFKL_JIMWLK_RowWise.pdf`, checked against an exact operator model of the wave function. Its rows are organized by intermediate state, as yours are. Group I is compared as a whole, because its rows are organized differently there. Pointwise in z at random points, for the symmetrized two- ρ and three- ρ parts:

row	charges	as written	corrected
Group I (all rows)	2	12 / 18	0 / 18 (max 2.3×10^{-13})
Group II, Row I	2	0 / 19	0 / 19 (max 2.9×10^{-14})
Group II, Row II	2	0 / 10	0 / 10 (max 2.0×10^{-15})
Group II, Row III	2	0 / 10	0 / 10 (max 4.4×10^{-16})
Group II, Row IV	2	0 / 1	0 / 1 (max 8.9×10^{-15})
Group III, Row I	2	0 / 18	0 / 18 (max 7.6×10^{-14})
Group III, Row II	2	0 / 18	0 / 18 (max 8.7×10^{-14})
Group III, Row III	2	0 / 9	0 / 9 (max 5.3×10^{-15})
Group III, Row IV	2	0 / 10	0 / 10 (max 3.9×10^{-14})
Group III, Row V	2	0 / 8	0 / 8 (max 5.3×10^{-15})
Group III, Row VI	2	0 / 8	0 / 8 (max 3.6×10^{-15})
Group I (all rows)	3	18 / 27	0 / 27 (max 3.5×10^{-14})
Group II, Row I	3	0 / 27	0 / 27 (max 1.0×10^{-14})
Group II, Row II	3	0 / 21	0 / 21 (max 1.6×10^{-15})
Group II, Row III	3	0 / 21	0 / 21 (max 3.3×10^{-16})
Group III, Row I	3	0 / 27	0 / 27 (max 1.1×10^{-14})
Group III, Row II	3	0 / 27	0 / 27 (max 1.2×10^{-14})
Group III, Row III	3	0 / 6	0 / 6 (max 9.4×10^{-16})
Group III, Row IV	3	0 / 21	0 / 21 (max 5.3×10^{-15})
Group III, Row V	3	0 / 6	0 / 6 (max 4.9×10^{-15})
Group III, Row VI	3	0 / 6	0 / 6 (max 1.6×10^{-15})

Table 3: Coefficient of $\log \vee$ of each row vs the independent region- P computation: number of differing kernel structures / total.

“corrected” means: Group I with P1 and with the α part of Row II removed from P (Sec. 4.1.2), i.e. fix F4 of the previous note. Each entry gives the number of kernel structures that differ, out of the total.

Cumulative fixes. Number of kernel structures (pointwise in z) where the $\log(\vee/k^+)$ terms differ from BFKL $\otimes$ LO–JIMWLK $\otimes$ LO:

state	two- ρ : differing / total	three- ρ (symmetrized): left over / total
as written	35 / 49	30 / 42
+ P1 (Group I Row III: sign of K_h)	31 / 49	24 / 42
+ F3 carried over (Row III part 2 $\rightarrow \mathcal{D}$)	11 / 49	0 / 30
+ F5 carried over (Group III Row III part 2: sign)	0 / 37	0 / 30

Table 4: The $\log(\vee/k^+)$ terms as written and after each fix.

Other configurations. At two other random configurations (seeds 21 and 33) the corrected two- ρ total again agrees in all structures (largest difference 8.3×10^{-14}). For a general (non-symmetric) adjoint U the notes do not apply, because they use $U^{bc} = U^{cb}$ throughout: 35 of 37 structures differ.

A Relations between colour networks

For the kernel structures in which the simplified colour structures are linearly dependent, each dependent one equals the following combination of basis structures (exact; checked for SU(3) and SU(4)):

Structure 7, $\mathcal{K}(x-w) \cdot \mathcal{K}(x'-w') K(x, x, z)$.

$$\begin{aligned}
& f^{abc} f^{deb} U^{hd}(z) U^{ha}(x) U^{lm}(w') U^{le}(x) \rho^m(x') \rho^c(x) \\
&= -f^{abc} f^{deh} U^{al}(w') U^{ce}(x) U^{bd}(z) \rho^l(x') \rho^h(x) - f^{abc} f^{dea} U^{hd}(z) U^{he}(x) U^{lm}(w') U^{lb}(x) \rho^m(x') \rho^c(x) \\
& f^{abc} f^{deb} U^{hd}(z) U^{ha}(x) U^{lm}(x') U^{le}(x) \rho^m(x') \rho^c(x) \\
&= -f^{abc} f^{deh} U^{al}(x') U^{ce}(x) U^{bd}(z) \rho^l(x') \rho^h(x) - f^{abc} f^{dea} U^{hd}(z) U^{he}(x) U^{lm}(x') U^{lb}(x) \rho^m(x') \rho^c(x)
\end{aligned}$$

Structure 8, $\mathcal{K}(x'-w) \cdot \mathcal{K}(x'-w') K(x, x', z)$.

$$\begin{aligned}
& f^{abc} f^{cde} U^{hb}(w') U^{ha}(w) \rho^d(x) \rho^e(x') \\
&= -f^{abc} f^{bde} U^{hd}(w') U^{ha}(w) \rho^e(x') \rho^c(x) - f^{abc} f^{dbe} U^{hd}(w) U^{ha}(w') \rho^c(x) \rho^e(x') \\
& f^{abc} f^{cde} U^{hd}(w') U^{ha}(x') \rho^b(x) \rho^e(x') \\
&= f^{abc} f^{dbe} U^{hd}(x') U^{ha}(w') \rho^c(x) \rho^e(x') + f^{abc} f^{cde} U^{hb}(w') U^{ha}(x') \rho^d(x) \rho^e(x') \\
& f^{abc} f^{dce} U^{ha}(x') U^{hd}(w) \rho^b(x) \rho^e(x')
\end{aligned}$$

$$= f^{abc} f^{bde} U^{hd}(x') U^{ha}(w) \rho^e(x') \rho^c(x) + f^{abc} f^{cde} U^{hb}(x') U^{ha}(w) \rho^d(x) \rho^e(x')$$

Structure 10, $\mathcal{K}(x-w) \cdot \mathcal{K}(x'-w') K(x', x', z)$.

$$\begin{aligned} & f^{abc} f^{deb} U^{hd}(z) U^{ha}(x') U^{le}(x') U^{lm}(w) \rho^c(x') \rho^m(x) \\ &= f^{abc} f^{deh} U^{al}(w) U^{cd}(x') U^{be}(z) \rho^h(x') \rho^l(x) - f^{abc} f^{dea} U^{hd}(z) U^{he}(x') U^{lb}(x') U^{lm}(w) \rho^c(x') \rho^m(x) \\ & f^{abc} f^{deb} U^{hd}(z) U^{ha}(x') U^{le}(x') U^{lm}(x) \rho^c(x') \rho^m(x) \\ &= f^{abc} f^{deh} U^{al}(x) U^{cd}(x') U^{be}(z) \rho^h(x') \rho^l(x) - f^{abc} f^{dea} U^{hd}(z) U^{he}(x') U^{lb}(x') U^{lm}(x) \rho^c(x') \rho^m(x) \end{aligned}$$

Structure 11, $\mathcal{K}(x-w) \cdot \mathcal{K}(x'-w') K(w, w, z)$.

$$\begin{aligned} & f^{abc} f^{deb} U^{hd}(z) U^{ha}(w) U^{lm}(w') U^{le}(w) \rho^m(x') \rho^c(x) \\ &= -f^{abc} f^{deh} U^{al}(w') U^{be}(z) U^{ch}(w) \rho^l(x') \rho^d(x) - f^{abc} f^{dea} U^{hd}(z) U^{he}(w) U^{lm}(w') U^{lb}(w) \rho^m(x') \rho^c(x) \\ & f^{abc} f^{deb} U^{hd}(z) U^{ha}(w) U^{lm}(x') U^{le}(w) \rho^m(x') \rho^c(x) \\ &= -f^{abc} f^{deh} U^{al}(x') U^{be}(z) U^{ch}(w) \rho^l(x') \rho^d(x) - f^{abc} f^{dea} U^{hd}(z) U^{he}(w) U^{lm}(x') U^{lb}(w) \rho^m(x') \rho^c(x) \end{aligned}$$

Structure 12, $\mathcal{K}(x-w) \cdot \mathcal{K}(x'-w') K(w', w', z)$.

$$\begin{aligned} & f^{abc} f^{deb} U^{hd}(z) U^{ha}(w') U^{le}(w') U^{lm}(w) \rho^c(x') \rho^m(x) \\ &= -f^{abc} f^{deh} U^{al}(w) U^{be}(z) U^{ch}(w') \rho^d(x') \rho^l(x) - f^{abc} f^{dea} U^{hd}(z) U^{he}(w') U^{lb}(w') U^{lm}(w) \rho^c(x') \rho^m(x) \\ & f^{abc} f^{deb} U^{hd}(z) U^{ha}(w') U^{le}(w') U^{lm}(x) \rho^c(x') \rho^m(x) \\ &= -f^{abc} f^{deh} U^{al}(x) U^{be}(z) U^{ch}(w') \rho^d(x') \rho^l(x) - f^{abc} f^{dea} U^{hd}(z) U^{he}(w') U^{lb}(w') U^{lm}(x) \rho^c(x') \rho^m(x) \end{aligned}$$